Quantum computation, quantum chemistry and Hamiltonian moments

by

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Abstract

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While sufficiently large and accurate quantum computers are expected to exploit quantum effects such as superposition and entanglement to gain an advantage over their classical counterparts for specific problems, it is unclear whether near- and medium-term quantum devices can provide such an advantage. For the problem of ground-state energy estimation, which is of particular interest as a first step into quantum chemistry, small-scale proof-of-principle experiments have been performed on current noisy devices, but scaling to larger problem instances has been hindered by noise inherent in today's quantum hardware making it difficult to achieve the required precision.

In this thesis, we investigate the use of Hamiltonian moments for improving on ground-state energy estimates from quantum devices, finding that the additional information incorporated into moments-corrected ground-state energy estimates allows accuracy beyond what would normally be possible for a given quantum circuit depth. This, combined with the inherent error robustness of the moments-based method, allowed us to perform calculations for Hydrogen chains beyond the Hartree-Fock approximation (Chapter 2) , and subsequently, the largest quantum chemical calculation to date to achieve millihartree accuracy on existing quantum hardware (Chapter 3) – computing the ground-state energy of the water molecule on 8 qubits. We then consider extensions to the moments-based method allowing for estimation of non-energetic properties, obtaining the electric dipole moment of the water molecule to an error of 2% (Chapter 4) and qualitatively accurate molecular forces over a range of molecular geometries from a single quantum calculation (Chapter 5).

Finally, we examine the scaling of the Hamiltonian moments evaluation through three different methods and verify the analytic results with numerical simulation (Chapter 6) While the methods do not scale as well as quantum phase estimation in the asymptotic limit, and further work is needed to provide concrete resource estimates, the hope is that the required circuit depths will be more reasonable for early fault-tolerant quantum devices.

Declaration of Authorship

I, Michael Jones, declare that this thesis titled, ‘Quantum computation, quantum chemistry and Hamiltonian moments’ and the work presented in it are my own. I confirm that:

- The thesis comprises only my original work towards the Doctor of Philosophy except where indicated in the preface;
- due acknowledgement has been made in the text to all other material used; and
- the thesis is fewer than the maximum word limit in length, exclusive of tables, maps, bibliographies and appendices as approved by the Research Higher Degrees Committee.

Signed:

Date:

Preface

During my enrolment in this degree, two first-author articles were published in peer-reviewed journals and another article has been submitted to a peer-reviewed journal. These articles, form the basis of Chapters 2, 3 and 4 of this thesis. In each case the paper abstract and introductory sections have been rewritten to avoid repetition and improve coherence between chapters. The main body of each article has been edited to maintain consistent definitions, spelling and notation. In Chapters 2 and 3, the Methods and Supplementary Information of the corresponding papers have been combined to reduce repetition. The papers corresponding to Chapters 2 and 3 appear in the bibliography, but are referenced only where they were referenced in later publications, elsewhere, reference is made directly to the thesis chapters.

Additionally, Chapter 5 is based on work in preparation that is intended for publication. Contribution statements from each chapter are as follows:

- Chapter 1: This chapter serves as an introduction and literature review and contains no original research completed during my candidature.
- Chapter 2: This chapter is based on the article published in *Scientific Reports* on 28th May 2022, 'Chemistry beyond the Hartree-Fock energy via quantum computed moments,' Michael A. Jones, Harish J. Vallury, Charles D. Hill and Lloyd C. L. Hollenberg.
  - Contribution statement: LCLH conceived the project. MAJ set up the computational framework and performed the calculations, with inputs from all authors.
- Chapter 3: This chapter is based on the article published in *Physical Review Applied* on 7th June 2024, 'Ground-state energy calculation for the water molecule on a superconducting quantum processor,' Michael A. Jones, Harish J. Vallury, Lloyd C. L. Hollenberg.
  - Contribution statement: LCLH conceived the project. MAJ set up the computational framework and performed the calculations and data analysis, with input from all authors. We acknowledge useful discussions with Gary Mooney, Harry Quiney and Andy Martin.

- Chapter 4: This chapter is based on the article submitted to the *Journal of Chemical Theory and Computation* on 10th April 2025, 'Quantum computation of the electric dipole moment of molecular systems,' Michael A. Jones, Harish J. Vallury, Manolo C. Per, Harry M. Quiney, Lloyd C. L. Hollenberg.
  - Contribution statement: MAJ and LCLH conceived the project. MAJ set up the computational framework and performed the calculations and data analysis, with input from all authors.
- Chapter 5: This chapter contains joint work with Manolo C. Per and Lloyd C. L. Hollenberg
  - Contribution statement: MCP conceived the project. MAJ set up the computational framework and performed the calculations and data analysis, with input from all authors.
- Chapter 6: This chapter contains joint work with Alana K. Fiusco, Gary J. Mooney and Lloyd C. L. Hollenberg
  - Contribution statement: LCLH and MAJ conceived the project. MAJ performed the analytical calculations and numerical simulation with input from all authors.
- Chapter 7: This chapter serves as a summary and conclusion and does not contain any original research.

During my enrolment two second-author articles were published in peer-reviewed journals. These articles are referenced as background material but do not otherwise appear in this thesis:

- 'Quantum computed moments correction to variational estimates,' Harish J. Vallury, Michael A. Jones, Charles D. Hill and Lloyd C. L. Hollenberg, published in *Quantum* on 15th December 2020.
- 'Noise-robust ground state energy estimates from deep quantum circuits,' Harish J. Vallury, Michael A. Jones, Gregory A. L. White, Floyd M. Creevey, Charles D. Hill and Lloyd C. L. Hollenberg, published in *Quantum* on 11th September 2023.

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Acknowledgements

Personally, I feel I have come a long way in the four and a half years since the beginning of my PhD and I owe thanks to many people for making this such a fruitful and fulfilling journey. Firstly, to my supervisory panel Lloyd Hollenberg, Charles Hill (who was my advisor for the first two years, prior to his departure from the University), Harry Quiney and Andy Martin for their wealth of knowledge but also for guiding me, not only in the process of conducting research but also in learning to communicate my results clearly to different audiences. Lloyd deserves extra appreciation for his tireless enthusiasm for science and for facilitating many great opportunities and connections over the past four years. Thank you also to Mary and Kate for their efforts organising things behind the scenes.

I would also like to thank a number of students (and former students) whose journeys have overlapped with my own, including: Floyd Creevey, Spiro Gicev, Ben Harper, Jamie Heredge, Haiyue Kang, Younghun Kim, Gary Mooney, Chris Nakhl, Myron Srikumar, Di Wang, Peiyong Wang, Max West, Greg White and Shaobo Zhang. Special thanks go to Harish Vallury and Alana Fiusco as those with whom I've worked most closely, but each of you has made my journey special in some small (or not-so-small) way.

Further acknowledgement goes to Manolo Per for providing a fresh perspective and a reservoir of knowledge and ideas. Thanks also to the team at IPAM, both the staff for coordinating and hosting a fantastic program and to the participants for many engaging discussions and memorable experiences.

And finally, to my family and friends outside of academia. Through your love and support you have been as much a part of this journey as the people listed above. Thank you.

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Chapter 1

Introduction and overview

The accurate and efficient simulation of quantum systems is of great interest in both the research community and in industry – it is estimated, as of 2019, that 35% of all supercomputing time is dedicated to molecular simulation [1]. The fundamental bottleneck to simulating quantum systems on a conventional computer is that the Hilbert space of the system grows exponentially with the system size, requiring exponential computational resources. While approximations can extend the reach of classical simulation algorithms, achieving the required accuracy remains hard.

Quantum computers on the other hand are able to access an exponentially large Hilbert space using a linear number of quantum bits (qubits) and so provide a potential speed-up for these simulation problems. As a first application, there has been significant interest in the estimation of molecular ground-state energies in the Born-Oppenheimer approximation (i.e. the electronic-structure problem) using present-day quantum computers to perform proof-of-principle demonstrations in anticipation of scaling to problems of interest as algorithms and hardware improve.

Since second-quantised problems map more easily onto present-day quantum computers, we will briefly review techniques for (approximately) solving the ground-state energy estimation problem for second-quantised electronic structure problems on classical computers in Section 1.1 and on quantum computers in Section 1.2. In particular, Section 1.2.6 summarises experimental demonstrations of ground-state energy estimation on quantum hardware, including our results showing that computation of the Hamiltonian moments can allow for more accurate computations when using limited trial-states, enabling us to perform the largest (at the time) quantum chemistry calculation to achieve millihartree accuracy on quantum hardware (Chapters 2 and 3). The moments-based method that we use is introduced in Section 1.3, and in Chapters 4 and 5 we extend the method to computation of non-energetic molecular properties. While these proof-of-principle demonstrations indicate the promise of using Hamiltonian moments, the classical pre-computation and number of quantum measurements required pose an obstacle to scaling to larger problems. In Chapter 6 we consider methods for moment estimation that potentially circumvent these issues and investigate the required scaling of quantum resources both analytically and numerically. Finally Chapter 7 summarises the work presented in this thesis and describes directions for future investigation.

1.1 Quantum chemistry on a classical computer

1.1.1 Hydrogen-like atoms

The simplest possible electronic structure problem is that of the Hydrogen-like atom i.e. an atom with nuclear charge Z (in units of the electronic charge magnitude) and one electron, which has the Hamiltonian (in atomic units):

$$\mathcal{H} = -\frac{\nabla_{\vec{r}}^2}{2} - \frac{Z}{|\vec{R} - \vec{r}|}, \tag{1.1}$$

where $\vec{r}$ and $\vec{R}$ are the positions of the electron and nucleus respectively. The first term corresponds to the kinetic energy of the electron and the second term to the electron-nuclear attraction, the Born-Oppenheimer approximation neglects the nuclear kinetic energy[1]. For this simple system, the eigenvalue problem;

$$\mathcal{H}\Psi_j = E_j\Psi_j, \tag{1.2}$$

for eigenstates Ψ_j and corresponding eigenenergies E_j, can be solved analytically by decoupling the radial and angular components, and the solutions written as real-valued wave-functions known as atomic orbitals and plotted in Figure 1.1. Each orbital is specified by three quantum numbers n, l, and m, which define the energy, angular momentum magnitude and z-projection of the angular momentum quantum levels respectively.

[1]For the Hydrogen-like atom this can be handled by using the reduced electron mass, but we'll need to make this approximation later anyway.

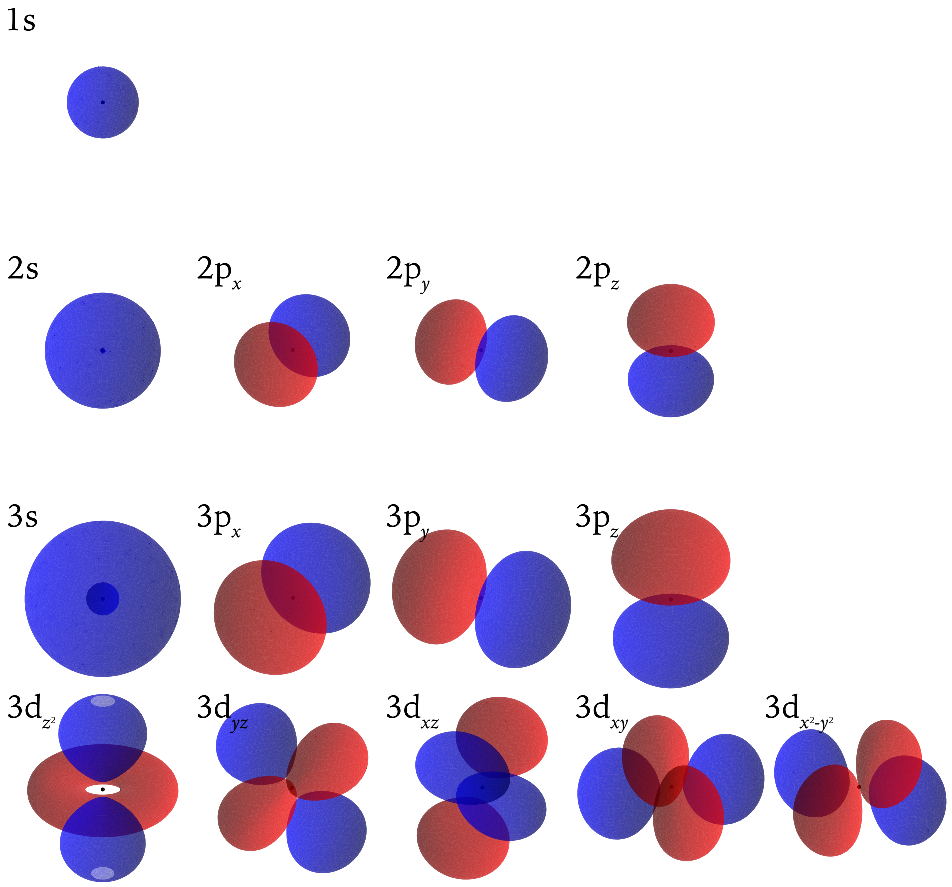


FIGURE 1.1: Hydrogen-like atomic orbitals. The magnitude of the electronic wavefunction is equal at any point on the surface(s), the probability that the electron is within the volume contained by the surface is 30%. The colour of the surface represents the phase of the wavefunction at that point (positive or negative). The atomic nucleus is the black dot in the center of each orbital, and the scale is the same for all orbitals.

1.1.2 Hartree-Fock theory and second-quantisation

Generalising to the case of many electrons and nuclei, the Hamiltonian of the system becomes;

$$\mathcal{H} = \sum_{J,K>J} \frac{Z_J Z_K}{|\vec{R}_J - \vec{R}_K|} - \sum_j \frac{\nabla^2_{\vec{r}_j}}{2} - \sum_{j,J} \frac{Z_J}{|\vec{R}_J - \vec{r}_j|} + \sum_{j,k>j} \frac{1}{|\vec{r}_j - \vec{r}_k|}, \tag{1.3}$$

where lower-case summation indices sum over electrons and capitals sum over nuclei, so the terms represent the nuclear-nuclear repulsion, electronic kinetic energy, electron-nuclear attraction and electron-electron repulsion respectively. In the general case, the system cannot be solved analytically and approximations or numerical methods are required. The simplest approach is to express the multi-electron wavefunction as a product of single electron wavefunctions,

$$\Psi(\vec{r}_1, \vec{r}_2) = \psi_1(\vec{r}_1)\psi_2(\vec{r}_2) = \psi_{n_1 l_1 m_1}(\vec{r}_1)\psi_{n_2 l_2 m_2}(\vec{r}_2). \tag{1.4}$$

However, there are several subtleties that must be addressed; firstly, the atomic orbitals are not orthogonal and are usually converted to molecular orbitals; secondly, an additional quantum number, s, must be introduced to account for the electronic spin degree of freedom to produce spin-orbitals; and thirdly, product states do not obey the correct fermionic exchange statistics, i.e. Equation 1.4 does not satisfy

$$\Psi(\vec{r}_1, \vec{r}_2) = -\Psi(\vec{r}_2, \vec{r}_1). \tag{1.5}$$

As such, product states should be replaced by antisymmetrised products, known as Slater determinants,

$$\Psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{2}}\Big[\psi_1(\vec{r}_1)\psi_2(\vec{r}_2) - \psi_2(\vec{r}_1)\psi_1(\vec{r}_2)\Big] = \frac{1}{\sqrt{2}}\begin{vmatrix} \psi_1(\vec{r}_1) & \psi_1(\vec{r}_2) \\ \psi_2(\vec{r}_1) & \psi_2(\vec{r}_2) \end{vmatrix}. \tag{1.6}$$

The Hartree-Fock method identifies the molecular orbitals that lead to the optimal single Slater determinant state, known as the Hartree-Fock state. While the Hartree-Fock method is computationally inexpensive in practice, and can account for additional electrons and nuclei, it makes several key approximations;

1. The Born-Oppenheimer approximation: It is assumed that the nuclei are stationary charges. Evidently this is not the case for chemical reactions, as static nuclei cannot form or break molecular bonds. However, comparing calculations at different nuclear geometries is often sufficient to overcome this limitation.

2. The mean-field approximation: The electron-electron repulsion is only accounted for in an average way, as such there is no correlated motion of electrons. This leads to the theory being unable to predict (for example) the liquefaction of nitrogen, through London dispersion forces.

3. The finite basis-set approximation: The molecular orbitals are assumed to be a linear combination of a finite number of atomic orbitals (and therefore there are a finite number of linearly independent molecular orbitals). This can be mitigated by performing calculations in increasingly large bases and extrapolating to the infinite basis limit.

Additionally, relativistic effects are usually neglected. While the finite basis-set and Born-Oppenheimer approximations can be accounted for, to address the mean-field approximation, we require so-called post Hartree-Fock methods. The discretisation of space into a set of molecular orbitals (termed second quantisation) also allows for a more convenient representation of Slater determinant states as Fock states;

$$\frac{1}{\sqrt{2}}\begin{vmatrix}\psi_1(\vec{r}_1) & \psi_1(\vec{r}_2)\\ \psi_2(\vec{r}_1) & \psi_2(\vec{r}_2)\end{vmatrix} \to |1100\ldots\rangle, \tag{1.7}$$
$$\frac{1}{\sqrt{2}}\begin{vmatrix}\psi_1(\vec{r}_1) & \psi_1(\vec{r}_2)\\ \psi_3(\vec{r}_1) & \psi_3(\vec{r}_2)\end{vmatrix} \to |1010\ldots\rangle.$$

In this form, the jth binary value, $x_j \in \{0,1\}$, in the ket corresponds to a molecular spin-orbital ψ_j, with a value of 1 indicating an occupied spin-orbital and 0 indicating an unnoccupied spin-orbital. These states form a basis for the 2^{N_s} dimensional space defined by N_s spin-orbitals. In this notation, the antisymmetry relation (Equation 1.5) leads to anticommutation of the creation and annihilation operators, $a_j^\dagger$ and a_j; i.e.

$$\begin{aligned}
&a_j^\dagger|x_0\ x_1\ldots x_{j-1}\ 0\ x_{j+1}\ldots\rangle = (-1)^{\sum_{k=0}^{j-1} x_k}|x_0\ x_1\ldots x_{j-1}\ 1\ x_{j+1}\ldots\rangle,\\
&a_j^\dagger|x_0\ x_1\ldots x_{j-1}\ 1\ x_{j+1}\ldots\rangle = 0,\\
&a_j|x_0\ x_1\ldots x_{j-1}\ 0\ x_{j+1}\ldots\rangle = 0,\\
&a_j|x_0\ x_1\ldots x_{j-1}\ 1\ x_{j+1}\ldots\rangle = (-1)^{\sum_{k=0}^{j-1} x_k}|x_0\ x_1\ldots x_{j-1}\ 0\ x_{j+1}\ldots\rangle,
\end{aligned} \tag{1.8}$$

leads to the anticommutators ($\{a,b\} = ab + ba$);

$$\{a_j^\dagger, a_k^\dagger\} = 0, \qquad \{a_j, a_k\} = 0, \qquad \{a_j^\dagger, a_k\} = \delta_{jk}, \tag{1.9}$$

with δ_{jk} as the Dirac-delta. The electronic Hamiltonian can then be expressed in terms of the creation and annihilation operators as

$$\mathcal{H} = \sum_{jk}^{N_s} h_{jk} a_j^\dagger a_k + \sum_{jklm}^{N_s} g_{jklm} a_j^\dagger a_k^\dagger a_l a_m, \tag{1.10}$$

where h_{jk} and g_{jklm} are the one- and two- electron integrals respectively and can be evaluated classically as

$$\begin{aligned}
h_{jk} &= \int_V \psi_j^*(\vec{r}_1)\left(-\frac{\nabla_{\vec{r}_1}^2}{2} - \sum_J \frac{Z_J}{|\vec{R}_J - \vec{r}_1|}\right)\psi_k(\vec{r}_1)d\vec{r}_1,\\
g_{jklm} &= \iint_V \psi_j^*(\vec{r}_1)\psi_m(\vec{r}_1)\frac{1}{|\vec{r}_1 - \vec{r}_2|}\psi_k^*(\vec{r}_2)\psi_l(\vec{r}_2)d\vec{r}_1 d\vec{r}_2,
\end{aligned} \tag{1.11}$$

where the integrals are taken over all space.

1.1.3 Configuration interaction and coupled cluster methods

There are a wide variety of methods to address the shortcomings of the Hartree-Fock method, here we focus on the configuration interaction (CI) and coupled cluster (CC) methods. In general, the ground-state wavefunction of the second-quantised Hamiltonian can be written as a linear combination of Fock-states, $|x\rangle$,

$$|\Psi\rangle = \sum_x b'_x |x\rangle, \tag{1.12}$$

where x is an integer used as shorthand for the corresponding binary expansion $x_0 x_1 x_2 \dots$ and b'_x are real coefficients (provided the spin-orbitals are real-valued as they are in Figure 1.1). However, such a description has many more degrees of freedom than required. For example, since the electron number is fixed, only basis states with the correct number of electrons need to be considered.

In CI methods, further assumptions can be made, such as the assumption that contributions from states closely related to the Hartree-Fock state, $|\Psi_{\mathrm{HF}}\rangle$ dominate, e.g.

$$|\Psi_{\mathrm{FCI}}\rangle = b_0 |\Psi_{\mathrm{HF}}\rangle + \sum_{jk} b_{jk} a_j^\dagger a_k |\Psi_{\mathrm{HF}}\rangle + \sum_{jklm} b_{jklm} a_j^\dagger a_k^\dagger a_l a_m |\Psi_{\mathrm{HF}}\rangle + \dots \tag{1.13}$$

where $a_j^\dagger a_k$ implements a single excitation, $a_j^\dagger a_k^\dagger a_l a_m$ indicates a double excitation and full configuration interaction (FCI) incorporates all orders of excitation. b_0, b_{jk}, b_{jklm} $\dots$ are real scalars. In practice, the order of excitations needs to be truncated for all but the smallest systems (e.g. CISD includes single and double excitations). The truncated CI wavefunction is variational but is not size extensive, i.e. the energy of two systems at infinite separation is not equal to the sum of their individual energies.

Alternatively, CC methods express the ground-state as an exponential ansatz

$$|\Psi_{\mathrm{FCC}}\rangle = \exp\left(\sum_{jk} t_{jk} a_j^\dagger a_k + \sum_{jklm} t_{jklm} a_j^\dagger a_k^\dagger a_l a_m + \dots\right) |\Psi_{\mathrm{HF}}\rangle = e^T |\Psi_{\mathrm{HF}}\rangle, \tag{1.14}$$

where T is the cluster operator and the coeffcients t_{jk}, t_{jklm} $\dots$ can be found by solving the coupled cluster equations. Again, the order of excitations needs to be truncated for practicality (e.g. CCSD includes single and double excitations), which leads to the ansatz becoming non-variational, though it remains size-extensive.

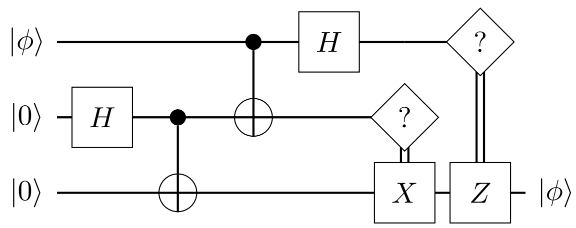
In the limit that all excitations are considered, the FCI and FCC (full coupled cluster, including all levels of excitation) wavefunctions are both variational and size-extensive, the lack of these properties is due to the truncation of the excitation order. Additionally, FCI and FCC are capable of expressing the ground-state exactly. The key issue preventing application of CI and CC methods to larger and more complex molecules is the need to estimate ground-state energies to chemical accuracy – within a tolerance of 1.6 millihartree of the infinite basis limit. Though coupled cluster calculations using perturbative triple-excitations, CCSD(T), are considered the 'gold-standard' of computational quantum chemistry, there are systems for which they are either insufficient or prohibitively expensive. There are also other post Hartree-Fock methods (such as Møller-Plesset perturbation theory) that lie beyond the scope of this thesis.

1.2 Quantum chemistry on a quantum computer

Since the difficulty of quantum chemistry arises mainly from the exponentially large Hilbert-space in which the many-electron wavefunction can exist, it is natural to think that quantum computers, which can access an exponentially large Hilbert-space, might provide a computational advantage. More simply put, it has been suggested that it should be easier to simulate a quantum system using another quantum system instead of using a classical one [2]. Here we review the framework of gate-based quantum computation and the mapping from a second-quantised electronic system to a qubit based one.

1.2.1 Gate-based quantum computation

A gate-based quantum computation can be written as a quantum circuit, such as the one below:



This circuit performs 'teleportation' of the input state, $|\phi\rangle$, from the first qubit to the third. While this circuit is not relevant to ground-state energy estimation, it serves as a self-contained example without requiring explanation of how chemical problems are mapped to the quantum device – which will be presented in Section 1.2.2. The initial state of the qubits, each of which can be expressed as a length 2, complex, unit vector, is given on the left of the circuit and the combined state of the system can be written as a tensor (Kronecker) product of the individual states

$$|\Psi_0\rangle = |\phi\rangle|00\rangle \equiv |\phi\rangle \otimes |0\rangle \otimes |0\rangle = \begin{pmatrix} b_0 \\ b_1 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix} \otimes \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \tag{1.15}$$

where $|\phi\rangle = b_0|0\rangle + b_1|1\rangle$ is an arbitrary single-qubit state with b_0 and b_1 as complex numbers. As time progresses from left to right, quantum logic gates such as the Hadamard (H) and CNOT gate act as unitary matrices on the statevector. While in principle any unitary operation can be implemented as a quantum gate, it is necessary in practice to decompose operations into a convenient set of universal base gate. The choice of base gates is predominantly determined by what operations are convenient for the quantum hardware in question and generally include only 1- and 2-qubit gates. When gates act on only a subset of the qubits, their action on the entire state can be computed by padding the extra dimensions with identity matrices

$$|\Psi_1\rangle = |\phi\rangle \otimes (H|0\rangle) \otimes |0\rangle = (I \otimes H \otimes I) \cdot |\phi\rangle|00\rangle. \tag{1.16}$$

The number of non-parallel layers of unitary operations is referred to as the circuit depth. Finally, a quantum algorithm requires measurement, represented in the circuit by the diamond shaped operations. Unlike in classical systems, the measurement outcome is probabilistic and causes the system to collapse into the measured state. For example, on measuring the state $|\phi\rangle$, the measurement probablities and resulting states are

$$|\phi\rangle = b_0|0\rangle + b_1|1\rangle \rightarrow \begin{cases} |0\rangle, & \text{with probability } |b_0|^2 \\ |1\rangle, & \text{with probability } |b_1|^2 \end{cases}. \tag{1.17}$$

Thus, the requirement that quantum states be unit vectors and (non-measurement) operations be unitary matrices follows directly from conservation of probability.

The power of quantum computing comes from the ability to generate and manipulate entanglement, i.e. non-trivial correlations between the measurement outcomes of two (or more) qubits. Consider for example, the two-qubit Bell state

$$\frac{|00\rangle + |11\rangle}{\sqrt{2}}. \tag{1.18}$$

Upon measuring the first qubit, the combined system will collapse into either the $|00\rangle$ state or the $|11\rangle$ state, and we immediately know with certainty the state of the second qubit. While this may appear to violate causality, there is no transfer of information between the qubits. In order to transfer information, such as by the qubit teleportation circuit presented above, additional classical communication is required, as represented by the double vertical lines. The transfer of information is then limited by the speed of this classical communication.

There are many resources which provide a more in depth introduction to quantum computation (one such comprehensive source is Neilsen and Chuang [3]), however, the topics covered here should provide enough background for the purposes of this thesis.

1.2.2 Mapping spin-orbitals to qubits

Given that computational basis states of a quantum computer can be written as bit-strings, the expression of second-quantised basis states as Fock states (Equation 1.7) suggests a straightforward mapping between fermionic states and qubit states; each qubit simply represents a spin-orbital, with $|1\rangle$ and $|0\rangle$ representing occupied and un-occupied orbitals respectively. This encoding is known as the Jordan-Wigner encoding [4]. The non-trivial aspect of the encoding is the mapping of the operators, in order to obey the anticommutation relations (Equation 1.9), the fermionic operators must map to qubit operators as

$$\begin{aligned} a_j^\dagger &\rightarrow \frac{1}{2}\left[\prod_{k=0}^{j-1} Z_k\right](X_j - iY_j), \\ a_j &\rightarrow \frac{1}{2}\left[\prod_{k=0}^{j-1} Z_k\right](X_j + iY_j), \end{aligned} \tag{1.19}$$

where X, Y and Z are the Pauli matrices

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \qquad Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \tag{1.20}$$

This mapping leads to highly non-local operators (acting on $\mathcal{O}(N_s)$ sites, where N_s is the number of spin-orbitals). Despite this, the Jordan-Wigner transform is the most commonly employed encoding due to its simplicity. Other mappings such as the parity mapping, and the Bravyi-Kitaev encoding [5, 6] can reduce computational costs by reducing the number of qubits through symmetry arguments and/or by reducing the non-locality of the fermionic operators, but will not be discussed here.

Once the electronic structure Hamiltonian has been mapped to a qubit Hamiltonian, we require an algorithm that can estimate the ground-state energy of the Hamiltonian to high precision. For fault-tolerant quantum computers the archetypal algorithm is quantum phase estimation [7], while for near-term devices, it is the variational quantum eigensolver [8].

1.2.3 Chemical accuracy

A large focus in the application of quantum computers to electronic structure problems is on the pursuit of 'chemical accuracy', defined to be a result within 1.6 mHa of the infinite basis limit *or* within the same margin of accurate experimental methods. However, in quantum computing literature, the phrase is often misused to mean 1.6 mHa from a less demanding target such as FCI in a minimal basis, or from noiseless simulation of the quantum device. These targets may also be referred to as 'chemical precision' though this too is potentially ambiguous, given that precision refers to the variance of results, where accuracy refers to the average. To minimise ambiguity, we use the above definition of chemical accuracy and define 'FCI accuracy' or 'millihartree accuracy from FCI' (also known as 'algorithmic accuracy' [9]) to be when a result is within 1.6mHa of the FCI result in a minimal basis, i.e. a basis composed of 1s orbitals for hydrogen and helium atoms and 1s, 2s, and 2p orbitals for elements in the second row of the periodic table. We do not consider elements beyond the second row. Further, we define 'trial-state accuracy' as when the measured energy is within 1.6 mHa of the exactly computed trial-state energy.

1.2.4 Fault-tolerant quantum computed chemistry

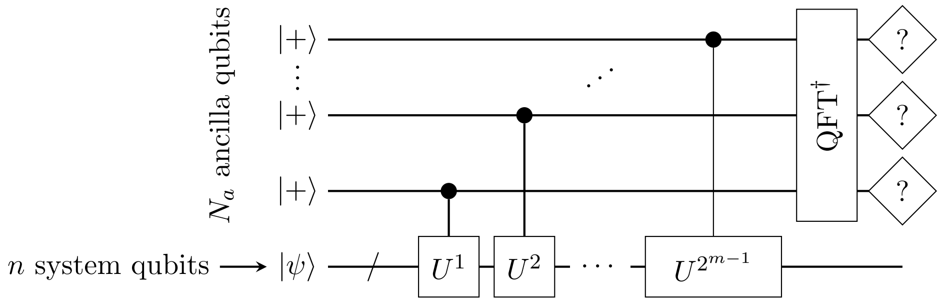
In practice computation is noisy, and quantum computation is significantly noisier than classical computation owing, in part, to the inherent fragility of quantum systems. In addition, quantum error correction is hampered by the existence of a 'no-cloning theorem', which states that no operation is capable of copying an arbitrary quantum state

$$\nexists U \text{s.t.} \forall |\psi\rangle,\ U|\psi\rangle|0\rangle = |\psi\rangle|\psi\rangle. \tag{1.21}$$

Nevertheless, quantum error correction codes exist and their study and improvement is a highly active field of research.

In the fault-tolerant regime, i.e. once quantum devices have enough qubits with sufficiently low error rates (generally quoted to be about 1% [10, 11] though it will vary based on the specific code and error model considered), the ground state energy estimation problem will be solvable[2] by quantum phase estimation (QPE, [7]), performed by the circuit below;

[2] Given a trial-state with sufficient ground-state overlap



where $|\psi\rangle$ is a trial-state, the Hamiltonian is encoded through the evolution operator,

$$U = e^{i\mathcal{H}}, \tag{1.22}$$

which can be implemented by techniques such as Trotterisation [12], and $\mathrm{QFT}^\dagger$ is the inverse Quantum Fourier Transform. At a high level, QPE can be viewed as performing an approximate measurement in the energy eigenbasis:

$$\mathrm{QPE}_{\mathcal{H}}|+^{\otimes N_a}\rangle|\psi\rangle \rightarrow |E_j\rangle|\phi_j\rangle, \text{ with probability } |\langle\phi_j|\psi\rangle|^2, \tag{1.23}$$

where the probability of collapsing the system into the eigenstate $|\phi_j\rangle$ is $|\langle\phi_j|\psi\rangle|^2$ and $|E_j\rangle$ is a binary encoding of the corresponding eigenenergy. By running the circuit multiple times, and taking the minimum measured energy value, the ground-state energy can be obtained approximately provided the state $|\psi\rangle$ has sufficient overlap with the ground-state. The degree of approximation is controlled by the number of ancilla qubits used and provides a straightforward route to approaching millihartree error levels. The key roadblock to implementing QPE is the algorithm's sensitivity to noise – both due to deep circuits and since errors in the computation can cause the algorithm to fail due to the behaviour of the 'minimum' function. As such QPE is not suited to present-day and near-term devices. Current estimates for solving a problem of industrial relevance – the iron-molybdenum cofactor (FeMoco) of the nitrogenase enzyme responsible for catalysing the nitrogen fixation process that converts atmospheric nitrogen into ammonia – suggest that around 10^5-10^8 physical qubits with error rates of around 10^{-9} would require between hundreds of hours to hundreds of days assuming 10ns gate times to estimate the ground-state energy to a precision of 0.1mHa [13].

1.2.5 Present-day quantum computed chemistry

While fault-tolerant quantum devices promise to provide an asymptotic speed-up over classical devices, it is unclear if Noisy Intermediate-Scale Quantum (NISQ [14]) devices can provide a computational advantage. Many candidate near-term quantum algorithms are based on the Variational Quantum Eigensolver (VQE, [8]), which aims to solve the ground-state energy estimation problem by constructing a trial-wavefunction (ansatz) on the quantum device and performing measurement by decomposing the Hamiltonian into fragments such as Pauli strings, i.e. tensor products of the single-qubit Pauli matrices. Due to the tensor product nature of these Pauli strings, they can be diagonalised – and therefore measured – efficiently and the contribution from each term, weighted and summed to obtain the energy, i.e.

$$\begin{aligned}
\langle \mathcal{H} \rangle &= \sum_j \omega_j \langle \psi | P_{j,1} \otimes P_{j,2} \otimes \ldots P_{j,n} | \psi \rangle \\
&= \sum_j \omega_j \langle \psi | U_{j,1}^\dagger D_{j,1} U_{j,1} \otimes U_{j,2}^\dagger D_{j,2} U_{j,2} \otimes \ldots U_{j,n}^\dagger D_{j,n} U_{j,n} | \psi \rangle \\
&= \sum_j \omega_j \langle \psi_j | D_{j,1} \otimes D_{j,2} \otimes \ldots D_{j,n} | \psi_j \rangle,
\end{aligned} \tag{1.24}$$

where ω_j is the scalar weight associated with the jth Pauli string; $P_{j,k}$ are Pauli matrices or the identity, I, acting on the kth qubit; $U_{j,k}$ are single qubit unitary matrices that diagonalise $P_{j,k}$ so $D_{j,k}$ are single qubit diagonal matrices i.e. Pauli-Z or the identity. Since the eigenbasis of a diagonal matrix is just the computational basis, the energy can be computed by summing over measurements of different states $|\psi_j\rangle = U_{j,1} \otimes U_{j,2} \otimes \ldots U_{j,n} |\psi\rangle$ (or equivalently, we can view this as measuring the same state, $|\psi\rangle$, in different bases). Further discussion of the energy measurement will be deferred to Chapters 2 and 3, with a more theoretical analysis in Section 6.1.1. For now, we note several features of the VQE that make it better suited than QPE to NISQ devices.

Firstly, since measurement of Pauli operators requires minimal additional operations, the circuit depth is drastically reduced compared to QPE allowing calculations to be completed within the coherence time of noisy qubits. Secondly, since the trial-state is optimised variationally, any coherent error, e.g. systematic error in the application of a gate, will be absorbed into the variational ansatz and, provided a sufficiently expressive ansatz, will not affect the final energy estimate. Finally, since the output from the quantum computer is an expectation value, error mitigation can be used to improve the quality of the results, usually by reducing bias at the expense of averaging over more measurements.

The key issue with the VQE algorithm is in achieving the required precision in the energy estimates, since this requires more expressive trial-states, and therefore more expensive minimisation – sometimes prohibitively so [15] – as well as more measurements to control the sampling error.

1.2.6 Experimental implementations of VQE and QPE

To date, there have been many experimental implementations of the VQE algorithm and its variants and a few implementations of the QPE algorithm for small-scale problems. Here we describe a small selection of these results that target the electronic structure problem.

Following early proof-of-principle experiments of VQE [8, 16–18], sub-millihartree precision was demonstrated for two-electron systems using a density matrix purification method [19]. The purification technique was later extended to any number of electrons provided the trial-state is a single Slater-determinant state, which necessarily limits the accuracy but allows greater resilience to noise for larger numbers of qubits [20]. We then showed that using Hamiltonian moments could improve the estimated ground-state energy beyond the trial-state limit (Chapter 2) while still allowing single Slater-determinant states and therefore the use of purification. At around the same time, the utility of moments-based methods was also demonstrated by other works [21, 22]. VQE was also applied to larger molecules using multi-determinant states [23] however, it is quickly limited by the depth of the circuit required to construct accurate trial-states. While we showed that again, Hamiltonian moments can be used to extend the reach of shallower trial-circuits (Chapter 3), much interest has turned to methods for reducing the quantum resource cost of variational quantum methods such as by entanglement forging [24–26], simulating spatial orbitals via the unitary paired coupled-cluster ansatz [27, 28], defining subspaces based on contextuality [29], transcorrelated methods [30], or embedding regions treated by quantum computation within larger classically simulated ones [31–37] including through Hamiltonian downfolding schemes [38]. Methods based on perturbation theory have also been implemented [39]. There have also been proposals to dramatically reduce the qubit requirements through configuration encoding [40], however it is not clear whether the required classical processing can be handled efficiently. On the other hand, sample-based quantum diagonalisation (or quantum selected configuration interaction) methods [36, 41–46], wherein the quantum computer is used to select Slater-determinants for a classical solver, is also an area of interest as much of the error introduced in the quantum computation does not influence the classical computation. Quantum subspace expansion methods [26, 47], where elements of a reduced Hamiltonian, that can be diagonalised classically, are extracted from the quantum computer, have also gained attention. More recently, true chemical accuracy, relative to experimentally measured values was obtained for the ionisation energy of the Helium atom using a moments-based technique [48].

The initial implementations of QPE were performed before the introduction of VQE and were able to reach high precision early on [49, 50]. However, extending to larger systems is hindered by the increased number of terms in the Hamiltonian leading to deeper circuits as well as the method's inherent fragility to noise. By contrast VQE methods required more circuits as the number of Hamiltonian terms is increased, but not deeper ones, and could leverage error mitigation techniques allowing demonstration on larger systems more readily. However, later QPE-based demonstrations have shown increased robustness to errors [51] and application to larger systems [52, 53].

Both VQE and QPE algorithms have also recently demonstrated the performance of error detecting codes [54, 55]. Additionally, there have been many theoretical developments improving quantum algorithms for chemistry. An incomplete list includes the use of explicitly correlated methods [30, 56–59], tensor hypercontraction and Hamiltonian factorisation [60–63], Hamiltonian downfolding [38, 64–68], perturbative methods [39, 69], and specific choices of basis and basis-set-free approaches [70–72].

1.3 Hamiltonian moments

We now formally introduce the Hamiltonian moments, $\langle \mathcal{H}^p \rangle$, which can be used to correct ground-state energy estimates in a number of ways such as via the connected moments expansion [73, 74], or the infimum theorem [75] used in this work. To gain an intuition as to why the moments allow such a correction, we first consider the Lanczos iteration method [76] which forms the basis of techniques used to determine eigenvalues of large sparse matrices in classical scientific computing packages such as *SciPy* [77]. Given a (Hermitian) Hamiltonian, $\mathcal{H}$, of size $N \times N$ and a trial-state, $|\psi_0\rangle$, let $|\psi_j\rangle$ be a set of orthonormal basis states computed by

$$
\begin{aligned}
\mathcal{H}|\psi_0\rangle &= \alpha_0|\psi_0\rangle + \beta_0|\psi_1\rangle, \\
\mathcal{H}|\psi_1\rangle &= \alpha_1|\psi_1\rangle + \beta_1|\psi_2\rangle + \beta_0|\psi_0\rangle, \\
&\vdots \\
\mathcal{H}|\psi_j\rangle &= \alpha_j|\psi_j\rangle + \beta_j|\psi_{j+1}\rangle + \beta_{j-1}|\psi_{j-1}\rangle, \\
&\vdots \\
\mathcal{H}|\psi_{N-2}\rangle &= \alpha_{N-2}|\psi_{N-2}\rangle + \beta_{N-2}|\psi_{N-1}\rangle + \beta_{N-3}|\psi_{N-3}\rangle, \\
\mathcal{H}|\psi_{N-1}\rangle &= \alpha_{N-1}|\psi_{N-1}\rangle + \beta_{N-2}|\psi_{N-2}\rangle,
\end{aligned}
\tag{1.25}
$$

then in the $|\psi_j\rangle$ basis, the Hamiltonian has the tri-diagonal form

$$\mathcal{H}_L = \begin{pmatrix} \alpha_0 & \beta_0 & & \\ \beta_0 & \alpha_1 & \ddots & \\ & \ddots & \ddots & \beta_{N-2} \\ & & \beta_{N-2} & \alpha_{N-1} \end{pmatrix}. \tag{1.26}$$

Provided the trial-state has good overlap with the ground-state, the ground-state energy will depend mostly on the first few α_j and β_j coefficients and therefore truncating the tri-diagonal Hamiltonian introduces only a small amount of error. It can be shown that the ground-state energy estimate converges exponentially with the size of this reduced Hamiltonian, or equivalently, with the number of times Equation 1.25 is applied [78]. In practice there are details involving orthogonalisation of basis states and floating point precision but the above description covers the basic version of the algorithm. However, taking a step further, it is possible to derive a cluster expansion [79] for the coefficients in terms of the cumulants,

$$c_p = \langle \mathcal{H}^p \rangle - \sum_{j=0}^{p-2} \binom{p-1}{j} c_{j+1} \langle \mathcal{H}^{p-j-1} \rangle. \tag{1.27}$$

Note that we begin enumerating the cumulants at c_1 (so that $c_1 = \langle \mathcal{H}^1 \rangle$) while the coefficients begin at α_0, β_0. The cluster expansion is then given by

$$\begin{aligned} \alpha_j &= c_1 + j\frac{c_3}{c_2}\frac{1}{V} + \frac{1}{2}j(j-1)\frac{3c_3^3 - 4c_2c_3c_4 + c_2^2c_5}{2c_2^4}\frac{1}{V^2} + \dots \\ \beta_j^2 &= (j+1)c_2\frac{1}{V} + \frac{1}{2}(j+1)j\frac{c_2c_4 - c_3^2}{c_2^2}\frac{1}{V^2} \\ &\quad + \frac{1}{6}(j+1)j(j-1)\frac{-12c_3^4 + 21c_2c_3^2c_4 - 4c_2^2c_4^2 - 6c_2^2c_3c_5 + c_2^3c_6}{2c_2^5}\frac{1}{V^3} + \dots \end{aligned} \tag{1.28}$$

and serves to link the Lanczos coefficients, α_j and β_j, to the connected moments, c_j. Since the expansion was initially formulated for extensive systems, V is the number of plaquettes i.e. the size of the system, and these coefficients relate to the Hamiltonian density $\mathcal{H}/V$. Using results from orthogonal polynomial theory [80, 81] it is possible to express the ground state energy, even for non-extensive systems [82], through an infimum theorem [75]

$$E_0 = \inf_{z>0} \left[\alpha(z) - 2\beta(z)\right], \tag{1.29}$$

where z is a continuous positive parameter related to the recursion index j and the system size V, and

$$\alpha(z) = c_1 + \frac{c_3}{c_2} z + \frac{3c_3^2 - 4c_2c_3c_4 + c_2^2c_5}{2c_2^4}\frac{z^2}{2} + \ldots \tag{1.30}$$
$$\beta^2(z) = c_2 z + \frac{c_2c_4 - c_3^2}{2c_2^2}\frac{z^2}{2} + \frac{-12c_3^4 + 21c_2c_3^2c_4 - 4c_2^2c_4^2 - 6c_2^2c_3c_5 + c_2^3c_6}{2c_2^5}\frac{z^3}{6} + \ldots$$

Note that unlike the truncation of the Lanczos matrix, the infimum theorem includes the contributions from all α_j and β_j. While the expressions for $\alpha(z)$ and $\beta^2(z)$ must be truncated at a given order in z, limiting the order of cumulants that contribute, the resulting approximation includes low order coefficients exactly and all higher order coefficients approximately. The standard Lanczos method neglects the higher order coefficients completely.

Since the cumulants can be computed from the Hamiltonian moments, $\langle \mathcal{H}^p \rangle$, then by evaluating the moments up to a given order, truncating $\alpha(z)$ and $\beta^2(z)$, plotting on the interval $z \in [0, 1)$ and locating the minimum, a corrected ground-state energy estimate can be obtained. The restriction to the interval $z \in [0, 1)$ is necessary since beyond this region the neglected higher order terms will dominate. Conveniently, truncating to fourth order in the moments gives an analytical expression for the minimum

$$E_{\text{inf}}^{(4)} = c_1 - \frac{c_2^2}{c_3^2 - c_2c_4}\left(\sqrt{3c_3^2 - 2c_2c_4} - c_3\right), \tag{1.31}$$

which can be computed without the use of numerical minimisation. However, for other order truncations, the missing terms may contribute significantly even within the region $z \in [0, 1)$, potentially leading to the lack of a minimum. In practice then, we locate the first stationary point or inflection point at z_0 and compute the approximate ground-state energy as $E_{\text{inf}} = \alpha(z_0) - 2\beta(z_0)$. Examples of the infimum theorem are given in Figure 1.2 for a random dense Hamiltonian.

In implementing this on a quantum computer [83–85], the task we assign to the quantum device is to evaluate Hamiltonian moments with respect to some suitably prepared trial-state, allowing us to apply the infimum theorem to improve upon the variational energy estimate, $\langle \mathcal{H} \rangle$, without requiring prohibitively deep trial-circuits. The Lanczos method has also been considered in relation to quantum computing in other contexts [86–90]. Initially [84] moment evaluation was performed by operator averaging, where the moment operators were evaluated 'term-by-term' and was dubbed the Quantum Computed Moments (QCM) [85] method. In application to quantum chemistry, we were able to

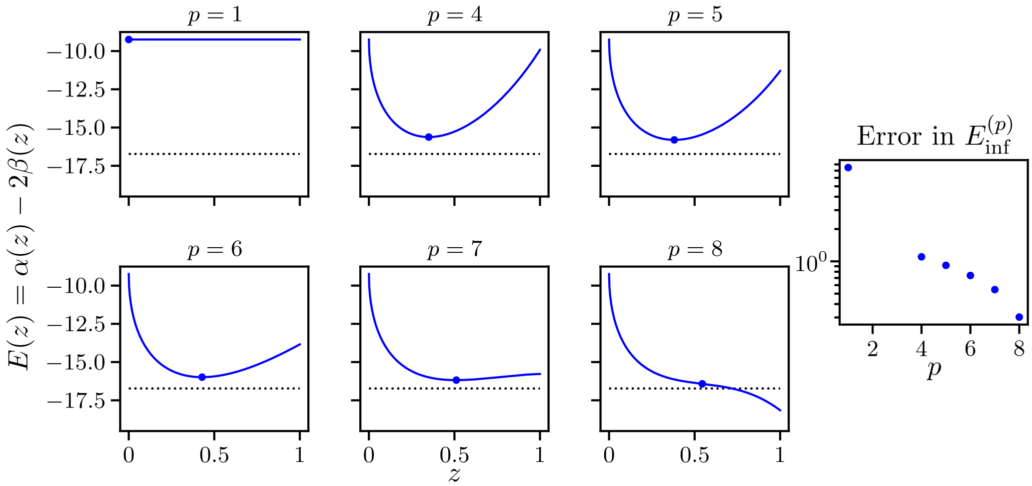


FIGURE 1.2: Example curves for $\alpha(z)-2\beta(z)$ (blue lines), approximated using up to the pth moment. The infimum estimate at each order is marked by a blue circle. Note that $p = 2$ and $p = 3$ are not shown as they do not possess a minimum or inflection point and so do not give ground-state energy estimates. The 8th order does not have a stationary point and instead exhibits an inflection point. The dashed black line is the true ground-state energy. The Hamiltonian is a dense 4 qubit Hamiltonian chosen specifically to demonstrate both local minima and inflection points in $E(z)$. Right: convergence of the error in the ground-state energy estimate $E_{\mathrm{inf}}^{(p)}$ towards 0 with increasing moment order, p.

demonstrate this by using single Slater-determinant trial-states, yet producing energy estimates with greater accuracy than the Hartree-Fock state (Chapter 2). Later we extended this to multi-determinant states, performing the largest (at the time) quantum computed chemistry calculation to achieve millihartree precision using noisy quantum hardware (Chapter 3). Additionally, we show that the moments-based method can correct molecular properties such as dipole moments (Chapter 4) and molecular forces (Chapter 5), with possible further extension to computing energy gaps [91].

These proof-of-principle demonstrations were performed using a 'term-by-term' operator averaging technique. While this method is viable for small systems and low-order moments, the required number of measurements, as well as the classical preprocessing cost scale rapidly with the size of the chemical system being investigated. As such, Chapter 6, explores 'circuit-based' methods of moment estimation [83, 92], i.e. methods where the Hamiltonian is encoded into the quantum circuit. There, we will derive analytical expressions for the error in moment estimation and compare with numerical simulation as well as indicating possible directions for improvement.

Chapter 2

Chemistry beyond the Hartree–Fock energy via quantum computed moments

2.1 Introduction

In this chapter, which is based on the peer-reviewed paper "Chemistry beyond the Hartree-Fock energy via quantum computed moments" published in *Scientific Reports*, we will investigate the application of the infimum theorem to electronic structure Hamiltonians using single Slater-determinant trial-states, including results from IBM superconducting quantum devices.

At the time the paper was written, recent advances had been made in extending the size of chemical systems that VQE could be applied to through use of density matrix purification [20, 93]. However this required restricting the trial-state to single Slater-determinant states, thus limiting the accuracy of the method, since these states are unable to incorporate correlated electronic behaviour. On the other hand, our previous work [85] had shown that the fourth-order infimum energy correction (Equation 1.31), could make use of moments computed from noisy quantum computers to improve the ground-state energy estimates for Heisenberg spin models. This method was referred to as the Quantum Computed Moments (QCM) method and was seen empirically to be highly robust against noise in the quantum computation. The obvious question then, was if QCM could be used to push ground-state energy estimates beyond the limit imposed by single Slater-determinant states while retaining the advantage of density matrix purification. In particular, we aimed to achieve 'FCI accuracy' (Section 1.2.3),

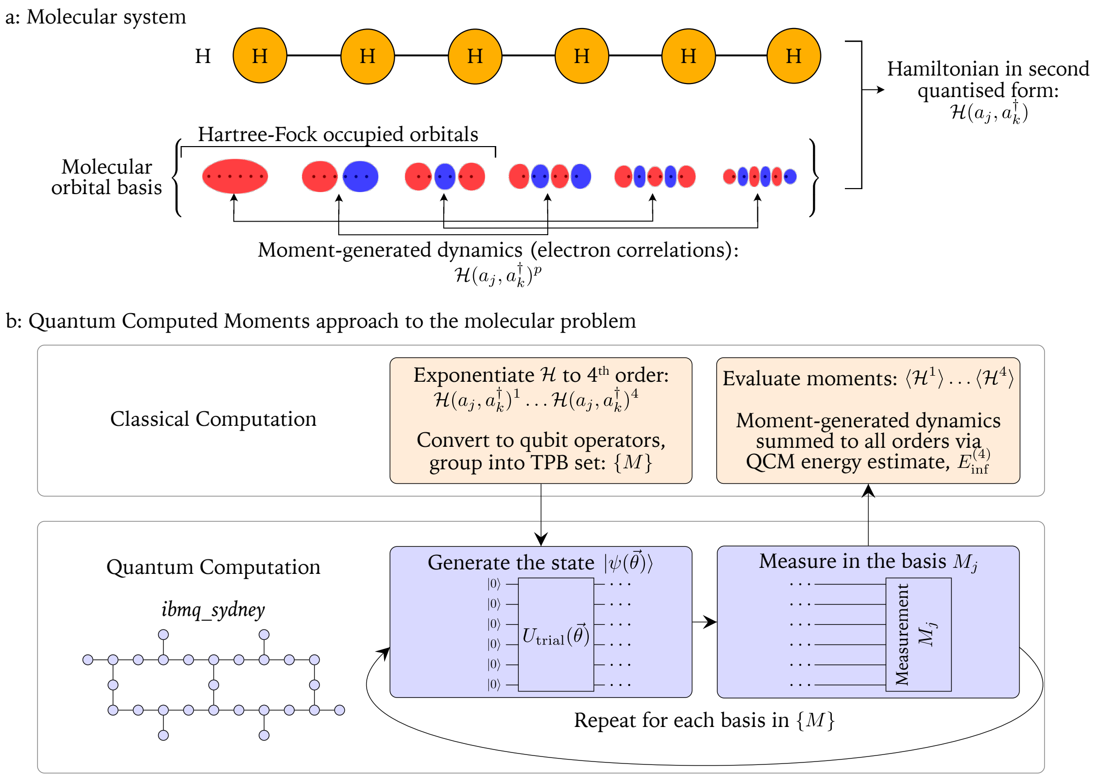


FIGURE 2.1: Overview of the quantum computed moments (QCM) approach applied to problems in chemistry. a: The molecular system H_6 is represented by a second-quantised Hamiltonian over a set of molecular orbitals. The trial-state is the Hartree-Fock state, i.e. the occupation of the indicated orbitals. Application of the Hamiltonian moments allows for the generation of electronic correlation effects that the Hartree-Fock state cannot otherwise incorporate. b: Overview of hybrid quantum/classical aspects of the QCM approach including a device map for the quantum processor *ibmq_sydney* used in this work.

i.e. an error of less than 1.59 mHa from the FCI result in the chosen basis, in the ground-state energy estimation for hydrogen molecular chains (H_2, H_4, H_6). The overall method is outlined in Figure 2.1, and Section 2.2 summarises the procedure and presents the key results. Section 2.3 provides further discussion of the results and method in comparison to the literature and Section 2.4 provides additional detail on the methods employed.

2.2 Results

Chemistry via Quantum Computed Moments

In order to compute the Hamiltonian moments from the quantum device, we first need to exponentiate $\mathcal{H}$ to produce $\{\mathcal{H}^1, \mathcal{H}^2, \mathcal{H}^3, \mathcal{H}^4\}$. Here we keep $\mathcal{H}$ in second quantised form so the multiplications can be performed, for example, by using Wick's theorem. After conversion to qubit operators, the growth of the number of terms in the exponentiated forms of $\mathcal{H}$ is controlled by forming tensor product basis (TPB) sets [17] in a classical pre-processing step (as discussed in Section 2.4.5). The trial-circuit employed in this work is based on that used in reference [20], where each qubit represents the occupation state of a molecular orbital, classically pre-computed in the STO-3G minimal basis using the *Python* package *PySCF* [94]. (See Section 2.4.4 for additional details.)

To reduce the computational burden on the quantum processor an extension of the spin-symmetry reduction of [20] was employed to reduce the number of qubits by a factor of 2. While this qubit reduction (see Section 2.4.3 for details) requires several restrictions on the problem, most notably that the trial-state must be a single Slater-determinant, the inclusion of electron dynamics introduced to the system by the QCM method allows the computation to achieve accuracy beyond what would normally be possible for such a trial-state. This technique also allows for a reduction in the number of Tensor Product Basis (TPB) elements that need to be measured in the Hamiltonian averaging procedure by a factor of N_s^{14} from the naive method of measuring the (worst case) $\mathcal{O}(N_s^{16})$ terms in $\mathcal{H}^4$ individually, where N_s is the number of spin-orbitals. The results here were obtained using the conventional $\mathcal{O}(N_s^4)$ scaling of the Hamiltonian terms for which classical pre-processing on modest computing resources limited the molecular system size. However, our results together with the wide applicability of the QCM method to molecular systems in general, provides the tantalising possibility that with alternative Hamiltonian representations the QCM approach, coupled with error mitigation schemes, has the potential to provide accurate and strongly error-robust results for the ground-state energy of larger chemical systems on near-term quantum computers.

Circuit parameter sweep

With respect to the single Slater-determinant trial-state defined in Figure 2.2 (parameter set $\vec{\theta}$), all observables contained in the TPB sets were measured directly on the *ibmq_sydney* device for the molecular systems H_2, H_4 and H_6 with all bond lengths set to 0.74Å (the equilibrium bond distance of molecular hydrogen). Each TPB set was measured using 8192 shots. To illustrate variational behaviour we present the data for values of the opening trial-state parameter θ_1 around the Hartree-Fock state compared to the Hartree-Fock energy and the Full Configuration Interaction (FCI) results in Figure 2.3a. (see Section 2.4.7 for ensemble data over the variational parameter sets). The error robustness of the QCM results is quite evident for all three molecular systems – while the pink triangles representing the direct measurement result (with respect to θ_1) move significantly away from the solid pink line indicating the trial-state limit (HF energy), the upward shift of the QCM due to device errors (green triangles shifting away

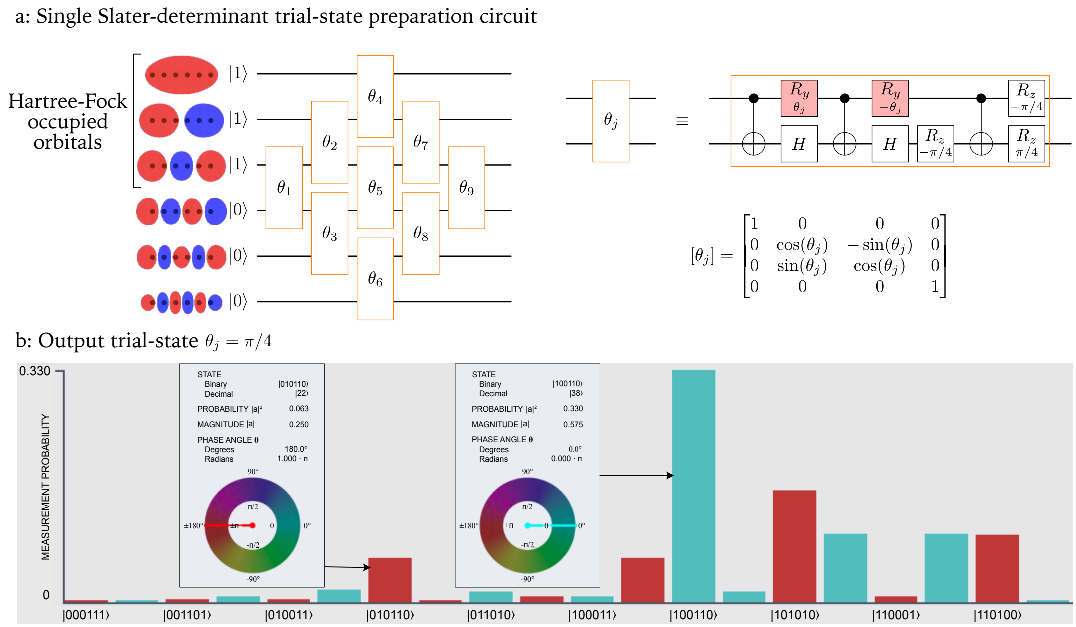


FIGURE 2.2: Trial-state construction and implementation: illustrated case – the 6 atom hydrogen chain. a: Qubit representation of the orbital basis, the trial-circuit expressed in terms of Givens rotations [95–97] and the Givens rotation expressed in terms of CNOT and single qubit gates and its matrix representation. Parameterised gates are shaded in red. b: The trial-state produced when $\theta_j = \pi/4$ visualised using the Quantum User Interface (QUI) system [98]. Only the states representing the correct number of electrons are included on the horizontal axis since only these states can have non-zero amplitudes.

from the green line) is suppressed by an order of magnitude. Despite the large number of observables required to be measured on the QC device in the determination of the moments, the QCM correction term consistently suppresses the device errors contained in the uncorrected computation $c_1 = \langle \mathcal{H} \rangle$. In fact, for the largest system, H_6, the raw QCM data already recovers 99.7% of the minimised trial-state energy at the HF point.

Density matrix purification

To investigate the potential of the method to reach beyond the Hartree-Fock energy we perform McWeeny purification on the 1-body reduced density matrix (1-RDM) [93] – this procedure has been shown to improve the direct energy measurement result at the Hartree-Fock point [20]. Details on the purification procedure can be found in Section 2.4.6. The results of the RDM purification procedure are shown in Figure 2.3b. As observed in [20], the purification procedure applied to the direct energy measurement (pink inverted triangles) goes a long way to recovering the trial-state energy (pink line). However, the key observation is that the RDM purification is also highly effective in correcting the remaining device errors in the QCM calculation (green inverted triangles),

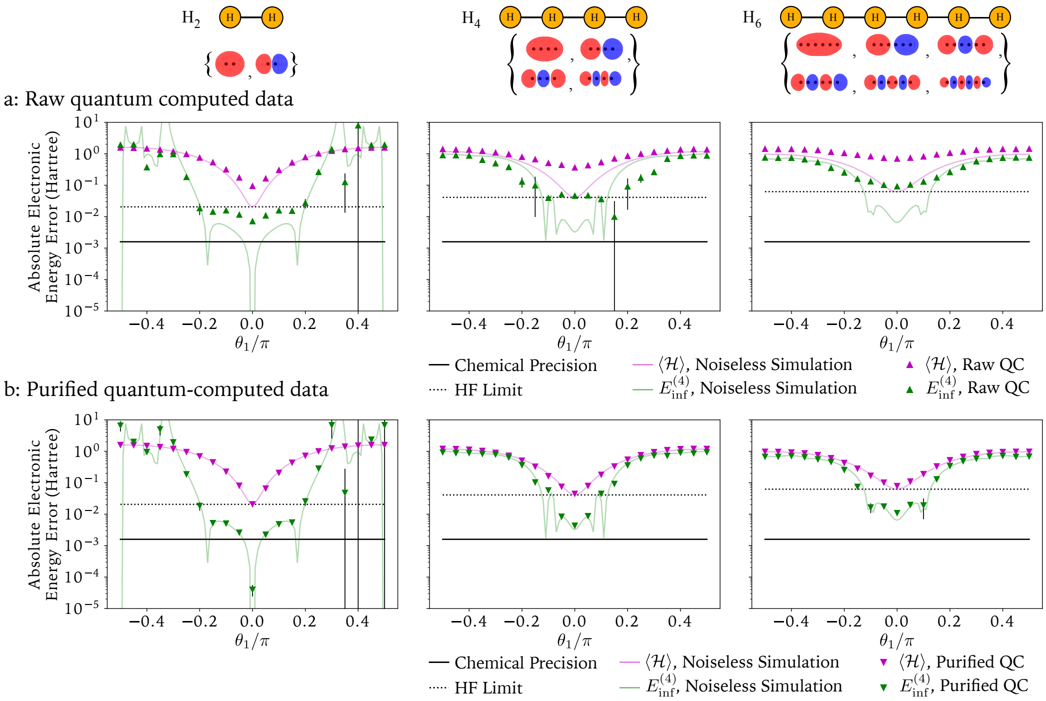


FIGURE 2.3: Results of varying the leading circuit parameter, θ_1 for (from left to right) H_2, H_4, H_6. The vertical axis is the absolute error in electronic energy from the FCI result $|E - E_{\text{FCI}}|$ on a logarithmic scale. Ideal simulations were performed without errors or shot noise while QC data was averaged over 4 runs (total 8192 shots). Statistical error bars on the QC data represent one standard deviation and are often smaller than the data points. a: Comparison of raw data results to noiseless simulation. b: Comparison of purified data to noiseless simulation.

explicitly recovering a portion of the electron correlation and pushing the result beyond the HF energy (black dashed line).

The high level of accuracy (relative to the FCI result) in the H_2 case is in part due to the fact that the Hilbert space spanned by the minimal basis is reasonably small, so the Hartree-Fock state is a better approximation to the true ground-state. For the H_4 and H_6 cases the accuracy of the Hartree-Fock method is decreased and so the QCM results based on the HF state are also reduced in accuracy. Note that due to the logarithmic vertical axes in Figure 2.3, the shift in the QCM results is magnified compared to the shift in the Hartree-Fock energy. Systematic improvement of the trial-state (of the form usually used in VQE calculations) is expected to allow the QCM method to more accurately estimate the ground-state energy.

Dissociation curves

Finally, we repeat the calculations at a range of atomic separations to compute the dissociation curves shown in Figure 2.4. Note that until now we have neglected the

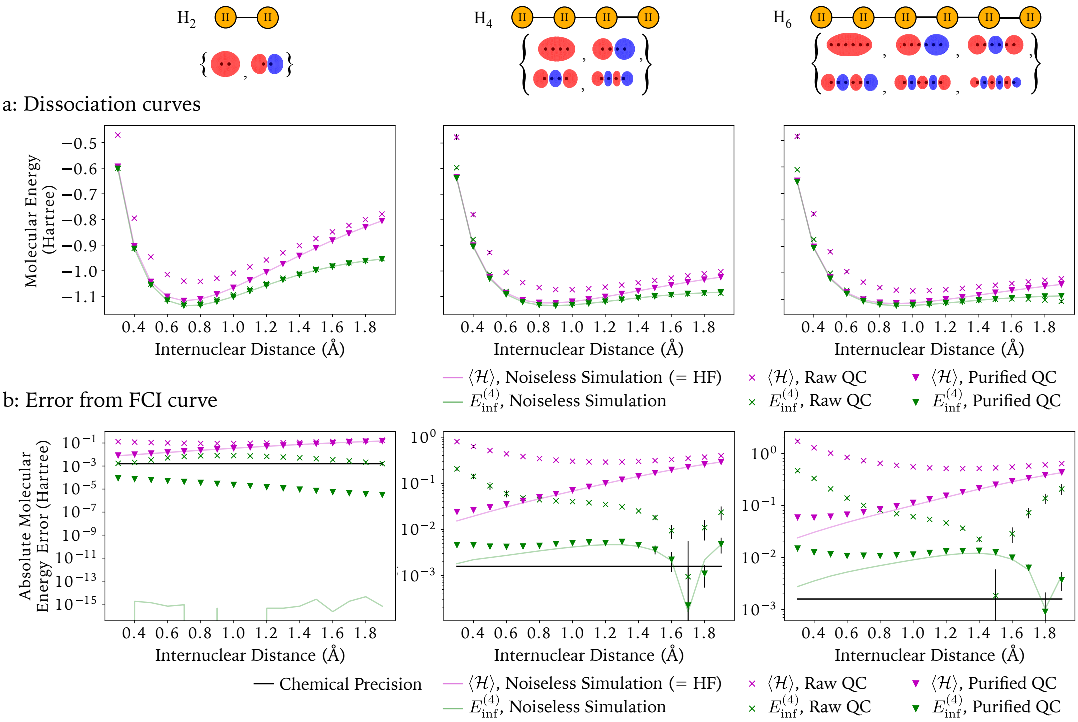


FIGURE 2.4: Dissociation curves for hydrogen chains. The energy values at each point correspond to a single energy evaluation (8192 shots) performed at parameter value $\theta = (0, 0, \dots)$. a: Molecular energy as a function of inter-nuclear distance. The purple line is the noiseless simulation result which reproduces exactly the Hartree-Fock energy. b: The absolute value of the error in molecular energy relative to the FCI results.

nuclear repulsion energy as, for a given nuclear geometry, this is just a constant added to the electronic Hamiltonian. When computing dissociation curves, however, this must be taken into account:

$$\mathcal{H} = \sum_{jk} h_{jk} a_j^\dagger a_k + \sum_{jklm} g_{jklm} a_j^\dagger a_k^\dagger a_l a_m + V(\vec{R}). \tag{2.1}$$

We refer to these two cases (without and with the nuclear repulsion) as the electronic energy and molecular energy respectively. As expected, in the absence of noise, the QCM correction (green line) incorporates the electronic correlations at a sufficient level to recover energies below the Hartree-Fock energy (pink line) for all bond distances examined for all three molecules. For most cases, especially at longer bond lengths where the Hartree-Fock energy diverges more significantly from the FCI results, the moments-based correction without any additional error mitigation (green crosses) is sufficient to calculate energies below the Hartree-Fock energy, even when performed on a noisy quantum device.

With the application of 1-RDM purification, the moments based method on the quantum device *ibmq_sydney* (green inverted triangles) was able to outperform the noiseless

Hartree-Fock results for all molecular geometries considered, reaching error thresholds relative to the FCI results of order 10 mH for the longest chain, H_6, and 0.1 mH for molecular hydrogen. The latter is below the FCI accuracy threshold of 1.59 mH (black line in Figure 2.4b)). We note that this result is within millihartree accuracy of the FCI energy calculated in the minimal STO-3G basis and that a significantly larger basis is required to claim true chemical accuracy [99]. We also note that the results presented in [20] are likewise restricted by the choice of basis and are in fact within millihartree accuracy only of the Hartree-Fock energy and not the FCI ground-state. Additionally, the results presented here do not rely on the symmetry of the system and could, in principle, be calculated for any spatial arrangement of the atoms. This is in contrast to the results of [100] where millihartree error relative to FCI calculations was achieved for the H_{10} ring by exploiting the high level of rotational symmetry in the system. Other results reporting FCI accuracy include work reported in [19] and [23]: The former simulates NaH on 4 superconducting qubits using a frozen core approximation and a UCC inspired circuit, a problem of similar size to the H_2 molecule simulated here, and reaches millihartree error relative to the FCI energy. The latter simulates H_2O on 11 trapped-ion qubits by carefully selecting the excitation operators to be included in a UCC-style ansatz circuit and achieves errors relative to the noiseless trial-state energy and standard deviations at the millihartree level for the first few determinants. Given the precision of the QCM method when performed using the less accurate single Slater-determinant trial-state it is expected that application of the method to UCC-style ansatz circuits would provide a further improvement on the precision of the ground-state energy estimates.

2.3 Discussion

We have applied the Quantum Computed Moments method to the quantum-chemical problem of computing the ground-state energy of linear chains of hydrogen atoms and found that, even for restrictive single Slater-determinant trial-states, use of the QCM method allows for recovery of electron correlation and therefore of energies below the Hartree-Fock threshold. Though computation of the Hamiltonian moments may seem expensive, the scaling of the number of terms in $\mathcal{H}^4$ is significantly better than the worst-case as seen in Section 2.4.5 and could be further reduced by transformation into an alternate basis. Additionally the number of measurements can be controlled by grouping mutually commuting operators for simultaneous measurement.

With the addition of McWeeny purification, we demonstrate that the method is capable of outperforming (noise-free) Hartree-Fock calculations, even when the moments are computed on noisy present-day quantum hardware, for chains of up to 6 atoms, the largest system studied here requiring up to 27 CNOT gates. For molecular hydrogen, H_2, we achieve results that are within millihartree error of the FCI result calculated with respect to the minimal basis set for a range of inter-nuclear distances around the equilibrium bond length. For the H_6 chain at 0.74Å, with no error mitigation the QCM method is able to recover 97.1% of the molecular energy while the usual direct measurement of $\langle \mathcal{H} \rangle$ is able to return only 78% of the energy due to noise in the trial-circuit preparation.

Although the McWeeny purification technique is not easily generalised to states that cannot be represented as a single Slater-determinant, the QCM method performs well even without the purification and would likely benefit from any improvement in the trial-state. Such improvement could come in the form of a UCCD circuit ansatz, for example, and should allow the QCM result to approach the FCI energy more closely while maintaining the QCM method's resilience to noise. Experimentation with improved circuit ansätze is left to future work as is a detailed time-cost analysis. Furthermore it is possible to adapt the QCM method for the computation of excited state energies [91] and to properties other than the energy [101]. With improvement of the trial-state, combined with error mitigation techniques and alternative Hamiltonian representations to control the scaling of the problem the QCM method is a promising technique for the pursuit of chemical accuracy on present-day quantum hardware.

2.4 Methods

2.4.1 The chemical Hamiltonian

The chemical Hamiltonians used in this work were computed using the *Python* package *PySCF* [94] to optimise the molecular orbitals and were converted to qubit Hamiltonians using the Jordan-Wigner transform [4].

2.4.2 Measurement details

To reduce the required number of state preparations the Pauli strings required for measurement of the 1-RDM are grouped into $\mathcal{O}(N_s^2)$ mutually commuting tensor product basis sets [17] as described in Section 2.4.5. For each TPB, the trial-circuit was executed and the output state was measured 8192 times to obtain the average results. To estimate uncertainties, the 8192 measurement results were randomly assigned to one of 4 bins. Each bin (of roughly 2000 results each) was processed individually and the standard deviation of these 4 results was calculated. Quantum computed data for the graphs in Figures 2.3 and 2.4 were taken from the device *ibmq_sydney* and simulated results were calculated using a statevector simulator without shot noise. Dissociation curves were also calculated on the *ibmq_toronto* and *ibmq_guadalupe* devices and found to be consistent with the *ibmq_sydney* results.

2.4.3 Spin-degeneracy qubit reduction

Although trial-state depth is potentially the most precious quantum resource in a VQE calculation, any reduction in the number of qubits required for the simulation, such as through qubit tapering [102], will allow reduction of the circuit depth since fewer gates will be needed to generate a sufficiently entangled quantum state over the reduced number of qubits. In addition, any reduction in the number of qubits will likely decrease the number of measurements and therefore, the number of trial-state preparations required per energy estimation.

There are several properties of the systems studied here, the linear hydrogen atom chains, that allow the reduction technique used in [20] to be applied. It should be noted that these restrictions apply only to the qubit-reduction technique not to the QCM method in general. The restrictions are;

- The Hamiltonian of the system is not explicitly dependent on spin, while this is true for all molecules in a vacuum, it is not true in the presence of an external magnetic field, which could potentially limit the applicability of this technique.
- The trial-state used to simulate the system consists of a single Slater determinant. The Slater determinant is often used in quantum chemistry as a starting point from which more accurate states can be constructed. However, as the objective of the QCM method is to reduce the complexity of the trial-state, the use of a single determinant trial-state is a reasonable choice. Additionally, it is seen that the use of the QCM method allows for the recovery of energies below the Hartree-Fock energy.
- The system is simulated in a singlet state, while this can be done for any molecular system, the ground-state of paramagnetic materials, such as oxygen gas (O_2), is not a singlet. In these cases, application of the qubit reduction would lead to reduced accuracy.

The Slater determinant trial-state can be written in second-quantised form as;

$$|\Psi_{\text{trial}}\rangle = |\psi_P\rangle = \prod_{p\in P} b_p^\dagger |0\rangle, \tag{2.2}$$

where P is a list of the basis states from which the determinant is constructed, $b_p^\dagger$ is the fermionic creation operator for state p and $|0\rangle$ is the vacuum state. In this form, the antisymmetrisation of the state is encoded in the anticommutation of the fermionic operators. The single excitation operator $b_q^\dagger b_r$ will have vanishing expectation value with respect to the trial-state unless $q, r \in P$, in which case

$$\begin{aligned}\langle\Psi_P|b_q^\dagger b_r|\Psi_P\rangle &= \delta_{qr}\langle\Psi_P|\Psi_P\rangle - \langle\Psi_P|b_r b_q^\dagger|\Psi_P\rangle \\ &= \delta_{qr},\end{aligned} \tag{2.3}$$

where the last step uses $q \in P$ and $(b_q^\dagger)^2 = 0$ to show that the second term vanishes. From this point on, it will be assumed that expectation values are taken with respect to the trial-state $|\Psi_P\rangle$. Using the anticommutation relations it can similarly be shown that the expectation value of a double-excitation operator is;

$$\langle b_q^\dagger b_r^\dagger b_{r'} b_{q'}\rangle = \delta_{qq'}\delta_{rr'} - \delta_{rq'}\delta_{qr'} = \begin{vmatrix}\delta_{qq'} & \delta_{qr'} \\ \delta_{rq'} & \delta_{rr'}\end{vmatrix}, \tag{2.4}$$

and an arbitrary level excitation operator has the expectation value;

$$\langle b_q^\dagger b_r^\dagger b_s^\dagger \dots b_{s'} b_{r'} b_{q'}\rangle = \begin{vmatrix}\delta_{qq'} & \delta_{qr'} & \delta_{qs'} & \cdots \\ \delta_{rq'} & \delta_{rr'} & \delta_{rs'} & \cdots \\ \delta_{sq'} & \delta_{sr'} & \delta_{ss'} & \cdots \\ \vdots & \vdots & \vdots & \ddots\end{vmatrix}. \tag{2.5}$$

It is worth noting that;

- Swapping two creation (annihilation) operators is equivalent to swapping two rows (columns) in the determinant. Both actions introduce a factor of -1 as expected.
- If two creation (annihilation) operators act on the same state, then two rows (columns) in the determinant are equal and both sides of the equation vanish.
- If a creation (annihilation) operator does not act on a state in the Slater determinant, the equation does *not* hold. In this case, a modified form of the equation is required;

$$\langle b_q^\dagger b_r^\dagger \dots b_{r'} b_{q'}\rangle = \begin{vmatrix}\langle b_q^\dagger b_{q'}\rangle & \langle b_q^\dagger b_{r'}\rangle & \cdots \\ \langle b_r^\dagger b_{q'}\rangle & \langle b_r^\dagger b_{r'}\rangle & \cdots \\ \vdots & \vdots & \ddots\end{vmatrix}. \tag{2.6}$$

When all indices are $\in P$, Equation 2.3 reduces this to Equation 2.5, but if any creation (annihilation) index is $\notin P$, then every entry in that row (column) will be 0 and the determinant will vanish, as required.

To allow for a more general state preparation, the basis states used in the construction of the Slater determinant are allowed to vary from the basis in which the second-quantised Hamiltonian is written. The trial-state basis will be represented by the creation (annihilation) operators $b^\dagger$ (b) while the Hamiltonian basis will be represented by the operators $a^\dagger$ (a). The relation between these bases can be written as;

$$a_j^\dagger = \sum_q A_{jq} b_q^\dagger, \tag{2.7}$$

for appropriately normalised complex amplitudes A_{jq}. The expectation value for an excitation operator in the Hamiltonian basis is then

$$\begin{aligned}
\langle \Psi_P | a_j^\dagger a_k^\dagger \dots a_{k'} a_{j'} | \Psi_P \rangle &= \sum_{qr\dots r'q'} A_{jq} A_{kr} \dots A^*_{k'r'} A^*_{j'q'} \langle \Psi_P | b_q^\dagger b_r^\dagger \dots b_{r'} b_{q'} | \Psi_P \rangle \\
&= \sum_{qr\dots r'q'} A_{jq} A_{kr} \dots A^*_{k'r'} A^*_{j'q'} \begin{vmatrix} \langle b_q^\dagger b_{q'} \rangle & \langle b_q^\dagger b_{r'} \rangle & \cdots \\ \langle b_r^\dagger b_{q'} \rangle & \langle b_r^\dagger b_{r'} \rangle & \cdots \\ \vdots & \vdots & \ddots \end{vmatrix} \\
&= \begin{vmatrix} \sum_{qq'} A_{jq} A^*_{j'q'} \langle b_q^\dagger b_{q'} \rangle & \sum_{qr'} A_{jq} A^*_{k'r'} \langle b_q^\dagger b_{r'} \rangle & \cdots \\ \sum_{rq'} A_{kr} A^*_{j'q'} \langle b_r^\dagger b_{q'} \rangle & \sum_{rr'} A_{kr} A^*_{k'r'} \langle b_r^\dagger b_{r'} \rangle & \cdots \\ \vdots & \vdots & \ddots \end{vmatrix} \\
&= \begin{vmatrix} \langle a_j^\dagger a_{j'} \rangle & \langle a_j^\dagger a_{k'} \rangle & \cdots \\ \langle a_k^\dagger a_{j'} \rangle & \langle a_k^\dagger a_{k'} \rangle & \cdots \\ \vdots & \vdots & \ddots \end{vmatrix}.
\end{aligned} \tag{2.8}$$

Since the Hamiltonian contains only single- and double-excitations, the qubit reduction as applied in [20] required only the expression for double-excitation operators, which had previously been determined in [93]. The more general form here is required for use with the Hamiltonian moments which can include up to 8th level excitation operators.

Equation 2.8 allows the measurement of an ith level excitation operator to be reduced to the measurement of i^2 single-excitation operators and the calculation of the determinant of an $i \times i$ matrix (as previously derived in [103]). Since the system is being simulated in the singlet state and the Hamiltonian does not depend explicitly on the spin, any spin crossing 2-mode terms must have vanishing expectation values and all down-spin terms will have expectation values identical to the corresponding spin-up terms:

$$\begin{aligned}\langle a^\dagger_{(j,\uparrow)} a_{(k,\downarrow)} \rangle &= 0,\\ \langle a^\dagger_{(j,\uparrow)} a_{(k,\uparrow)} \rangle &= \langle a^\dagger_{(j,\downarrow)} a_{(k,\downarrow)} \rangle.\end{aligned} \tag{2.9}$$

Using Equations 2.8 and 2.9 any expectation value required for estimating the Hamiltonian can be calculated by simulating and measuring only the spin-up states (or equivalently only the spin-down states), allowing the number of qubits required for simulation to be reduced by a factor of 2.

While this procedure may seem expensive at first glance, it is important to note that the level of the expectation values and therefore the size of the determinants to be computed is dependant only on the level of the moments correction used and not on the system size. Additionally, there are only $N_s^2/4$ single-excitation expectation values from which the determinants can be constructed, for a simulation involving N_s spin-orbitals. Each of these expectation values only needs to be measured once and Section 2.4.5 discusses ways in which the number of trial-state preparations can be reduced further. From these $\mathcal{O}(N_s^2)$ expectation values, the 1-body reduced density matrix can be constructed and all non-vanishing fermionic expectation values can be written as minors (determinants of a submatrix) of the 1-RDM.

2.4.4 The trial-state

When applying the VQE algorithm to examples from quantum chemistry it makes sense to map the molecular orbitals, constructed as linear combinations of atomic orbitals (LCAOs), to each qubit. The molecular integrals of Equation 2.1 can then be calculated with respect to these orbitals. In this work, the STO-3G basis is used to represent each atomic orbital by a linear combination of 3 Gaussian orbitals, the molecular orbitals and their corresponding integrals are then calculated using the *Python* package *PySCF* [94] through the *qiskit* package [104]. Because the molecular orbitals are pre-optimised by *PySCF*, the solution to the Hartree-Fock minimisation should trivially be the occupation of the η lowest energy spin-orbitals, where $\eta = N_s/2$ (for the hydrogen atom chains) is the number of electrons. However if the molecular orbitals were not pre-optimised (for example, the use of alternate bases can reduce the number of terms in the Hamiltonian, see Section 2.4.5) then the optimal trial-circuit parameters will need to be found. The trial-circuit used here is based on that used in [20].

From the initial state, a series of parameterised Givens rotations [95] is applied between neighbouring qubits following the optimal layout determined in [96]. The resulting circuit (see Figure 2.2) has depth $\mathcal{O}(N_s)$ and gate count $\mathcal{O}(N_s^2)$. The rotations used here are defined in such a way that when the gate parameter, θ, goes to 0, the Givens rotation reduces to the identity operation and therefore a circuit with the parameter set $(0, 0, 0, \dots)$ will return the classically pre-optimised Hartree-Fock state. In addition, when the rotation parameters are non-zero, the Givens rotations effectively implement a rotation of the single-particle basis and thus necessarily preserve the particle number symmetry of the chemical Hamiltonian as demonstrated for a particular choice of parameters in Figure 2.2c.

2.4.5 Problem scaling

While the QCM method provides a technique for the suppression of noise for VQE, which in turn was developed to take advantage of near-term quantum hardware while reducing the impact of errors, a possible limitation of VQE and, by extension, the QCM method is the potentially rapid growth of the number of terms present in the Hamiltonian which determines the number of measurements and hence the number of state preparations required of the quantum processor. In the usual molecular orbital basis the number of terms in the quantum chemical Hamiltonian scales as $\mathcal{O}(N_s^4)$ (where N_s is the number of spin-orbitals) which is then amplified by the exponentiation required for the moments correction. The scaling problem in VQE has led to various methods designed to reduce the number of measurements required. These can be split into two categories. The first class of methods are those that aim to reduce the number of terms in the Hamiltonian, for example by transforming the operator into a different basis in which the number of terms scales better than the usual $\mathcal{O}(N_s^4)$ [70, 72, 105] or by neglecting Hamiltonian terms with a weight below a certain threshold, such as interaction terms between spatially separated wavefunctions (orbitals). The second class of methods are those that aim to reduce the number of measurements (but not the number of terms) by partitioning the Hamiltonian terms into groups of mutually commuting, and therefore simultaneously measurable, operators. There are a variety of methods through which this grouping can be performed such as mapping to a minimum clique cover problem [106] or using an identity-operator sorting algorithm [85].

In this work, we make use of the fact that only the single-excitation operators need to be measured. For a system represented by N_s spin-orbitals this already limits the number of expectation value measurements to $\mathcal{O}(N_s^2)$. A refinement performed in [20] further reduces the necessary number of measurements to $N_s/2 + 1$. Here, since the exponentiation of the Hamiltonian scales as $\mathcal{O}(N_s^{16})$ at worst, the $\mathcal{O}(N_s)$ method is not performed and an alternative that scales as $\mathcal{O}(N_s^2)$ but that avoids any need for virtual swapping or circuit recompilation, is employed instead. In comparison to the cost of calculating the moments formulae, the $\mathcal{O}(N_s^2)$ measurement cost is deemed acceptable.

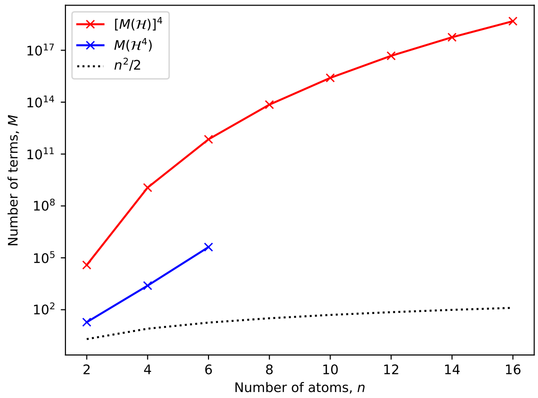


FIGURE 2.5: Scaling of the number of terms, M, in the fourth moment for the one-dimensional hydrogen chains. The blue line is the observed number of terms in $\mathcal{H}^4$ (before grouping commuting operators) while the red line represents the worst-case scenario of raising the number of terms in $\mathcal{H}$ to the power 4. The dotted black line is the number of measurements required by the method described in Section 2.4.5, $n^2/2$ where n is the number of atoms.

To form TPB groupings the 2-mode operators are first transformed to qubit operators using the Jordan-Wigner transform [4]. For real-valued orbitals and $j < k$;

$$\langle a_j^\dagger a_k \rangle = \frac{1}{4}\big(\langle \dots I_{j-1} X_j Z_{j+1} \dots Z_{k-1} X_k I_{k+1} \dots \rangle + \langle \dots I_{j-1} Y_j Z_{j+1} \dots Z_{k-1} Y_k I_{k+1} \dots \rangle\big). \tag{2.10}$$

For the $j > k$ case, j and k can be exchanged on the right-hand side of the equation. Each 2-mode operator can be characterised by a distance, $d = |j - k|$. By measuring the Pauli strings consisting of X on every qubit and Y on every qubit, all distance, $d = 1$, operators can be reconstructed. Likewise measuring the strings $XZXZ\dots$, $YZYZ\dots$, $ZXZX\dots$ and $ZYZY\dots$ allows for the reconstruction of all distance, $d = 2$, operators. Continuing this pattern, it can be shown that every $d \in (0, N_s/4]$ operator can be reconstructed by measuring $2d$ strings. For operators with $d \in [N_s/4, N_s/2]$ the operators are less local but there are fewer possible operators and the number of string measurements required is $N_s - 2d$. For the cases considered here, $N_s/4$ is always an integer since the singlet state requires an even number of hydrogen atoms and the spin degeneracy introduces a further factor of 2. In the special case of $d = 0$ the expectation value becomes $\langle a_j^\dagger a_j \rangle = \frac{1}{2}(1 - \langle Z_j \rangle)$ and can be reconstructed from the four measurements made for the $d = 2$ case. The total number of measurements required is then,

$$M = \sum_{d=1}^{N_s/4-1} 2d + \sum_{d=N_s/4}^{N_s/2-1} (N_s - 2d) = \frac{N_s^2}{8},$$
$$(N_s > 4,\ N_s/4 \in \mathbb{Z}). \tag{2.11}$$

Compared to the naive method of measuring each term in the moments formulae individually which scales at worst as $\mathcal{O}(N_s^{16})$, the $\mathcal{O}(N_s^2)$ cost is a significant improvement.

In terms of computation time, the code used to perform the calculations will require further development and optimisation before meaningful comparisons to classical methods can be drawn.

2.4.6 Reduced density matrix purification

In the Hartree-Fock approximation, the one-body reduced density matrix (1-RDM), R, is expected to be idempotent ($R = R^2$). In practice, the presence of errors in the application of gates or in readout, as well as the statistical effects of shot noise lead to the calculated 1-RDM being only nearly idempotent. In this work, the McWeeny iterative purification method [20, 93] is used to correct for these errors. A brief motivation of the procedure is given below;

The error in the RDM (in the sense of its distance from idempotency) can be quantified by $D = R^2 - R$ in which case the objective of the purification is to reduce all elements of the matrix D to 0. By framing this as a minimisation problem with cost function

$$C = \sum_{ij} (D_{ij})^2,$$

the RDM can be purified by gradient descent with a step size δ according to;

$$R' = R - \delta \nabla_R(C), \tag{2.12}$$

where R and R' are the current and updated 1-RDMs respectively and $\nabla_R(C)$ is the gradient of the cost function with respect to the matrix R;

$$\nabla_R(C) = \begin{bmatrix} \frac{\partial C}{\partial R_{11}} & \frac{\partial C}{\partial R_{12}} & \cdots \\ \frac{\partial C}{\partial R_{21}} & \frac{\partial C}{\partial R_{22}} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

In the case of real-valued orbitals, the error matrix D inherits the symmetry of the 1-RDM and so the cost function can be rewritten using the identity

$$\sum_{ij}(D_{ij})^p = \mathrm{Tr}(D^p), \tag{2.13}$$

for symmetric matrix D and non-negative integer p:

$$C = \mathrm{Tr}(R^4 - 2R^3 + R^2).$$

So the gradient evaluates to

$$\nabla_R(C) = 4R^3 - 6R^2 + 2R.$$

Substituting this into Equation 2.12 and choosing the step size to be $\delta = 1/2$, the expected McWeeny purification formula is recovered

$$R' = 3R^2 - 2R^3. \tag{2.14}$$

Once the 1-RDM has been measured

$$R = \begin{bmatrix} \langle a_0^\dagger a_0 \rangle & \langle a_0^\dagger a_1 \rangle & \cdots \\ \langle a_1^\dagger a_0 \rangle & \langle a_1^\dagger a_1 \rangle & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \tag{2.15}$$

Equation 2.14 can be applied iteratively to obtain a corrected 1-RDM that more accurately represents a single Slater-determinant state. Using the results of Section 2.4.3, the energy can then be calculated from the elements of this corrected 1-RDM.

2.4.7 Sample measurement data

Data for the H_2 molecular Hamiltonian calculated at a bond length of 0.74Å using the minimal STO-3G basis is presented in Table 1. Table 2 contains sample data from the quantum processor *ibmq_sydney* and from which the 1-RDM for molecular hydrogen (with circuit parameter $\theta = 0$) can be constructed. Figure 2.6 presents estimated energies for the H_4 chain for 97 sets of random circuit parameters for both direct and moments-based measurement.

Fermionic operator	$\langle\mathcal{H}\rangle$	$\langle\mathcal{H}^2\rangle$	$\langle\mathcal{H}^3\rangle$	$\langle\mathcal{H}^4\rangle$
$a_0^\dagger a_0$	-1.2533	1.5708	-1.9687	2.4674
$a_1^\dagger a_1$	-0.4751	0.2257	-0.1072	0.0509
$a_2^\dagger a_2$	-1.2533	1.5708	-1.9687	2.4674
$a_3^\dagger a_3$	-0.4751	0.2257	-0.1072	0.0509
$a_1^\dagger a_0^\dagger a_1 a_0$	-0.4825	0.2443	-0.142	0.1089
$a_2^\dagger a_0^\dagger a_2 a_0$	-0.6748	-0.247	2.3385	-6.6903
$a_2^\dagger a_0^\dagger a_3 a_1$	-0.1812	0.3777	-0.7094	1.3164
$a_2^\dagger a_1^\dagger a_2 a_1$	-0.6637	0.6301	-0.7642	1.009
$a_2^\dagger a_1^\dagger a_3 a_0$	-0.1812	0.3859	-0.6222	0.9001
$a_3^\dagger a_0^\dagger a_2 a_1$	-0.1812	0.3859	-0.6222	0.9001
$a_3^\dagger a_0^\dagger a_3 a_0$	-0.6637	0.6301	-0.7642	1.009
$a_3^\dagger a_1^\dagger a_2 a_0$	-0.1812	0.3777	-0.7094	1.3164
$a_3^\dagger a_1^\dagger a_3 a_1$	-0.6977	0.3548	-0.1216	-0.0501
$a_3^\dagger a_2^\dagger a_3 a_2$	-0.4825	0.2443	-0.142	0.1089
$a_2^\dagger a_1^\dagger a_0^\dagger a_2 a_1 a_0$		1.3926	-3.913	8.7428
$a_3^\dagger a_1^\dagger a_0^\dagger a_3 a_1 a_0$		0.6637	-1.1088	1.4847
$a_3^\dagger a_2^\dagger a_0^\dagger a_3 a_2 a_0$		1.3926	-3.913	8.7428
$a_3^\dagger a_2^\dagger a_1^\dagger a_3 a_2 a_1$		0.6637	-1.1088	1.4847
$a_3^\dagger a_2^\dagger a_1^\dagger a_0^\dagger a_3 a_2 a_1 a_0$		2.4195	-5.4784	10.9158

TABLE 2.1: Fermionic operator weights for molecular hydrogen calculated in the STO-3G basis at equilibrium bond length (0.74Å). Molecular orbitals are ordered from 0 to 3 as spin-up bonding, spin-up anti-bonding, spin-down bonding, spin-down anti-bonding.

$\langle a_i^\dagger a_j\rangle$	$j=0,2$	$j=1,3$
$i=0,2$	0.96	-0.0114
$i=1,3$	-0.0114	0.0236

$\langle a_i^\dagger a_j\rangle$	$j=0,2$	$j=1,3$
$i=0,2$	0.9619	-0.0082
$i=1,3$	-0.0082	0.0229

$\langle a_i^\dagger a_j\rangle$	$j=0,2$	$j=1,3$
$i=0,2$	0.9647	-0.0036
$i=1,3$	-0.0036	0.023

$\langle a_i^\dagger a_j\rangle$	$j=0,2$	$j=1,3$
$i=0,2$	0.9675	-0.0095
$i=1,3$	-0.0095	0.0262

TABLE 2.2: Four sets of 1-RDM elements for the hydrogen molecule ($\theta = 0$) calculated on *ibmq_sydney* with $\approx$ 2000 shots each. The ordering of orbitals matches that used in Table 2.1.

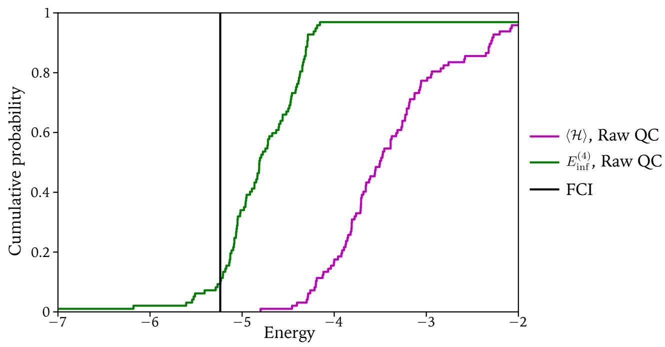


FIGURE 2.6: Cumulative frequency plot for the H_4 molecular chain. 100 sets of uniform random parameters were chosen and the energy was evaluated both directly and by the QCM method on the quantum processor *ibmq_montreal*. The vertical axis is the cumulative frequency, i.e. the fraction of points with a measured energy below a given value. In the ideal case (that the FCI energy is recovered regardless of the parameter values) the resulting plot would be the step function at the FCI energy (black line). Of the 100 random points, only 97 are represented in the plot. The remaining 3 points were rejected due to large variations in the QCM energy when repeated, e.g. standard deviations of the same scale or larger than the energy measurements. These anomalous results are likely due to the combination of a poorly conditioned point and errors in the device leading to non-physical density matrices. The large deviations are seen to vanish when McWeeny purification is applied.

Chapter 3

Ground-state energy calculation for the water molecule on a superconducting quantum processor

3.1 Introduction

In this chapter, which is based on the peer-reviewed paper "Ground-state energy calculation for the water molecule on a superconducting quantum processor" published in *Physical Review Applied*, we now consider multi-determinant trial-states, specifically selected double-excitation unitary coupled-cluster (UCCD) states.

At the time of writing a large focus in the application of quantum computers to chemistry problems was on the pursuit of 'chemical accuracy' (Section 1.2.3). However, despite several claims, the most accurate calculations performed on quantum hardware achieved only FCI accuracy [17, 19, 22, 24, 25, 40, 100, 107–110] or trial-state accuracy [20, 23]. The results presented in this chapter achieve FCI accuracy, though the FCI computation is performed over 14 spin-orbitals while the quantum computation is performed over 8 spin-orbitals. Of the remaining 6 orbitals, 2 core orbitals are neglected completely and the 4 virtual orbitals are incorporated via the QCM approach. These were the largest (in qubit number) quantum computations carried out on noisy quantum hardware to achieve millihartree error from the FCI result at the time, more recently sample-based quantum diagonalisation has allowed investigation of larger systems [44, 45]. The calculations employ a circuit of 22 CZ gates and 25 depth over 8 qubits, achieving an error of 1.0 ±

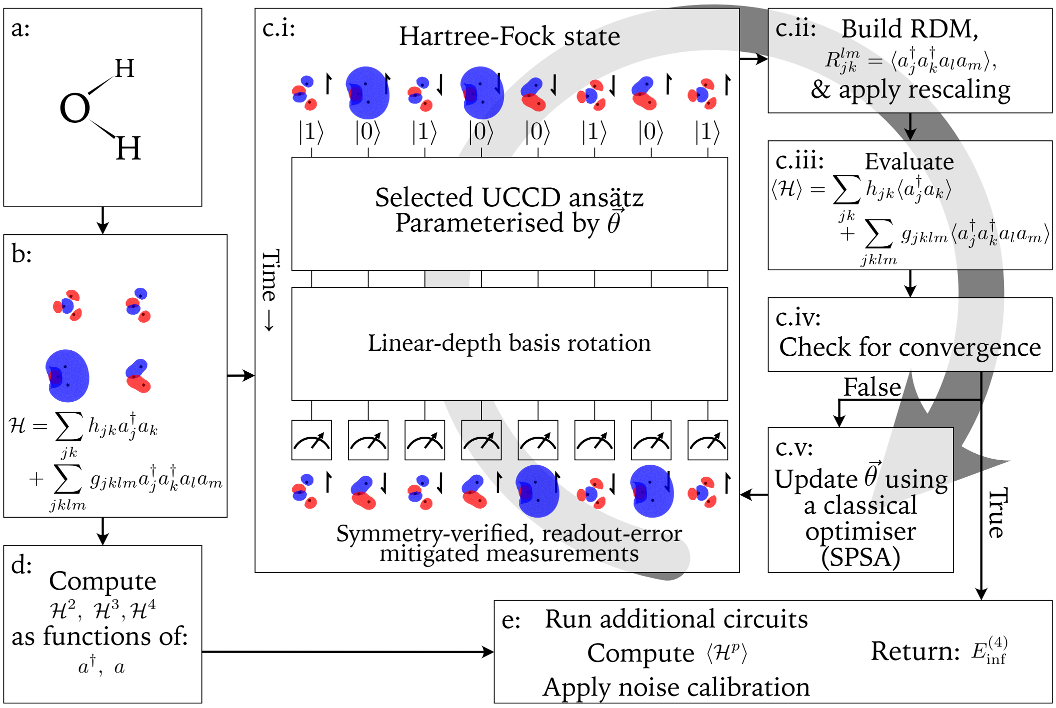


FIGURE 3.1: Flow chart of the overall approach employed in this work. a: Definition of the molecular system. b: Calculation of the molecular orbitals and electronic Hamiltonian. c: Hybrid minimisation loop to approximate the ground-state on the quantum device. d: Powers of the Hamiltonian can be evaluated independently of (i.e. in parallel with) the minimisation loop. e: Calculation of the Hamiltonian moments, $\langle \mathcal{H}^p \rangle$, and hence the Lanczos cluster expansion corrected energy estimate, $E_{\text{inf}}^{(4)}$. For additional detail on the above steps see Section 3.4.

0.8 mHa relative to FCI in the 14 spin-orbital minimal basis. While electronic structure computations have been performed with larger numbers of qubits, these computations were unable to achieve FCI accuracy due to noise in the quantum computation, either directly or indirectly by limiting circuit depth. In our case, in addition to error mitigation techniques, the noise is offset by the observed robustness of the moments-based correction [111].

3.2 Results

Calculations were performed following the overall procedure outlined in Figure 3.1. The parameters, $\hat{\theta}$, in the Trotterised UCCD trial-circuit were optimised using a classical statevector simulator; the problem of optimising the trial-state structure and parameters in the presence of noise is left to future work. The final computations were performed on the *ibmq_kolkata* and *ibm_torino* devices using 10^5 and 2.5×10^4 shots per basis respectively. Additional method details can be found in Section 3.4.

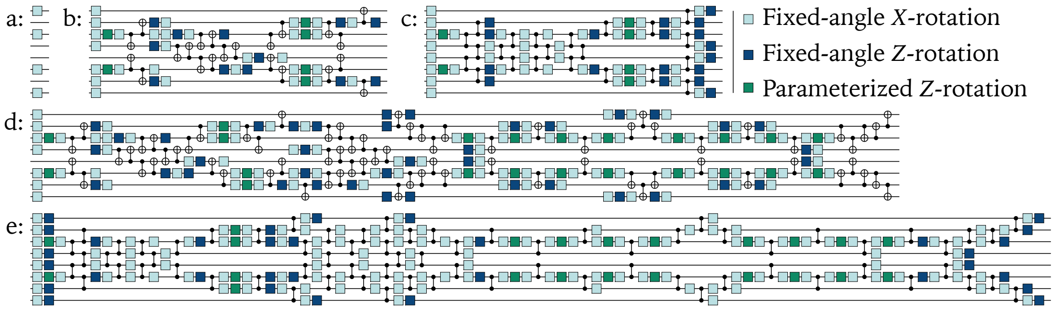


FIGURE 3.2: Structure of the trial-state circuits used in this work a: Minimal circuit to prepare the Hartree-Fock state. 4-excitation trial-state circuit compiled to the base gateset of b: *ibmq_kolkata* and c: *ibm_torino* each with depth $D = 25$. 6-excitation trial-state circuit compiled to the base gateset of d: *ibmq_kolkata* and e: *ibm_torino* with depths $D = 74$ and $D = 87$ respectively. For each circuit, there exist a set of parameters such that the Hartree-Fock state is produced. These circuits are followed by linear depth basis rotations (see Section 3.4.3), then computational basis measurements.

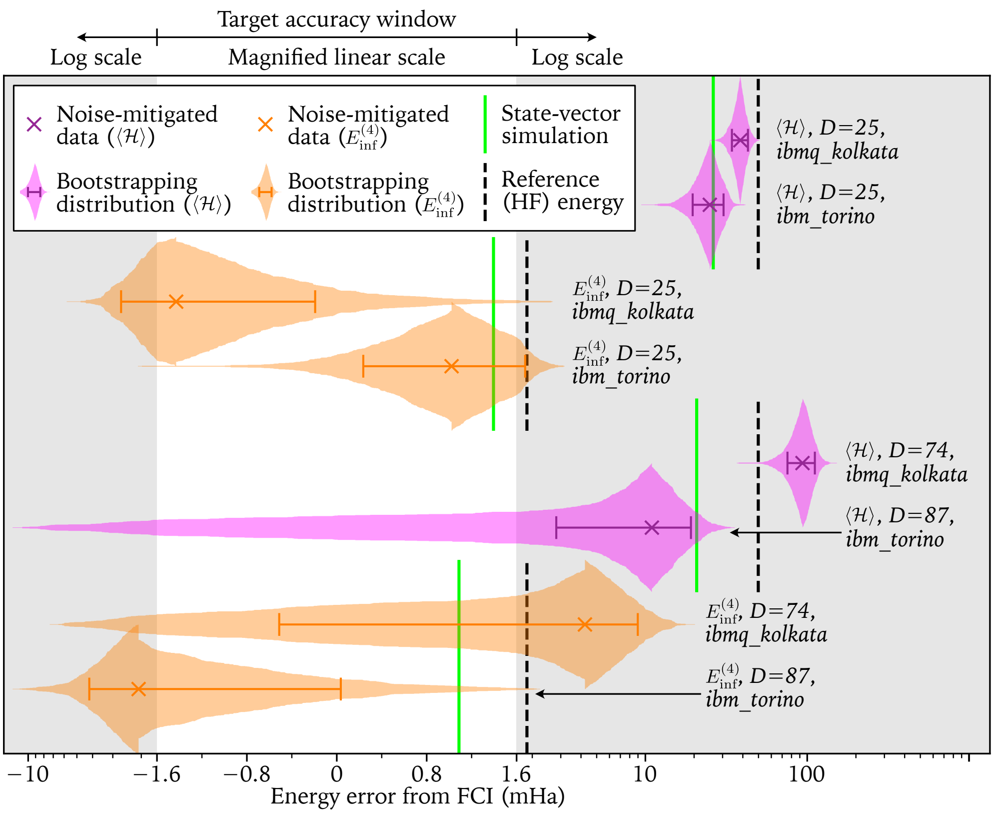


FIGURE 3.3: Energy estimation results for the water molecule based on measuring $\langle \mathcal{H} \rangle$ and $E^{(4)}_{\mathrm{inf}}$ for the 4-excitation ($D = 25$) and 6-excitation ($D = 74$ and $D = 87$) circuits presented in Figure 3.2b-e. The noiseless simulation results are shown for the trial-state (in green) and the reference-state (in dashed black). Noise-mitigated results, taken from the quantum devices *ibmq_kolkata* (10^5 shots per basis) and *ibm_torino* (2.5×10^4 shots per basis), are marked by magenta and orange crosses (for $\langle \mathcal{H} \rangle$ and $E^{(4)}_{\mathrm{inf}}$ respectively). The cumulative distributions indicate 500 statistical bootstrapping results and the error bars represent one standard deviation. Note that the horizontal axis uses a linear scale in the target accuracy window $[-1.6, 1.6]$ mHa and a logarithmic scale outside this window.

	HF reference-state		4-excitation UCC trial-state		6-excitation UCC trial-state	
	$\langle\mathcal{H}\rangle$	$E_{\mathrm{inf}}^{(4)}$	$\langle\mathcal{H}\rangle$	$E_{\mathrm{inf}}^{(4)}$	$\langle\mathcal{H}\rangle$	$E_{\mathrm{inf}}^{(4)}$
ibmq_kolkata			38.6 ± 4.4	-1.4 ± 1.2	93.5 ± 18.0	4.2 ± 4.7
ibm_torino			25.0 ± 5.4	1.0 ± 0.8	11.0 ± 8.2	-2.1 ± 2.1
Statevector simulation	49.8	1.8	26.2	1.4	20.8	1.1

Table 3.1: Energy estimation error in millihartree from FCI (-84.1813Ha, excluding nuclear repulsion) for $\langle\mathcal{H}\rangle$ and $E_{\mathrm{inf}}^{(4)}$ for different trial-states and devices. The uncertainties are one standard deviation as calculated using 500 bootstrapping samples.

Two different trial-state depths were used, the state preparation circuits in Figure 3.2b and c have a depth of 25 (22 CNOT/CZ gates) and implement 4 excitations from the truncated UCC ansatz (see Section 3.4.2) while the circuits in Figure 3.2d and e have depth 74 and 87 (72 CNOT and 70 CZ gates respectively) and implement 6 excitations. After performing the demonstration, a slightly more simplified form of the circuit 3.2d was found (70 CNOTs, 73 depth). It is expected that the improved circuit will have minimal impact on the final energy estimates. These depths and gate counts include only the state-preparation component of the circuit since different measurement bases increase these values by different amounts. The results for both energy estimates $\langle\mathcal{H}\rangle$ and $E_{\mathrm{inf}}^{(4)}$ and for both trial-states are presented in Figure 3.3 and numerical data is given in Table 3.1. As expected the results from *ibm_torino* outperform the results from *ibmq_kolkata* since the former is a newer device with lower reported error rates, this is true even though the *ibmq_kolkata* results use four times as many shots as the *ibm_torino* results. An important observation regarding the results presented is that the moments of the Hartree-Fock state are classically tractable (and are required for the reference-state calibration technique used for noise mitigation, see Section 3.4.7). Therefore, if the quantum computer is going to provide any utility in the computation, it must give expectation values that are (at least) competitive with this reference-state energy (black dashed lines in Figure 3.3). It is also important to note that the energy computed from Equation 1.31 is non-variational, so the FCI energy is not a lower bound which can lead to ‘negative errors’.

3.2.1 4-excitation trial-state

For the 4-excitation trial-state, all energy estimates ($\langle\mathcal{H}\rangle$ and $E_{\mathrm{inf}}^{(4)}$ on both devices) have lower error than the reference-state. $\langle\mathcal{H}\rangle$ in particular, is separated from the corresponding reference-state estimate by more than 2 standard deviations, indicating that the benefits of implementing the quantum circuit outweigh the cost (in accuracy) imposed by device noise. For $E_{\text{inf}}^{(4)}$, the separation between noise-mitigated and reference-state energies (in absolute error from FCI) is less than half a standard deviation for *ibmq_kolkata* and about one standard deviation for *ibm_torino*. The reduced resolution can be largely attributed to the high accuracy of the moments method applied to the reference-state, which has an error only 0.5mHa larger than the statevector simulation. Given that the target accuracy is 1.6mHa, the obtained standard deviations of $\leq$1.2mHa are acceptable and it is unnecessary to attempt to reduce this further. Instead, it may be of interest to repeat the demonstration for a system that is not well modelled by Hartree-Fock methods to determine if the resolution between reference and noise-mitigated energies can be increased.

3.2.2 6-excitation trial-state

Due to the significant depth increase to the trial-circuit, the 6-excitation trial-state energies are expected to be more susceptible to noise than the 4-excitation energies. On *ibmq_kolkata*, the noisier device, both $\langle \mathcal{H} \rangle$ and $E_{\text{inf}}^{(4)}$ fail to outperform the reference state. This is unexpected given that the reference-state calibration method should ensure that a trial-state that outperforms the reference-state in the absence of noise should remain competitive with the reference in the presence of noise. On the other hand, *ibm_torino* gives a value of $\langle \mathcal{H} \rangle$ that is significantly lower than the statevector simulation, which is also unexpected. A possible explanation for both of these observations is that the noise-model assumptions used in error mitigation (Section 3.4.7) may be invalid, possibly because of coherent errors in the state preparation. If this is the case, a solution would be to use randomised compiling [112] to convert coherent noise into unbiased stochastic noise. Alternatively, the unexpected behaviour may be due to a failure to satisfy the N-representability conditions that determine whether the computed reduced density matrices are physical (see Section 3.4.7).

3.3 Discussion

While larger electronic structure computations have been performed, these are mostly made possible by encoding multiple spin-orbitals into a single qubit [20, 27, 28, 100] and/or through exploiting the assumption of weak entanglement [24, 25]. This can allow simplification of paired double-excitations to circuit elements involving only two CNOT/CZ gates, however, it prohibits the implementation of single- and unpaired double-excitations, limiting the effectiveness of the trial-state. Here, knowledge of the initial state of the circuit is used to simplify the implementation of the first four excitations, however, the circuit is not restricted to paired coupled cluster states. After these four simplified excitations, the general form of the double excitation gate is required, which results in the comparatively deep 6-excitation circuit.

The accuracy obtained here for the water molecule compares well with the accuracy achieved for the benzene molecule using VQE for 8 qubits and advanced circuit optimisation on a trapped-ion quantum computer [113], though since the computations are performed for different molecules, a direct comparison is difficult. Additionally, the trapped-ion energy errors are reported relative to 8 spin-orbital complete active space calculations, while we compare to the more accurate 14 spin-orbital FCI. Notably, the trapped-ion experiment required fewer state preparations at the expense of deeper circuits. This trade-off is favourable for trapped-ion qubits which have longer coherence times but a lower repetition rate than superconducting qubits. The results obtained here also compare well with previous results for the F_2 molecule [22] computed using 12 superconducting qubits. While our results use fewer qubits, we achieve a higher accuracy. Additionally, the system considered here is at half-filling of the spin-orbitals (where the problem is hardest) while moving away from half-filling (as in the case of F_2) makes the problem somewhat easier to solve.

We demonstrate that up to circuit depths > 25 and widths of 8 qubits, there is sufficient information contained in the output probability distributions from IBM Quantum hardware to outperform the classical Hartree-Fock reference-state. It is important to note that the energy scale on which this comparison takes place is much smaller than the scale set by the maximally mixed state, which has error on the order of 1-2 Ha (500-800 mHa) for $\langle \mathcal{H} \rangle$ ($E_{\text{inf}}^{(4)}$). This is 2-3 orders of magnitude larger than the energy scales of interest. This result demonstrates that it is possible for moments-based methods to extend the reach of variational algorithms for quantum chemistry on present-day and near-term quantum hardware, pointing the way towards a potential for quantum utility.

3.4 Methods

3.4.1 Electronic Hamiltonian

The electronic Hamiltonian is calculated using the *PySCF* [94] *Python* package in the minimal STO-3G basis comprising 14 spin-orbitals. The core orbitals (predominantly the 1s orbitals of the O atom) are frozen and the Hamiltonian moments are computed via Wick's theorem in the 12 spin-orbital basis. The Hamiltonian and its moments are then reduced further, to 8 spin-orbitals by freezing the 4 orbitals that participate least in the bonding. By computing moments in the 12 spin-orbital basis, the moments-corrected energies are able to extend the computation into the otherwise unused virtual orbitals. The FCI energies are also computed by *PySCF* in the 14 spin-orbital basis.

3.4.2 Trial-state design

One of the most challenging aspects of implementing variational hybrid algorithms is choosing a trial-circuit that is expressive enough to contain (a good approximation to) the true ground-state and at the same time is shallow enough for the quantum device to remain coherent throughout the state preparation. In addition, care should be taken to avoid over-parameterisation of the ansatz which can lead to so-called 'barren plateaus' [15] – regions of exponentially vanishing gradient that cause classical optimisation to become infeasible. A common, physically motivated, starting point for creating trial-circuits for quantum chemistry is the Unitary Coupled Cluster (UCC,[114]) ansatz:

$$\begin{aligned} |\Psi_{\text{UCC}}\rangle &= e^{i\tau}|\Psi_{\text{init}}\rangle, \\ \tau &= i(T - T^{\dagger}) \end{aligned} \tag{3.1}$$

where T is a weighted sum over all possible electronic excitations, including both uncorrelated (single) and correlated (double, triple, etc.) excitations. The initial state, $|\Psi_{\text{init}}\rangle$, is an easily prepared state in the correct symmetry sector of Hilbert space and is usually taken to be the Hartree-Fock state, $|\Psi_{\text{HF}}\rangle$. While the UCC trial-state is expected to give a good approximation of the true ground-state as the number of excitations is increased (in the limit that all excitations are included the trial-state can exactly express the ground state), practical implementation requires both limitation to some maximum excitation order and Trotterisation. Here, we use the double-excitation cluster operator, T_2, and Trotterise to first order:

$$\begin{aligned} |\Psi_{\text{trial}}\rangle &= \prod_{jklm} G_{jklm}(\theta_{jklm}) \cdot |\Psi_{\text{HF}}\rangle, \\ G_{jklm}(\theta) &= \exp\left(\theta(a_j^{\dagger}a_k^{\dagger}a_l a_m - a_m^{\dagger}a_l^{\dagger}a_k a_j)\right), \end{aligned} \tag{3.2}$$

where $a_j^{\dagger}a_k^{\dagger}a_l a_m$ implements the electronic double-excitation (the hermitian conjugate ensures unitarity of the G operator) and θ_{jklm} are variational parameters (corresponding to the weights in the original cluster operator, T). Given sufficient circuit depth, the single- and double- excitation ansatz has been seen to be an effective trial-state [115]. In this case the exclusion of single-excitation operators is partially justified by the observation that leading single-excitations cannot introduce multi-reference character and therefore cannot improve on the Hartree-Fock state, while trailing single-excitations can be accounted for efficiently with classical processing [28, 116, 117]. In the Trotterisation of the cluster operator, there is a degree of freedom as to the order in which non-commuting component excitations should be performed. Here, the order is chosen based on the known contribution of each excitation term to the true ground-state [23] and, to limit circuit depth and permit implementation on present-day devices, only the first 4-6 excitations are considered. In practice this ordering would not be known a priori, however, it should be possible to make use of physical/chemical intuition and/or an adaptive ansatz [118, 119] to determine a suitable ordering. This choice of ordering further justifies the lack of single-excitations, as they are seen to have a smaller contribution than double-excitations to the true ground-state for the specific Hamiltonian considered here.

Since the trial-state preparation operator has now been written as a product of double-excitations, a quantum circuit to implement the operator can be separated into a number of 'blocks' each of which implements a double-excitation operation on a set of four fermionic modes. By considering the fermionic modes as logical qubits[1] on a linear array of physical qubits it can be shown that using the fermionic swap operator to permute the involved fermionic modes together converts the highly non-local excitation to a local interaction (shown in Figure 3.4). Decomposition of the local fermionic interaction leads to 19 CNOT gates and 19 single-qubit (H, S, $S^\dagger$ and $R_y(\theta)$) gates with linear connectivity as shown in Figure 3.5.

3.4.3 Reduced density matrices and measurement bases

Generally, measurement within the VQE framework involves forming mutually commuting partitions of Hamiltonian fragments [17]. An alternative is to obtain a classical description of the quantum state from which relevant operators can be estimated, such as via 'classical shadows' [120, 121] or similar techniques. Here, the 4-body Reduced Density Matrix (4-RDM) of the trial-state is measured:

$$R_{jklm}^{j'k'l'm'} = \langle a_j^\dagger a_k^\dagger a_l^\dagger a_m^\dagger a_{j'} a_{k'} a_{l'} a_{m'} \rangle. \tag{3.3}$$

The RDM is an efficient classical description of the state from which expectation values of lower order excitations can be reconstructed:

$$R_{j_1 \dots j_p}^{k_1 \dots k_p} = \frac{1}{N_e - p} \sum_l R_{j_1 \dots j_p l}^{k_1 \dots k_p l}. \tag{3.4}$$

[1]In this work, we take a 'logical' qubit to be the information that is stored on a physical qubit (sometimes also known as a 'virtual' qubit). This should not be confused with the terminology as used in quantum error correction.

a:

b:

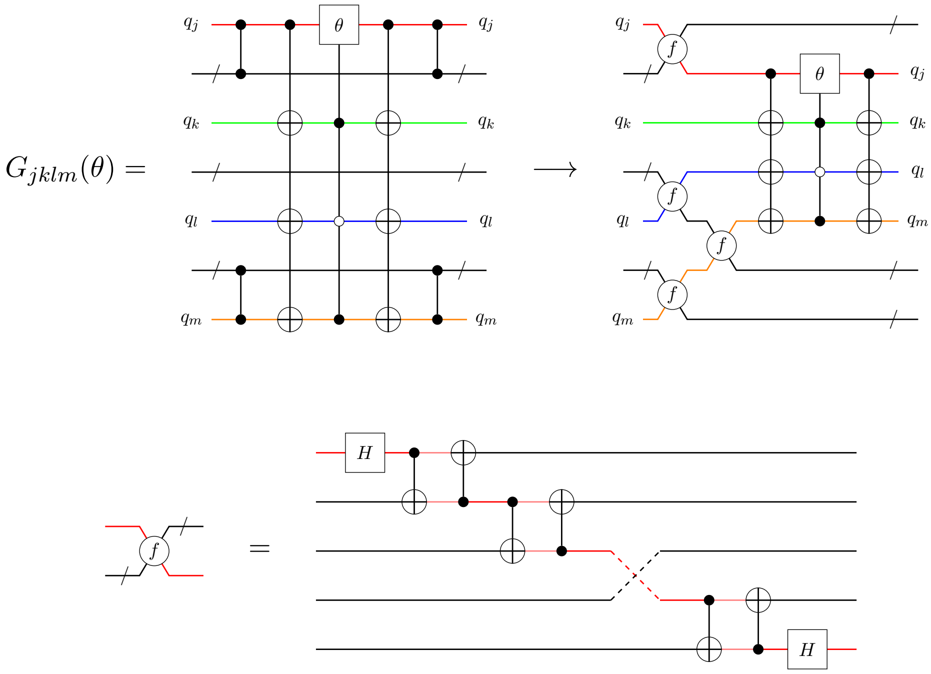


FIGURE 3.4: a: Conversion of the non-local fermionic double-excitation to a local operation using fermionic swap operations, f, on linear connectivity. The θ gate is a Y-rotation by angle θ and when combined with the CNOT gates, the controlled version can be decomposed to respect the linear connectivity as in Figure 3.5. The CZ interactions are between each qubit in the bundle and one of the target qubits. The output of the two circuits is identical up to reordering with fermionic swaps. b: Decomposition of the fermionic swap gate. Note that when many swaps are chained together in this way, the fermionic swap is cheaper to implement than the standard swap in both number of CNOTs and depth.

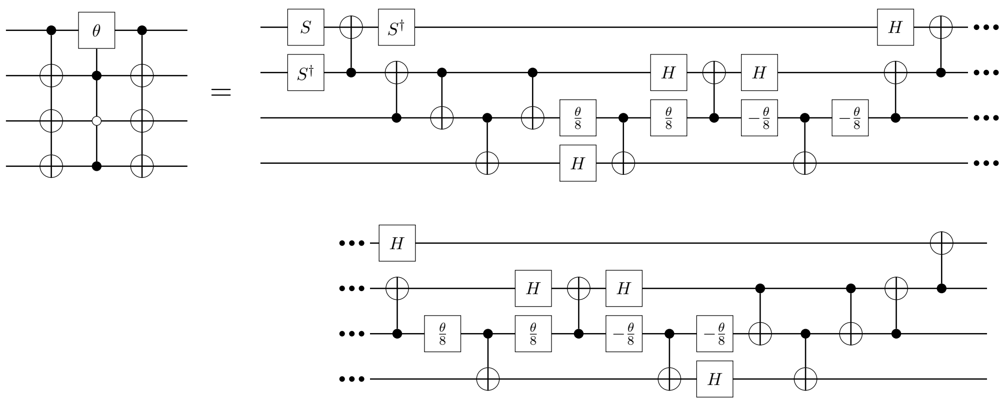


FIGURE 3.5: Decomposition of the local double excitation to linear connectivity. The $\pm\theta/8$ gates are Y rotations by $\pm\theta/8$. This can be improved by prior knowledge of the input state, e.g. correlations between input qubits can be used to remove the long range controls allowing for more efficient decomposition. Including the cost of performing fermionic swaps, the worst-case CNOT count is $4n + 3$ for n qubits/spin-orbitals. If it is required that the logical qubits return to their original physical qubit then the cost is $8n - 13$ CNOT gates.

Therefore, since any second quantised electronic structure Hamiltonian can be written in the form of Equation 1.10, measurement of the 2-RDM allows computation of *any* molecular energy (for a given trial-circuit). Additionally, other properties of interest can be written as sums of low order excitation operators and hence evaluated from the RDM. For example: the dipole moment is a sum over single-excitation operators

$$\mu = \sum_{jk} t_{jk} \langle a_j^\dagger a_k \rangle = \sum_{jk} t_{jk} R_j^k. \tag{3.5}$$

In general, the 4th Hamiltonian moment contains up to 8-body excitations and therefore would require estimation of the 8-RDM. However, due to the fact that there are only 4 electrons present in the trial-state, excitation operators beyond 4th order will have vanishing expectation values.

Below we present a method for grouping and measuring correlated excitation operators for the estimation of reduced density matrices.

Filtering RDM elements

In general the number of p-RDM elements for N_s spin-orbitals is

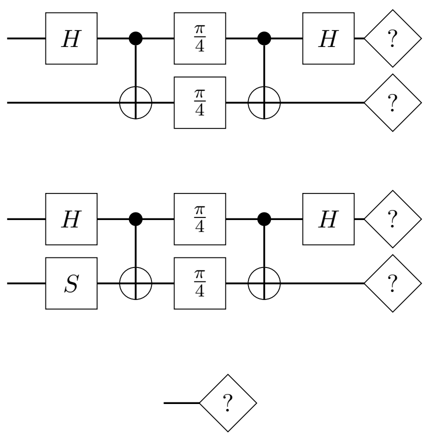
$$N_r = \frac{1}{2}\left(\frac{N_s!}{(N_s - p)!p!}\right)^2 = \mathcal{O}({N_s}^{2p}), \tag{3.6}$$

where it can be shown that

- repeated indices in either the superscript or subscript on the left-hand side of Equation 3.3 lead to the right-hand side vanishing,
- permutations of indices only introduce factors of -1 due to the fermionic anti-commutation relations,
- interchanging upper and lower indices is equivalent to complex conjugation.

For the 4-RDM considered here, this is 2485 operators but can be reduced further by considering spin-symmetry [108]. Since the trial-state is constructed within a specific spin-subspace of the full Hilbert space, any excitation that does not conserve spin must have vanishing expectation value. This restriction reduces the number of operators requiring measurement to 940, which (combined with the grouping method introduced below) is enough reduction to allow computation within a reasonable time.

Operator | Circuit | Bitstring | Eigenvalue



$\mathrm{Re}(a_j^\dagger a_{j+1})$

Bitstring	Eigenvalue
00	0
01	1/2
10	−1/2
11	0

$\mathrm{Im}(a_j^\dagger a_{j+1})$

Bitstring	Eigenvalue
00	0
01	−1/2
10	1/2
11	0

$a_j^\dagger a_j$

Bitstring	Eigenvalue
0	0
1	1

FIGURE 3.6: Measurement circuits for single-excitation operators, $a_j^\dagger a_k$. When $k \notin \{j, j \pm 1\}$, fermionic swap operations can be used to reorder the logical qubits, then the circuits above can be applied. The $\frac{\pi}{4}$ gates are rotations about the y-axis of the Bloch sphere by $\frac{\pi}{4}$.

Measurement of uncorrelated excitations

Before generalising to correlated excitations, we first seek to measure uncorrelated excitations, i.e. operators of the form $a_j^\dagger a_k$. One method to do this involves applying the Jordan-Wigner transformation to convert the operator to a sum of Pauli-strings and measuring each string separately. Unfortunately, this can lead to highly non-local strings which are less likely to qubit-wise commute and are more prone to bit-flip errors in the readout process. Additionally, it is not possible to perform post selection when measuring Pauli-strings as they do not necessarily commute with the total number operator or the spin-projection operator. Instead, consider the case $k = j \pm 1$. In this case, the excitation operator is a 2-local operator and its real and imaginary components are diagonalised (and hence measured) by the gate sequences given in Figure 3.6. For uncorrelated excitations on real-valued spin-orbitals, the imaginary component is guaranteed to have vanishing expectation but will be required when evaluating correlated excitations. In the case that $k \notin \{j, j \pm 1\}$, the fermionic swap operation (Figure 3.4) can be used to reorder the logical qubits so that they are adjacent. Finally, if $j = k$ then the operator is the number operator and its eigenvalue is simply the bit-value when measured in the computational basis.

A disadvantage of this measurement scheme is the necessity of fermionic swap operations which extends the circuit depth by $\mathcal{O}(N_s)$. On the other hand it avoids non-local measurement strings and allows symmetry verification by post-selection on the total electron number (since all circuits in Figure 3.6 commute with the total electron number operator). Additionally, the only single excitations that do not commute with the total spin-projection operator are those for which j and k have different spins. From the arguments made in Section 3.4.3, these operators vanish in expectation and their measurement is therefore unnecessary, allowing post selection based on the total spin-projection.

Measurement of correlated excitations

Measuring correlated excitations directly is more difficult than measuring uncorrelated ones as the circuit depth required for diagonalisation increases with excitation order. Instead, it is possible to decompose a correlated excitation into a sum of products of real and imaginary components of uncorrelated excitations, e.g. assuming unique subscripts:

$$\begin{aligned} a_{j_1}^\dagger a_{j_2}^\dagger a_{k_1} a_{k_2} &= -(a_{j_1}^\dagger a_{k_1})(a_{j_2}^\dagger a_{k_2}) \\ &= -\left(\mathrm{Re}(a_{j_1}^\dagger a_{k_1}) + i\ \mathrm{Im}(a_{j_1}^\dagger a_{k_1})\right) \\ &\quad \times \left(\mathrm{Re}(a_{j_2}^\dagger a_{k_2}) + i\ \mathrm{Im}(a_{j_2}^\dagger a_{k_2})\right) \\ &= -\mathrm{Re}(a_{j_1}^\dagger a_{k_1})\mathrm{Re}(a_{j_2}^\dagger a_{k_2}) \\ &\quad - i\mathrm{Re}(a_{j_1}^\dagger a_{k_1})\mathrm{Im}(a_{j_2}^\dagger a_{k_2}) \\ &\quad - i\mathrm{Im}(a_{j_1}^\dagger a_{k_1})\mathrm{Re}(a_{j_2}^\dagger a_{k_2}) \\ &\quad + \mathrm{Im}(a_{j_1}^\dagger a_{k_1})\mathrm{Im}(a_{j_2}^\dagger a_{k_2}), \end{aligned} \tag{3.7}$$

$$\begin{aligned} \langle a_{j_1}^\dagger a_{j_2}^\dagger a_{k_1} a_{k_2} \rangle &= -\left\langle \mathrm{Re}(a_{j_1}^\dagger a_{k_1})\mathrm{Re}(a_{j_2}^\dagger a_{k_2}) \right\rangle \\ &\quad + \left\langle \mathrm{Im}(a_{j_1}^\dagger a_{k_1})\mathrm{Im}(a_{j_2}^\dagger a_{k_2}) \right\rangle, \end{aligned} \tag{3.8}$$

where the final step discards imaginary components that vanish in expectation for real-valued spin-orbitals. Measurement of each term in the last line requires measuring the real and imaginary components (of different operators) simultaneously and multiplying their eigenvalues before averaging, in the same way that Pauli string expectation values can be computed from individual Pauli measurements. The number of components in the sum grows as 2^p for an order p operator. However, since the maximum order is fixed, the decomposition contributes only a constant factor to the required number of measurements.

Since this method utilises the same circuits as the uncorrelated method it can be made to allow post-selection based on both the total number and total spin-projection operators. To ensure the latter is possible it is necessary to note that there are multiple ways to pair the indices in the first step of the decomposition. By pairing indices with the same spin (if this is not possible then the excitation must have vanishing expectation value by the arguments in Section 3.4.3), the total spin-projection symmetry is preserved and post-selection is possible. The decomposition can be generalised to cases where there are non-unique indices by first factoring out all non-unique indices and measuring these logical qubits in the computational basis.

Grouping of mutually commuting excitations

Since there are 940 non-trivial excitations to measure each of which requires up to 8 different measurement bases, it is desirable to group mutually commuting excitations to be measured simultaneously. This grouping is performed by multiple applications of a greedy algorithm, a brief outline of which is given here, with an example given in Section 3.4.8.

First we define a representation of a measurement basis as a list of qubit interactions. Each qubit can interact with one other qubit (or no other qubit for a number measurement $a_j^\dagger a_j$). For example, a measurement basis might be written as: $(0, 0), (1, 2)$, indicating a number measurment on qubit 0 and a joint measurement on qubits 1 and 2. A basis measures an excitation if each creation index in the excitation is interacted with an annihilation index. Note that at this stage no distinction is made between the two different interactions given in Figure 3.6.

The first stage of the algorithm partitions excitations into commuting (though not yet measurable) bases. The second stage partitions each basis further to allow measurement and can be mapped to the more well-known tensor-product-basis grouping problem.

Each partitioning is performed by iterating over the operators to be measured, selecting the 'best' basis and updating it. If no compatible bases are found a new basis is appended to the set. Using this method the 940 excitations (with multiple components each) are partitioned into 200-204 measurement bases. Due to the greedy algorithm, the final number of bases is dependent on the order in which the excitations are presented to the algorithm and since the different trial-states result in different arrangements of spin-orbitals the ordering of excitations and hence number of bases is different between the two states.

Routing of logical qubits

An additional difficulty that has not been considered in the previous section is that of routing each pair of qubits together with as little circuit depth increase as possible. It is possible to show that each qubit can be interacted with each other qubit in linear depth [96], while the problem here is for each qubit to be interacted with a *specific* partner.

Therefore the additional circuit depth should grow at most linearly with the number of qubits but naively routing qubit pairs sequentially often generates prohibitively deep circuit extensions. Additionally it is not hard to design an example for which the brick-work circuit [96] also performs poorly compared to solutions generated 'by inspection'. To solve this systematically we convert the problem to an integer linear program [122] and solve using Gurobi [123]. The binary variables we optimise are $x^{(c)}_{jk,t}$ and $y^{(c)}_{jk,t}$ where the x-variables represent a logical qubit from pair c moving from physical qubit j to an adjacent (or the same) qubit k at time t and the y variables represent a logical qubit from pair c on physical qubit j interacting with its partner on adjacent qubit k at time t. Two constraints on the y variables immediately become apparent

- $y^{(c)}_{jk,t} = y^{(c)}_{kj,t}$, i.e. interactions must be between two logical qubits from the same pair, in the same timestep
- $y^{(c)}_{jj,t} = 0$, i.e. two interacting logical qubits cannot be on the same physical qubit

In addition we use the following constraints:

1. Enforce starting positions of logical qubits ($S^{(c)}_j$ is 1 if a logical qubit from pair c starts on qubit j and 0 otherwise),
$$\sum_k \left[x^{(c)}_{jk,0} + y^{(c)}_{jk,0}\right] = S^{(c)}_j, \tag{3.9}$$
2. The number of logical qubits entering (or remaining on) a physical qubit at a given timestep must equal the number of logical qubits leaving (or remaining on) that physical qubit in the next timestep,
$$\sum_k \left[x^{(c)}_{jk,t} - x^{(c)}_{kj,t+1} + y^{(c)}_{jk,t} - y^{(c)}_{kj,t+1}\right] = 0, \tag{3.10}$$
3. The number of logical qubits entering a physical qubit at a given timestep must not be greater than 1,
$$\sum_{kc} \left[x^{(c)}_{jk,t} + y^{(c)}_{jk,t}\right] \leq 1, \tag{3.11}$$
4. If a logical qubit moves from physical qubit j to k at timestep t, then any logical qubit on k at t must move to j (i.e. enforce swapping behaviour),
$$\sum_c \left[x^{(c)}_{jk,t} + \sum_{j'\neq j} x^{(c)}_{kj',t}\right] \leq 1, \tag{3.12}$$

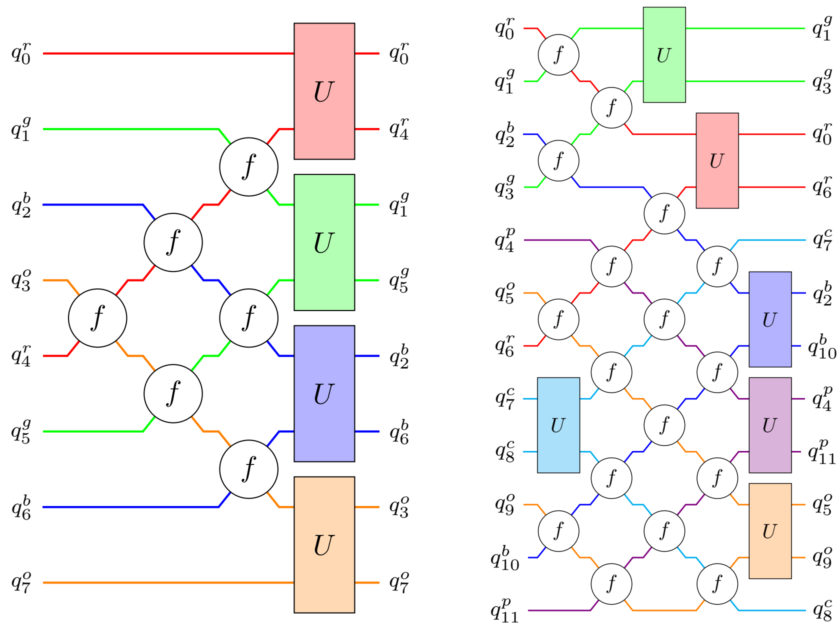


FIGURE 3.7: Example circuits for logical-qubit routing from the integer linear program for 8 (left) and 12 (right) qubits. The problem is to interact each qubit with its partner (matching colour/superscript) in as few timesteps as possible, with a secondary objective to minimise the number of swaps. The f gates are fermionic swaps while the U gates are the interactions given in Figure 3.6 for converting the joint measurement into single qubit measurements.

5. Interactions must occur exactly once for each pair,

$$\sum_{jkt} \left[y_{jk,t}^{(c)} \right] = 2. \tag{3.13}$$

Note that the conditions on y can be used to analytically simplify the implementation of the conditions above but have not been substituted here to maintain readability. Example outputs from the qubit routing optimisation are shown in Figure 3.7.

3.4.4 Energy minimisation

Energy minimisation is performed using a customised SPSA optimiser on noiseless, classical simulation. Minimisation was performed 5 times using different seeds for the stochastic optimiser and the optimised parameters from the best run were used for the hardware demonstration, though the difference between runs was minimal. The optimisation generally converged after 50-60 iterations.

3.4.5 IBM quantum backends

The original quantum computations in this work were performed on *ibmq_kolkata*, one of IBM's superconducting quantum devices [104], using 8 of the 27 available qubits. Later computations were performed on *ibm_torino*, a newer device with improved error rates and a different set of base gates. On *ibmq_kolkata* (*ibm_torino*), each measurement basis was measured 10^5 times (2.5×10^4 times); for 200 measurement bases at two different values of $\vec{\theta}$ (trial- and reference-states) and including the readout-noise calibration, this leads to approximately 42 million shots (10 million shots) and a total quantum computation time of 1.5 hours (50 minutes) per trial-state.

3.4.6 Statistical bootstrapping

Standard deviations are estimated using a bootstrapping method [124]. By resampling (classically) 500×10^5 ($500 \times 2.5 \times 10^4$ for *ibm_torino*) times from the probability distributions obtained from the quantum hardware, a distribution of energy estimates is obtained from which error estimates can be extracted.

3.4.7 Noise mitigation

In the near-term, fault tolerant quantum computation (i.e. error correction) is not possible due to hardware constraints. Instead, various schemes have been proposed to mitigate hardware noise, usually at the cost of running additional circuits. The methods used in this work are detailed below.

Quantum readout-noise mitigation

One potential source of errors occurs during the qubit-readout process, where the qubit state may be misclassified. Quantum Readout-Error Mitigation (QREM, [125]) aims to address these readout errors by calibrating the readout procedure on known computational basis states. By creating and measuring a selection of computational basis states, the assignment matrix of each qubit can be formed. This matrix represents the transformation from the ideal output probability vector to the noisy vector. By inverting this matrix the transformation from noisy to ideal vector can be obtained. In the implementation used here [126], readout error on qubit j is averaged over (a subset of) the states of the other qubits but is not explicitly treated as being correlated noise. Therefore the assignment matrices are 2×2 matrices and are efficiently invertible.

In practice QREM may lead to non-physical quasi-probabilities. To enforce physicality the 'negative probability' can be redistributed evenly between measurement outcomes with positive probability [127] (see also [128]).

Symmetry verification

Due to the physical motivation for the UCC-based trial-state, the electron number and total electronic spin are fixed for any value of the trial-state parameters. Additionally, since we are able to measure both of these quantities simultaneously with the RDM elements through the fermionic measurement procedure described in Section 3.4.3, they can be used for symmetry verification [129]. i.e. each time a bitstring is measured, if the corresponding electron number or total electronic spin is not the value expected from the trial-state design, then we can say with certainty that an error has occurred and discard the result. In this way only errors that commute with both symmetries are not detected and are able to influence the resulting energy.

Such a post-selection strategy necessarily leads to a reduction in the effective number of counts. In the demonstrations performed here, only 56% and 22% of measurement shots performed on *ibmq_kolkata* are accepted for the 4- and 6-excitation trial-circuits respectively, while the numbers for *ibm_torino* are 63% and 46%. Note that generating bitstrings at random would have a 14% acceptance rate.

Reduced density matrix rescaling

The reduced density matrices as given by Equation 3.3 are only well-defined for states with a fixed number of electrons, otherwise the use of Equation 3.4 can lead to rapid accumulation of error. Unfortunately, and somewhat counter-intuitively, symmetry verification alone is not sufficient to enforce the correct electron number for the RDM. This is because symmetry verification enforces the correct correlations only between elements of the same measurement basis, so properties that are reconstructed from measurements in multiple bases can violate symmetry constraints. To resolve this, it can be noted that the electron number can be directly related to the trace of the ideal p-RDM by

$$\begin{aligned} \mathrm{Tr}({}_{\mathrm{ideal}}R) &\equiv \sum_{j_1<j_2\cdots<j_p} {}_{\mathrm{ideal}}R^{j_1 j_2 \cdots j_p}_{j_1 j_2 \cdots j_p} \\ &= \frac{N_e!}{p!(N_e-p)!}, \end{aligned} \tag{3.14}$$

and to enforce the correct trace (and therefore electron number) it is possible to simply rescale the RDM by the ratio of the ideal trace to the measured trace [108], i.e.

$$_{\text{corrected}}R = \frac{N_e!}{p!(N_e-p)!}\frac{1}{\mathrm{Tr}(_{\text{noisy}}R)} \cdot {}_{\text{noisy}}R. \tag{3.15}$$

We note that the correct trace is a necessary, but not sufficient, condition for the p-RDM to represent a valid physical state. In general, the RDM needs to satisfy the N-representability conditions [130–134].

Using $\mathbf{j}$ to represent a list of indices, $j_1 j_2 \ldots$, for brevity, then a (still insufficient) list of necessary conditions for validity of the p-RDM is [133]:

1. Hermiticity:

$$R_{\mathbf{j}}^{\mathbf{k}} = (R_{\mathbf{k}}^{\mathbf{j}})^*, \tag{3.16}$$

2. Antisymmetry:

$$\begin{aligned} R_{P(\mathbf{j})}^{\mathbf{k}} &= \sigma_P R_{\mathbf{k}}^{\mathbf{j}}, \\ R_{\mathbf{j}}^{P(\mathbf{k})} &= \sigma_P R_{\mathbf{k}}^{\mathbf{j}}, \end{aligned} \tag{3.17}$$

where $P(\mathbf{j})$ is a permutation of the indices contained in $\mathbf{j}$ and $\sigma_P = \pm 1$ is the sign of the permutation.

3. Contraction:

$$R_{j_1 \ldots j_p}^{k_1 \ldots k_p} = \frac{1}{N_e - p}\sum_l R_{j_1 \ldots j_p l}^{k_1 \ldots k_p l}, \tag{3.18}$$

4. Trace:

$$\mathrm{Tr}(R) = \frac{N_e!}{p!(N_e-p)!}, \tag{3.19}$$

5. Positive-semidefiniteness:

$$R \succeq 0. \tag{3.20}$$

Further constraints can be derived from (for example) the requirement that the trial-state is an eigenstate of the spin-projection operator [131, 133]. Note that in this work, the Hermiticity and antisymmetry conditions are used to reduce the number of required measurements in Section 3.4.3, the contraction condition is exactly Equation 3.4 which is used to extract lower-order expectation values from high order RDMs, and the trace condition is enforced by the rescaling technique. In future work, additional constraints could potentially be implemented to further detect/mitigate errors as has been proposed [133] and performed [135] for 2-RDMs.

An interesting feature of measuring RDMs is that even with this rescaling technique, if no other error mitigation is applied, the energy estimates it produces have lower accuracy than estimating the energy directly from counts. On the other hand, we see that the variance of the energy estimates is significantly reduced by the RDM method. This is advantageous because many error mitigation techniques improve accuracy at the cost of an increased variance. In combination (in particular, with the noise calibration described below), the RDM method results in both more accurate and more precise (lower variance) estimates than calculating energy estimates directly from counts.

Reference-state noise calibration

Another powerful error mitigation technique is based on the existence of reference circuits [136, 137], circuits with similar overall structure to the trial-state but with a limited number of non-Clifford gates. The simplest reference-state for the circuit used here is the Hartree-Fock state, which can be prepared by setting all the variational parameters to 0. Since properties of the Hartree-Fock state can be efficiently evaluated classically, comparison between classical and quantum computations can be used to 'learn' the noise model of the device. Here, a global white noise model is assumed,

$$\begin{aligned} \rho \rightarrow \rho^{\text{noisy}} &= (1-q)\rho + qJ, \\ \Rightarrow \mathrm{Tr}(\mathcal{H}\rho^{\text{noisy}}) &= (1-q)\mathrm{Tr}(\mathcal{H}\rho) + q\mathrm{Tr}(\mathcal{H}J), \end{aligned} \tag{3.21}$$

where q represents the effective global error rate and J is related to the maximally mixed state (concretely it is the image of the maximally mixed state after application of other error mitigation techniques e.g. post-selection will first zero-out some entries of the mixed-state, $I/2^{N_s}$, then renormalise).

This model has been seen to be reasonable for certain classes of circuits [138] and circuits could be made to better obey the model through randomised compiling techniques [112]. Using the classical (noiseless) and quantum (noisy) expectation values from the Hartree-Fock state, an estimate, $\hat{q}$, for the error rate can be obtained by solving Equation 3.21. For the 2-excitation trial-state circuits, the calculated noise level on *ibmq_kolkata* (*ibm_torino*) ranges between 3-40% (5-20%), depending on the application of other noise-mitigation techniques. Following this, quantum computation of properties

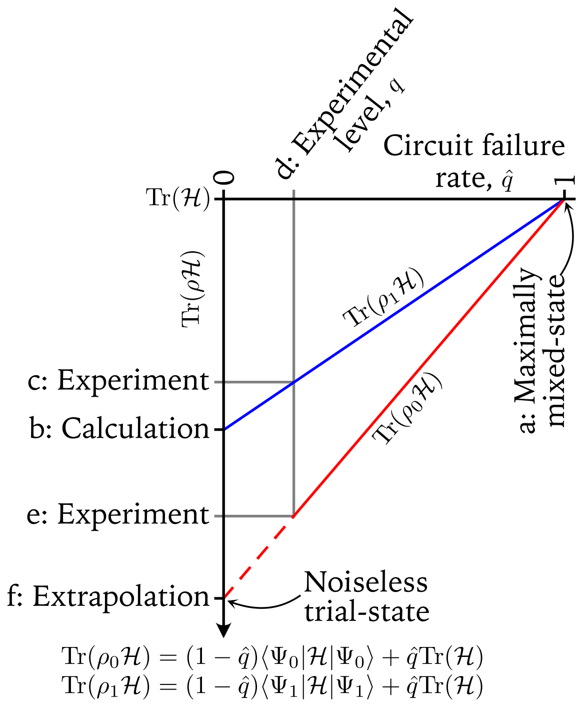


FIGURE 3.8: The reference-state calibration method can be visualised on a plot of expectation value vs. circuit failure rate. The maximally noisy (a) and noiseless (b) energies of the reference-state are classically tractable, so using quantum computation of the reference-energy (c) allows estimation of the effective noise level in the device (d). By evaluating the energy of the trial-state on a quantum computer (e) the noiseless energy of the trial-state (f) can be estimated by extrapolation. The linear relationship is due to the white-noise model, with more sophisticated noise models requiring more reference-state calculations to fit [137]. The reference-state error mitigation scheme differs from the usual zero-noise extrapolation method [139, 140] since the additional information is obtained through reference-states instead of artificially increasing the noise.

of the trial-state can be corrected using the noise estimate, $\hat{q}$. A graphical representation of this process is shown in Figure 3.8.

3.4.8 Basis grouping example

Fermionic measurement grouping (first level)

Consider measuring the spin-conserving excitations for the 2-RDM on 4 spin-orbitals:

$$\begin{array}{ccccc}
a_0^\dagger a_1^\dagger a_0 a_1 & & & & \\
 & a_0^\dagger a_2^\dagger a_0 a_2 & a_0^\dagger a_2^\dagger a_1 a_2 & a_0^\dagger a_2^\dagger a_0 a_3 & a_0^\dagger a_2^\dagger a_1 a_3 \\
 & & a_1^\dagger a_2^\dagger a_1 a_2 & a_1^\dagger a_2^\dagger a_0 a_3 & a_1^\dagger a_2^\dagger a_1 a_3 \\
 & & & a_0^\dagger a_3^\dagger a_0 a_3 & a_0^\dagger a_3^\dagger a_1 a_3 \\
 & & & & a_1^\dagger a_3^\dagger a_1 a_3 \\
 & & & & & a_2^\dagger a_3^\dagger a_2 a_3
\end{array}$$

where spin-orbitals 0 and 1 have spin-up and spin-orbitals 2 and 3 have spin-down. We begin with an empty basis set, $\{\}$, and for each excitation above, search for a commuting basis. If a basis is found we update it, if no basis is found we add a new one.

1. $a_0^\dagger a_1^\dagger a_0 a_1$
   - $\{\} \to \{(0,1)\}$,
   - No commuting basis. New basis $(0,1)$.
2. $a_0^\dagger a_2^\dagger a_0 a_2$
   - $\{(0,1)\} \to \left\{\begin{matrix} (0,1) \\ (0,0),(2,2) \end{matrix}\right\}$,
   - No commuting basis. New basis $(0,0),(2,2)$. Note that spin-orbitals 0 and 2 have opposite spin, so interacting them is not allowed (fails to commute with the spin-projection operator).
3. $a_0^\dagger a_2^\dagger a_1 a_2$
   - $\left\{\begin{matrix} (0,1) \\ (0,0),(2,2) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \end{matrix}\right\}$,
   - Commuting basis, $(0,1)$ found and updated to $(0,1),(2,2)$.
4. $a_0^\dagger a_2^\dagger a_0 a_3$
   - $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \\ (0,0),(2,3) \end{matrix}\right\}$,
   - No commuting basis. New basis $(0,0),(2,3)$.
5. $a_0^\dagger a_2^\dagger a_1 a_3$
   - $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \\ (0,0),(2,3) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \\ (0,0),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
   - No commuting basis. New basis $(0,1),(2,3)$.
6. $a_1^\dagger a_2^\dagger a_1 a_2$
   - $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(2,2) \\ (0,0),(2,3) \\ (0,1),(2,3) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2) \\ (0,0),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,

- Commuting basis, $(0,0),(2,2)$ found and updated to $(0,0),(1,1),(2,2)$.

7. $a_1^\dagger a_2^\dagger a_0 a_3$

- $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2) \\ (0,0),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
- Measuring basis $(0,1),(2,3)$ found.

8. $a_1^\dagger a_2^\dagger a_1 a_3$

- $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2) \\ (0,0),(2,3) \\ (0,1),(2,3) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
- Commuting basis $(0,0),(2,3)$ found and updated to $(0,0),(1,1),(2,3)$.

9. $a_0^\dagger a_3^\dagger a_0 a_3$

- $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2),(3,3) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
- Commuting basis $(0,0),(1,1),(2,2)$ found and updated to $(0,0),(1,1),(2,2),(3,3)$.

10. $a_0^\dagger a_3^\dagger a_1 a_3$

- $\left\{\begin{matrix} (0,1),(2,2) \\ (0,0),(1,1),(2,2),(3,3) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\} \to \left\{\begin{matrix} (0,1),(2,2),(3,3) \\ (0,0),(1,1),(2,2),(3,3) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
- Commuting basis $(0,1),(2,2)$ found and updated to $(0,1),(2,2),(3,3)$.

11. $a_1^\dagger a_3^\dagger a_1 a_3$

- $\left\{\begin{matrix} (0,1),(2,2),(3,3) \\ (0,0),(1,1),(2,2),(3,3) \\ (0,0),(1,1),(2,3) \\ (0,1),(2,3) \end{matrix}\right\}$,
- Measuring basis $(0,0),(1,1),(2,2),(3,3)$ found.

12. $a_2^\dagger a_3^\dagger a_2 a_3$

- $$\left\{\begin{matrix}(0,1),(2,2),(3,3)\\(0,0),(1,1),(2,2),(3,3)\\(0,0),(1,1),(2,3)\\(0,1),(2,3)\end{matrix}\right\},$$
- Measuring basis $(0,1),(2,2),(3,3)$ found. Note that $(0,0),(1,1),(2,2),(3,3)$ also measures this excitation. At this point the choice between the two is arbitrary.

Mapping to Pauli measurement grouping (second level)

Based on the results for the previous example, we now need to assign the interaction types (i.e. which circuit from Figure 3.6) to use for each interaction in each basis in the set. The assignment of interactions in a given basis are independent of the other bases and so we can take the bases in any order.

1. $(0,0),(1,1),(2,2),(3,3)$
   - Measures $a_1^\dagger a_3^\dagger a_1 a_3$, $a_0^\dagger a_3^\dagger a_0 a_3$, $a_1^\dagger a_2^\dagger a_1 a_2$ and $a_0^\dagger a_2^\dagger a_0 a_2$.
   - Since each qubit is "interacted" with itself, the only choice is to measure in the computational basis. This results in measuring the number operators $a_j^\dagger a_j$ (for $j \in [0,3]$) from which the products can be reconstructed. In this case no further partitioning is required.
2. $(0,1),(2,2),(3,3)$
   - Measures $a_2^\dagger a_3^\dagger a_2 a_3$, $a_0^\dagger a_3^\dagger a_1 a_3$, $a_0^\dagger a_2^\dagger a_1 a_2$ and $a_0^\dagger a_1^\dagger a_0 a_1$.
   - Only one of the interactions is between different qubits, therefore this can be mapped to a (trivial) 1 site tensor product basis grouping problem. We will use 'site' for these virtual qubits and 'qubits' for the actual qubits.
     - The interaction $(0,1)$ is mapped to the first (and only) site. We need to decide what basis to measure it in.
     - The excitation operators are mapped to the Pauli operators to be grouped.
       * $a_2^\dagger a_3^\dagger a_2 a_3$ does not act on the site, so maps to I.
       * $a_0^\dagger a_3^\dagger a_1 a_3$ acts on the site as $a_0^\dagger a_1 \to \mathrm{Re}(a_0^\dagger a_1) \to X$.
       * $a_0^\dagger a_2^\dagger a_1 a_2$ also maps to X.
       * $a_0^\dagger a_1^\dagger a_0 a_1$ is a special case and can be extracted from either of the interactions in Figure 3.6 (since they are both number conserving by design) so we can also map this to I.

  - As stated above, the grouping problem in this case is trivial, since there is only 1 site and only one non-identity operator (X).

3. $(0,0),(1,1),(2,3)$
   - Measures $a_1^\dagger a_2^\dagger a_1 a_3$, $a_0^\dagger a_2^\dagger a_0 a_3$
   - Similar to above, this maps to the problem of 1 site and one operator (X).

4. $(0,1),(2,3)$
   - Measures $a_1^\dagger a_2^\dagger a_0 a_3$ and $a_0^\dagger a_2^\dagger a_1 a_3$
   - In this case we map to a 2 site tensor product basis problem
     - $(0,1)$ and $(2,3)$ become the first and second sites respectively
     - The excitation operators map as
       * $a_1^\dagger a_2^\dagger a_0 a_3 \to \mathrm{Re}(a_1^\dagger a_0)\mathrm{Re}(a_2^\dagger a_3) + \mathrm{Im}(a_1^\dagger a_0)\mathrm{Im}(a_2^\dagger a_3) \to XX + YY$
       * $a_0^\dagger a_2^\dagger a_1 a_3 \to \mathrm{Re}(a_0^\dagger a_1)\mathrm{Re}(a_2^\dagger a_3) + \mathrm{Im}(a_0^\dagger a_1)\mathrm{Im}(a_2^\dagger a_3) \to XX + YY$
     - This then leads to the problem of finding a tensor product grouping for XX and YY (which trivially requires 2 bases)

In non-trivial cases i.e. for higher number of spin-orbitals, the Pauli measurement grouping problem is solved using a greedy algorithm analogous to the Fermionic grouping algorithm of the previous section.

Chapter 4

Moments-corrected quantum computation of the electric dipole moment of molecular systems

4.1 Introduction

In this chapter, which is based on the paper "Moments-corrected quantum computation of the electric dipole moment of molecular systems" submitted to the *Journal of Chemical Theory and Computation*, we aim to estimate non-energetic ground-state properties of molecular systems.

In the previous chapters, the problem we aimed to solve with the moments-based correction was the problem of ground-state energy estimation. While most experimental demonstrations of quantum chemical calculations on quantum devices have focused on the ground-state energy of the system, attention has recently shifted to other properties of the ground-state, in part due to the difficulty in achieving precision requirements for ground-state-energy estimation, but also due to the fact that they are properties that can be directly measured in a chemical experiment (while ground-state energies are not, since only energy differences can be measured). This provides the opportunity for a more direct comparison between potential quantum computations and experimental data. As such, progress has been made on estimation of excited states and band-structure calculations [44, 45, 107, 141–150], vibrational modes [151] and dipole moments [152, 153] on quantum hardware. Dipole moments and similar properties that do not depend on excited electronic states or motion of the atomic nuclei seem a straightforward extension to the ground-state energy estimation problem. To date, however, quantum hardware computation of these properties has been restricted to direct expectation value determination, and has not yet taken advantage of techniques used to improve energy estimation such as variations of the quantum phase estimation algorithm [52], imaginary time evolution [87, 154], Hamiltonian moments [21, 22, 48, 109, 155], subspace expansion [107, 143, 156], and selected configuration interaction [41–46, 157]. Methods designed specifically for the evaluation of such properties [158–160] have also yet to be demonstrated on quantum hardware, though techniques for the estimation of molecular response properties [159, 161], i.e. the response of properties such as the dipole moment to external fields due to excitation to higher energy states, have been implemented [162, 163]. However, our interest is in evaluating non-energetic properties of the ground-state in the absence of external potentials and in improving the accuracy of results when access to the exact ground-state is infeasible.

In this chapter, we show improvement over the direct estimation of ground-state observables, using the electric dipole moment of the water molecule as an example. Through use of the Hamiltonian moments and the fourth order infimum correction [85, 111] we obtain noise mitigated results that agree with FCI calculations to within 0.03 ± 0.007 debye ($2\% \pm 0.5\%$), compared to direct expectation value determination with errors on the order of 0.07 debye (5%), even in the absence of noise. The noise-robust, moments-based estimate utilises a Hellmann-Feynman approach [101, 164–166] as summarised in Figure 4.1. The method was previously applied to Heisenberg models in simulation [166], here we extend this by demonstrating implementation of the method using results from an IBM superconducting quantum device [104], providing experimental verification that the moment-based QCM method produces superior estimates for such quantities compared to direct evaluation when applied to chemical systems.

4.2 Quantum computed moments for arbitrary ground-state observable estimation

To estimate ground-state observables via the Hellmann-Feynman theorem in the context of moments-based methods [164–166], consider the modified Hamiltonian,

$$\mathcal{H}_\lambda \equiv \mathcal{H} + \lambda\mu, \tag{4.1}$$

where λ is a real scalar and μ is the observable of interest. For concreteness, let $\mu = \mu_x$ be the x-component of the electric dipole moment operator (the only component with

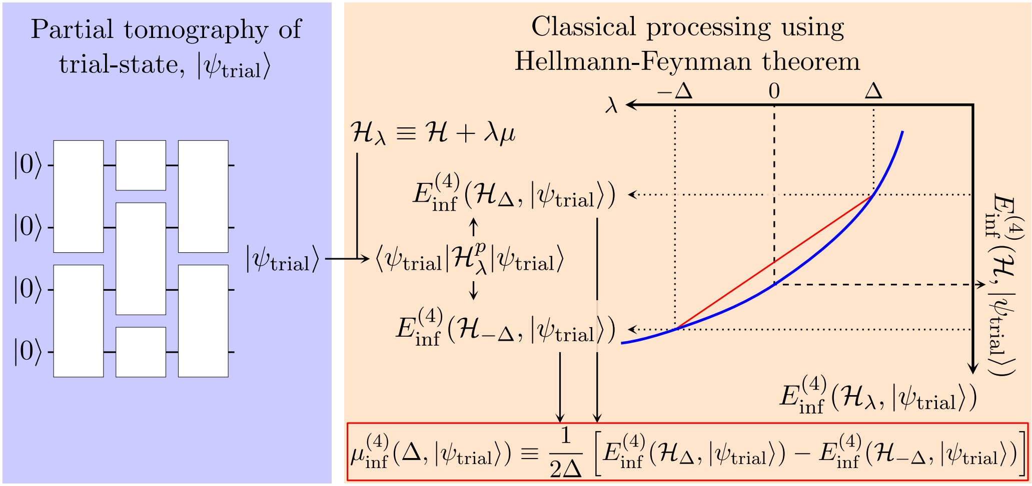


FIGURE 4.1: Method overview for estimating the ground-state observable, μ, using the quantum computed moments method. The first step is to perform partial tomography of the trial-state $|\psi_{\text{trial}}\rangle$, to obtain a classical description (reduced density matrix) of the quantum system. This step is independent of the choice of μ and λ. Next, the observable to be measured, μ, is selected and the modified Hamiltonian $\mathcal{H}_\lambda \equiv \mathcal{H}+\lambda\mu$ is constructed, λ may be a small constant Δ (Figure 4.3A) or may be symbolic (Figure 4.3B or C). The expectation value of μ with respect to the ground-state (not the trial-state) is the derivative of the ground-state energy of the modified Hamiltonian with respect to λ evaluated at $\lambda = 0$. To approximate this, we compute moments-corrected ground-state energy estimates at $\lambda = \pm\Delta$ and apply the finite difference method as indicated by the plot on the right and the formula in the red box.

non-vanishing expectation value by symmetry, Figure 4.2b),

$$\mu = \sum_{jk}^{N_s} f_{jk} a_j^\dagger a_k, \tag{4.2}$$

where f_{jk} are classically computed integrals. Let $|\Phi_\lambda\rangle$ be the ground-state of $\mathcal{H}_\lambda$ with energy E_λ, then the Hellmann-Feynman theorem states:

$$\frac{\partial E_\lambda}{\partial \lambda} = \langle\Phi_\lambda|\mu|\Phi_\lambda\rangle, \tag{4.3}$$

and evaluating at $\lambda = 0$ gives expectation values with respect to the ground-state of the unmodified Hamiltonian

$$\left.\frac{\partial E_\lambda}{\partial \lambda}\right|_{\lambda=0} = \langle\Phi_0|\mu|\Phi_0\rangle \equiv \mu_0. \tag{4.4}$$

While the Hellmann-Feynman theorem also holds for states other than energy eigenstates, subject to conditions on the energy derivative with respect to trial-state parameters, here the formulation in terms of the ground-state is sufficient, as we aim to estimate

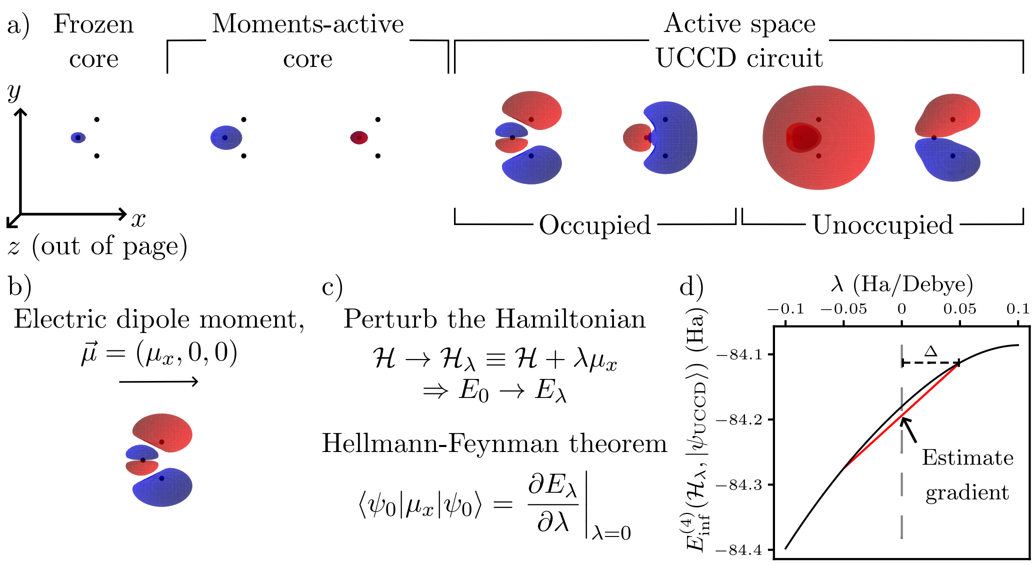


FIGURE 4.2: Details of the application of the method to the electric dipole moment of the water molecule. a) The STO-3G molecular orbitals used to represent the water molecule. b) Due to symmetry considerations, the electric dipole moment must be aligned with the x-axis. We use the convention that the dipole moment vector points from negative to positive charge. c) By perturbing the Hamiltonian with the dipole operator, the Hellmann-Feynman theorem can be applied to compute the dipole moment. d) As in Figure 4.1, E_λ is approximated by $E_{\mathrm{inf}}^{(4)}(\mathcal{H}_\lambda, |\psi_{\mathrm{UCCD}}\rangle)$ and the finite difference method is employed to estimate the derivative. Note that the trial-state $|\psi_{\mathrm{trial}}\rangle = |\psi_{\mathrm{UCCD}}\rangle$ is independent of λ, allowing for reuse of the same quantum measurements and thereby reducing the effects of stochastic error. Δ is the step-size used for the finite difference method.

non-energetic ground-state properties (right-hand side of Equation 4.4) from approximations to the ground-state energy (left-hand side of Equation 4.4). The applicability of various forms of the Hellmann-Feynmann theorem is a much studied topic [167–169].

To approximate the derivative, we can employ the finite difference method (Figures 4.1 and 4.2d),

$$\mu_0 \approx \frac{1}{2\Delta}\left(E_\Delta - E_{-\Delta}\right). \tag{4.5}$$

Further, using the moments-based correction to improve the evaluation of $E_{\pm\Delta}$ and therefore the estimation of the dipole moment relative to the standard variational estimate gives

$$\mu_0 \approx \frac{1}{2\Delta}\left[E_{\mathrm{inf}}^{(4)}(\lambda = \Delta) - E_{\mathrm{inf}}^{(4)}(\lambda = -\Delta)\right]. \tag{4.6}$$

Note that the notation used here is that λ is a variable (with respect to which we would like to take the derivative) while $\pm\Delta$ is the value assigned to λ in order to compute the finite difference approximation to the derivative.

When performing finite difference computations on a quantum device, often the accuracy of the difference formula is severely limited by stochastic noise. However, in this case the parameter with respect to which the derivative is taken, λ, appears in the post-processing rather than in the quantum computation which allows reuse of the quantum output for both $\lambda = \pm\Delta$ values and vastly reduces the accumulation of stochastic noise. In the absence of the moments-based correction, reusing the trial-state this way would essentially make the approximation that the ground-state is independent of the change in the Hamiltonian, leading to systematic error due to neglecting the wavefunction response terms. If the response terms are ignored completely and expectation values are taken at $\lambda = \pm\Delta$, then the estimate of the dipole moment is simply the expectation value of the dipole operator with respect to the trial-state,

$$\mu_0 \approx \frac{\langle\psi_{\text{trial}}|\mathcal{H}_\Delta|\psi_{\text{trial}}\rangle - \langle\psi_{\text{trial}}|\mathcal{H}_{-\Delta}|\psi_{\text{trial}}\rangle}{2\Delta} = \langle\psi_{\text{trial}}|\mu|\psi_{\text{trial}}\rangle. \tag{4.7}$$

While the equality in Equation 4.7 holds exactly (since the modified Hamiltonian is linear in λ, there is no error due to the finite difference method), it is not practically useful as the cost of evaluating the left- and right-hand sides are roughly equal. Additionally, the dipole-moment estimate obtained is the expectation value with respect to the trial-state, not the ground-state, so systematically approaching the ground-state expectation value requires improving the trial-state leading to additional quantum gates and therefore increased noise.

The Hellmann-Feynman theorem may be of practical use in cases where we have access to approximate ground-state energies, but not to the corresponding state. This is the case for a number of ground-state energy estimation techniques such as the moments-based correction [164].

In this case, the estimates are able to incorporate additional information from the Hamiltonian to circumvent the limit imposed by the trial-state, effectively incorporating some of the wavefunction response and improving on the dipole moment estimate. We denote the dipole moment estimate computed this way using the moments-based correction, $E^{(4)}_{\text{inf}}$, as $\mu^{(4)}_{\text{inf}}(\Delta, |\psi\rangle)$, so the final expression used for the dipole-moment estimation in this work is

$$\mu_0 \approx \mu^{(4)}_{\text{inf}}(\Delta, |\psi_{\text{trial}}\rangle) \equiv \frac{1}{2\Delta}\left[E^{(4)}_{\text{inf}}(\Delta, |\psi_{\text{trial}}\rangle) - E^{(4)}_{\text{inf}}(-\Delta, |\psi_{\text{trial}}\rangle)\right]. \tag{4.8}$$

To minimise the required classical computation, we express $\mathcal{H}^p_\lambda$ analytically as a polynomial in λ and compute the operator-valued coefficients (Methods B and C in Figure 4.3).

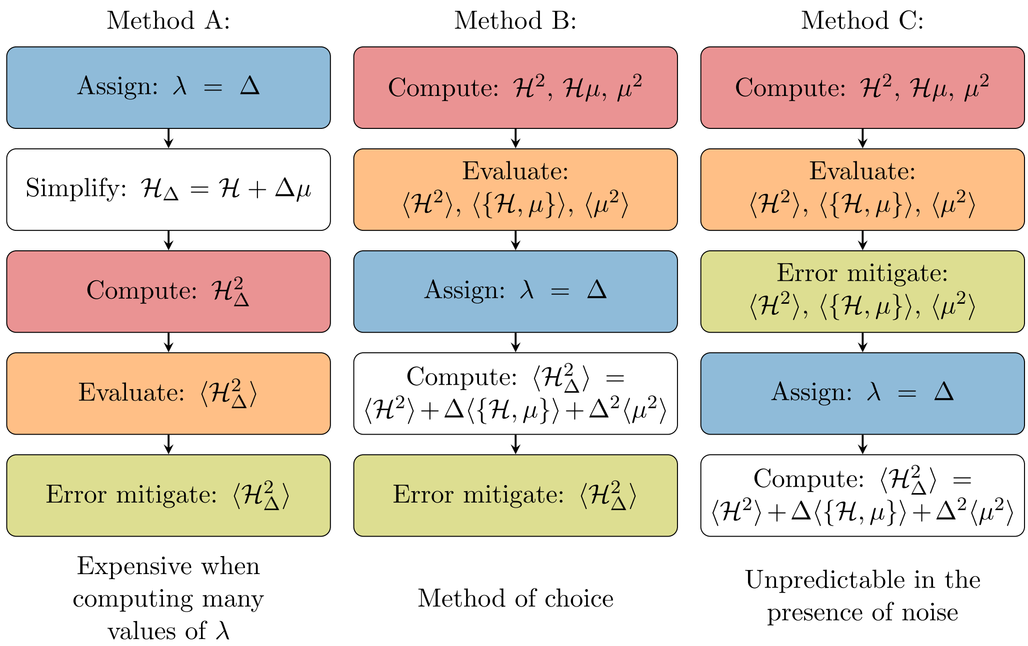


FIGURE 4.3: Three methods for evaluating the second moment, $\langle \mathcal{H}_\Delta^2 \rangle$. The computational bottleneck is the implementation of Wick's theorem to multiply the ferminionic operators (red boxes). Performing these steps in a way that is independent of Δ (methods B and C) allows for much more efficient evaluation of $\langle \mathcal{H}_\Delta^2 \rangle$ at many values of λ, since steps prior to assigning a value to λ (above the blue boxes) do not need to be repeated. However, for certain values of λ, method C was seen to be unstable to errors from the quantum computation. Note that $\langle \{\mathcal{H}, \mu\} \rangle \equiv \langle \mathcal{H}\mu + \mu\mathcal{H} \rangle = 2\mathrm{Re}\langle \mathcal{H}\mu \rangle$. The methods may be generalised to evaluate $\langle \mathcal{H}_\Delta^p \rangle$.

This allows evaluation of the moments-corrected ground-state energy estimate for many values of λ with minimal additional processing. However, as indicated in Figure 4.3, we apply the reference-state calibration to the modified moments, $\langle \mathcal{H}_\lambda^p \rangle$ (Method B), as we find this to be more resilient to noise compared to applying the correction to the coefficients, ($\langle \mathcal{H} \rangle$, $\langle \mu \rangle$ etc. as in Method C).

4.3 Results and Discussion

Using the same trial-state and measurement settings from Chapter 3, the dipole moment is calculated as described in Section 4.2 and the results are displayed in Figure 4.4. With the moments-based correction, the UCCD trial-state results from *ibm_torino* (green dots) reproduce statevector simulation results (green dashed line) to within 1 standard deviation (errorbars, determined from 50 bootstrapping samples) for small values of Δ, i.e. in the region of interest. On the other hand, direct evaluation of the expectation value (magenta dot) fails to reproduce the statevector simulation results (magenta dashed line). In addition to the moments-corrected $\mu_{\text{inf}}^{(4)}$ being closer to the target FCI value, its variance is also notably smaller than that of the expectation value, $\langle \mu \rangle$, demonstrating that

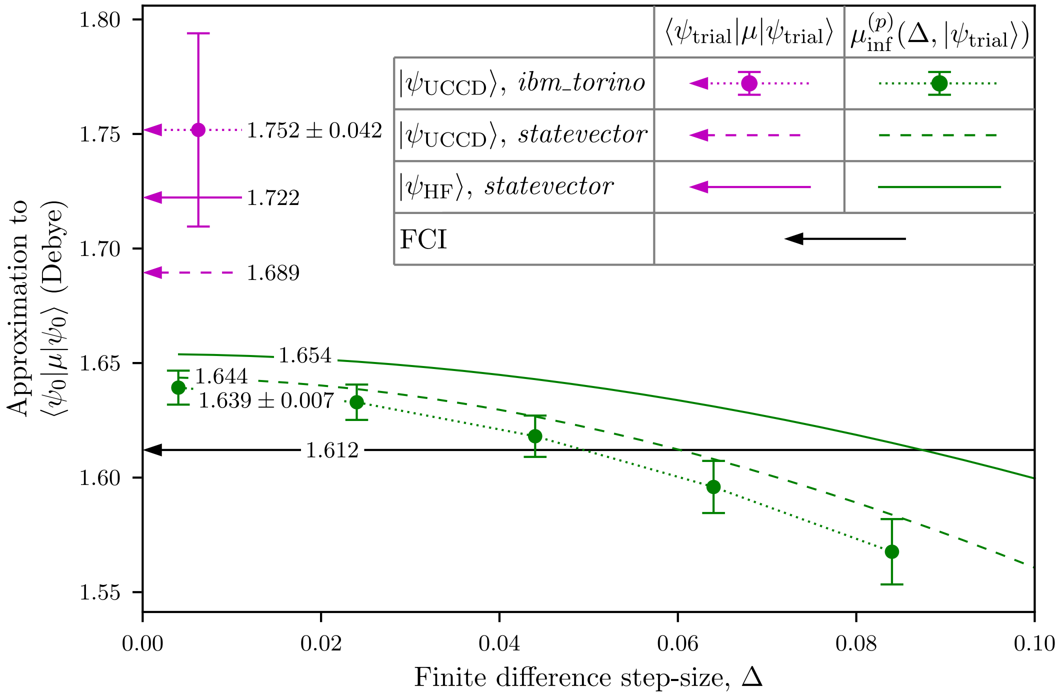


FIGURE 4.4: Results of evaluating the dipole moment with (green) and without (magenta) the moments-based correction compared with the FCI result (black). The results from *ibm_torino* are shown by the round markers, with 1 standard deviation errorbars (determined by 50 bootstrapping samples), the dotted lines are guides to the eye. Note that all markers are calculated from the same data from the quantum device, post-processed using different values of Δ. Dashed and solid lines show the results of statevector simulation of the trial-state and reference- (Hartree-Fock) state respectively. The finite difference error vanishes in the limit $\Delta \to 0$, the results for larger Δ values are plotted to demonstrate the convergence of $\mu_{\text{inf}}^{(4)}$. The value of $\mu_{\text{inf}}^{(4)}(\Delta = 0.005)$ is taken to be sufficiently converged and is given for each curve.

the observed error robustness property [111] of the moments-based energy correction transfers to the evaluation of non-energetic properties.

While the error-mitigated $\langle\mu\rangle$ results are represented as a single point in Figure 4.4, there is a very slight dependence on Δ which is caused by the reference-state calibration method, since the estimated error rate depends on the chosen value of Δ. However, on the relevant scales, this effect is insignificant, e.g. the difference in $\langle\mu\rangle$ between $\Delta = 0.004$ and $\Delta = 0.1$ is 0.1 millidebye, roughly 2-3 orders of magnitude below the differences seen in $\mu_{\text{inf}}^{(4)}$ and in the limit $\Delta \to \infty$, the change approaches 3.7 millidebye. The statevector simulations of $\langle\mu\rangle$ are completely independent of Δ as expected since the finite difference method is exact for the linear function $\langle\mathcal{H} + \lambda\mu\rangle$, whereas for the error-mitigated result, the function being differentiated is only an approximation to $\langle\mathcal{H} + \lambda\mu\rangle$ leading to a small non-linearity. On the other hand $E_{\text{inf}}^{(4)}(\mathcal{H}_\lambda, |\psi\rangle)$ is highly non-linear leading to Δ-dependence in its approximate derivative $\mu_{\text{inf}}^{(4)}$.

While the percentage error in $\mu_{\text{inf}}^{(4)}$ (1.7%) is smaller than that of $\langle\mu\rangle$ (8.7%), these are both much larger than the corresponding percentage errors in the energy estimation (0.05% and 1.2% for $E_{\text{inf}}^{(4)}$ and $\langle\mathcal{H}\rangle$ respectively). This is to be expected since for small rotations, $d\theta$, away from the true ground-state the change in the observables is

$$\begin{aligned} d\langle\mathcal{H}\rangle_\theta &= \mathcal{O}(d\theta^2), \\ d\langle\mu\rangle_\theta &= \mathcal{O}(d\theta), \end{aligned} \tag{4.9}$$

where the $\mathcal{O}(d\theta)$ term in the energy change vanishes since the trial-state is near a minimum. The same arguments hold for the moments $d\langle\mathcal{H}^p\rangle = \mathcal{O}(d\theta^2)$ and modified moments $d\langle\mathcal{H}_\lambda^p\rangle = \mathcal{O}(d\theta)$.

It is also possible that the reason the percentage error in the dipole operator is larger than that of the energy despite having fewer terms in Equation 4.2 compared to Equation Equation 1.10 is that in the Hamiltonian, the diagonal terms dominate, in particular the 1-body terms, g_{jj}, while for the dipole, there is also significant contribution from the off diagonal terms, f_{jk}. Since the off-diagonal elements of the reference-state RDM and the mixed-state RDM are the same

$$\mathrm{Tr}(\rho_{\mathrm{HF}}^{\mathrm{exact}} a_j^\dagger a_k) = \mathrm{Tr}(\rho^{\mathrm{mixed}} a_j^\dagger a_k) = 0, \qquad j \neq k, \tag{4.10}$$

they cannot be corrected directly using the reference state error-mitigation since Equation 3.21 does not have a unique solution for q. For this reason, the noise calibration is applied to $\langle\mathcal{H}^p\rangle$ instead of the individual RDM elements, which allows error in the off-diagonal elements to be corrected indirectly, essentially by assuming that the correct q value is related to the error level in the diagonal elements. However, a more effective method to correct these off-diagonal elements might be to use reference-states that have non-zero off diagonal terms.

4.4 Conclussion

While ground-state energies are important in the application of quantum algorithms to quantum chemistry, it is also important to be able to evaluate other chemical properties in an accurate, noise-robust way. In contrast to previous results that evaluate expectation values, we have demonstrated that moments-based ground-state energy corrections can be adapted to evaluate such properties (using the electric dipole moment of the water molecule as an example) using the Hellmann-Feynman theorem, and that such a method transfers the inherent accuracy and noise-robustness of the corrected energy value to the quantity of interest. These observations broaden the potential range of applications in chemistry to which quantum devices could provide a potential advantage over classical computation.

Chapter 5

Moments-corrected quantum computation of molecular forces

5.1 Introduction

Having now seen the application of the moments-based method to energies (Chapters 2 and 3) and properties of the electronic ground-state (Chapter 4) we now turn our attention to another application of interest; the computation of molecular forces

$$F_j = -\frac{\partial E(\vec{R})}{\partial R_j} = -\frac{\partial}{\partial R_j}\left(\min_{\psi}\langle\psi|\mathcal{H}(\vec{R})|\psi\rangle\right), \tag{5.1}$$

where $E(\vec{R})$ is the *molecular* ground-state energy (not the *electronic* ground-state energy), and R_j is a coordinate of the molecular geometry (e.g. a bond-length, bond-angle or dihedral angle). Computation of molecular forces can be important for molecular geometry optimisation, molecular dynamics simulations, and investigation of reaction pathways and rates which are key to the design of new catalysts or pharmaceuticals.

The most straightforward approach to evaluating an energy derivative of the form of Equation 5.1 on a quantum computer is to perform two variational computations; one at R_j and one at $R_j + h$, each using r measurements, and compute the finite difference:

$$F_j \approx -\frac{1}{h}\Big(\langle\psi_1|\mathcal{H}(R_j+h)|\psi_1\rangle - \langle\psi_2|\mathcal{H}(R_j)|\psi_2\rangle\Big). \tag{5.2}$$

Which gives error:

$$\delta F_j \approx \mathcal{O}\left(\frac{||\mathcal{H}(R_j+h)|| + ||\mathcal{H}(R_j)||}{h\sqrt{r}}\right) + \mathcal{O}(h), \tag{5.3}$$

where $||\cdot||$ denotes the spectral norm of an operator. The first term corresponds to stochastic error from energy estimation by sampling and the second is due to the finite difference method.

An alternative method to evaluate the forces is to make the approximation $|\psi_1\rangle = |\psi_2\rangle \equiv |\psi\rangle$, i.e. that the ground electronic state is independent of R_j. Under this assumption, the measurement outcomes may be reused between the two Hamiltonian evaluations, this correlated sampling helps avoid the accumulation of stochastic error.

$$\begin{aligned} F_j &\approx -\frac{1}{h}\Big(\langle\psi|\mathcal{H}(R_j+h) - \mathcal{H}(R_j)|\psi\rangle\Big), \\ \delta F_j &\approx \mathcal{O}\left(\frac{||\mathcal{H}(R_j+h) - \mathcal{H}(R_j)||}{h\sqrt{r}}\right) + \mathcal{O}(h) + \xi_\psi. \end{aligned} \tag{5.4}$$

Since the first error term now scales with the spectral norm of the difference between Hamiltonians rather than the sum of their individual norms, the number of shots needed to control the stochastic error term is dramatically reduced at the expense of introducing further systematic error, ξ_ψ, due to the method's failure to account for the wavefunction response, sometimes known as the Pulay forces [170]. Essentially, these are the terms involving the derivative of the state that are introduced by the product rule

$$\begin{aligned} F_j &= -\langle\psi|\frac{\partial\mathcal{H}(R_j)}{\partial R_j}|\psi\rangle - \frac{\partial\langle\psi|}{\partial R_j}\mathcal{H}(R_j)|\psi\rangle - \langle\psi|\mathcal{H}(R_j)\frac{\partial|\psi\rangle}{\partial R_j}, \\ \xi_\psi &= -\frac{\partial\langle\psi|}{\partial R_j}\mathcal{H}(R_j)|\psi\rangle - \langle\psi|\mathcal{H}(R_j)\frac{\partial|\psi\rangle}{\partial R_j}. \end{aligned} \tag{5.5}$$

To account for the wavefunction response, we can consider techniques that correct ground-state energy estimates when the trial-state is not the exact ground-state but that do not require additional quantum measurements. One such technique is orbital optimisation, which corresponds to inserting a sequence of single electron excitations,

$$U(\vec{\theta}) = \prod_{jk} \exp\left(\theta_{jk}(a_j^\dagger a_k - a_k^\dagger a_j)\right), \tag{5.6}$$

between the trial-state and the Hamiltonian that can be optimised classically by conjugating the Hamiltonian, i.e.

$$E(\vec{R}) = \min_\psi \langle\psi|\mathcal{H}(\vec{R})|\psi\rangle \approx \min_{\vec{\theta}} \langle\psi|U(\vec{\theta})^\dagger\mathcal{H}(\vec{R})U(\vec{\theta})|\psi\rangle = \min_{\vec{\theta}}\langle\psi|\mathcal{H}'(\vec{R},\vec{\theta})|\psi\rangle \tag{5.7}$$

where it can be shown that $\mathcal{H}'(\vec{R},\vec{\theta})$ can be efficiently calculated classically based on the fermionic anti-commutation relations. Alternatively, as seen in previous chapters, the moments-based correction can be used to account for insufficiencies in the ground-state approximation, incorporating additional information from the Hamiltonian into the ground-state energy estimate and offsetting the error due to the missing wavefunction response terms at the expense of also measuring the Hamiltonian moments.

A further improvement to the molecular force estimation would be to compute the Hamiltonian derivative analytically [171–173]. This would eliminate the finite difference error and can also be made to incorporate the response forces, provided access to the exact ground-state. However, since the moments-corrected estimates can use an approximate wavefunction, it is possible to compute forces at a range of nuclear geometries from a single quantum computation. This suggests that when approximate forces are acceptable, e.g. for geometry optimisation, or when an exact ground-state is not available, the moments-based technique may be preferable to the analytical treatment of the response forces. Conversely, in situations where high accuracy is important, such as in molecular dynamics simulations, the moments-based correction may be insufficient and more accurate treatment of the derivative may be required, provided that a sufficiently accurate representation of the ground-state can be produced. Other methods to estimate molecular forces on a quantum device have been proposed based on Bayesian phase difference estimation [174], and on the Hellmann-Feynman theorem [175].

Previously, molecular forces have been explicitly calculated on quantum hardware using one- and two- qubit representations of the electronic structure problem [176–178], though we note that computation of dissociation curves implicitly allows calculation of the forces as derivatives of the curve. Using Hamiltonian moments, we compute energy derivatives for the water molecule using the 8-qubit trial-state introduced in Chapter 3.

5.2 Results & Discussion

To investigate the feasibility of computing moments-corrected molecular forces on current quantum hardware, we compute the energy landscape and forces for the water molecule using both the variational energy, $\langle\mathcal{H}\rangle$, and the moments corrected energy, $E_{\text{inf}}^{(4)}$, for the Hartree-Fock reference-state and the 8-qubit selected UCCD trial-state used in Chapter 3, including the values computed from the quantum device *ibm_torino*. Figure 5.1 plots the error in the calculated energy landscape relative to the exact FCI result and Figures 5.2 and 5.3 plot the error in the radial and angular components of the force. Heat-maps of the potential energy surface itself are shown in Figure 5.4 while plots of the forces are not shown since the scale over which the forces vary is much larger than the differences between methods and trial-states and so these maps provide little information. We emphasise that in each case the same trial-state data is reused at all molecular geometries, even though the circuit is optimised only for one particular geometry. The errors are measured relative to the FCI results, which incorporate the

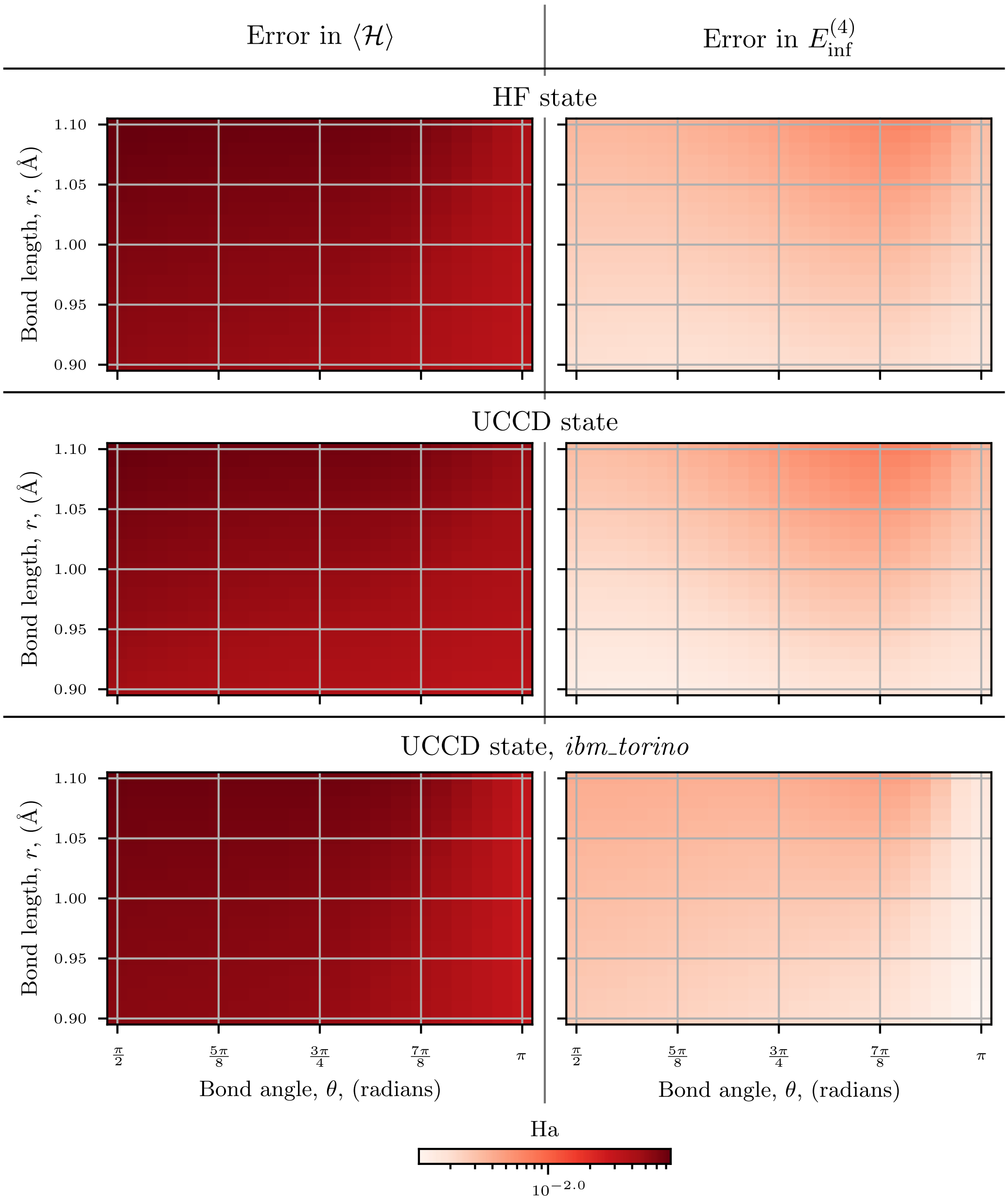


FIGURE 5.1: The error from FCI in the energy landscape computed using the Hartree-Fock state (top row), the optimised UCCD trial-state from Chapter 3 (middle row) and the noisy UCCD trial-state from *ibm_torino* (bottom row). Plots in the left-hand column use the variational energy, $\langle \mathcal{H} \rangle$, while plots in the right-hand column use the moments-corrected energy $E_{\text{inf}}^{(4)}$.

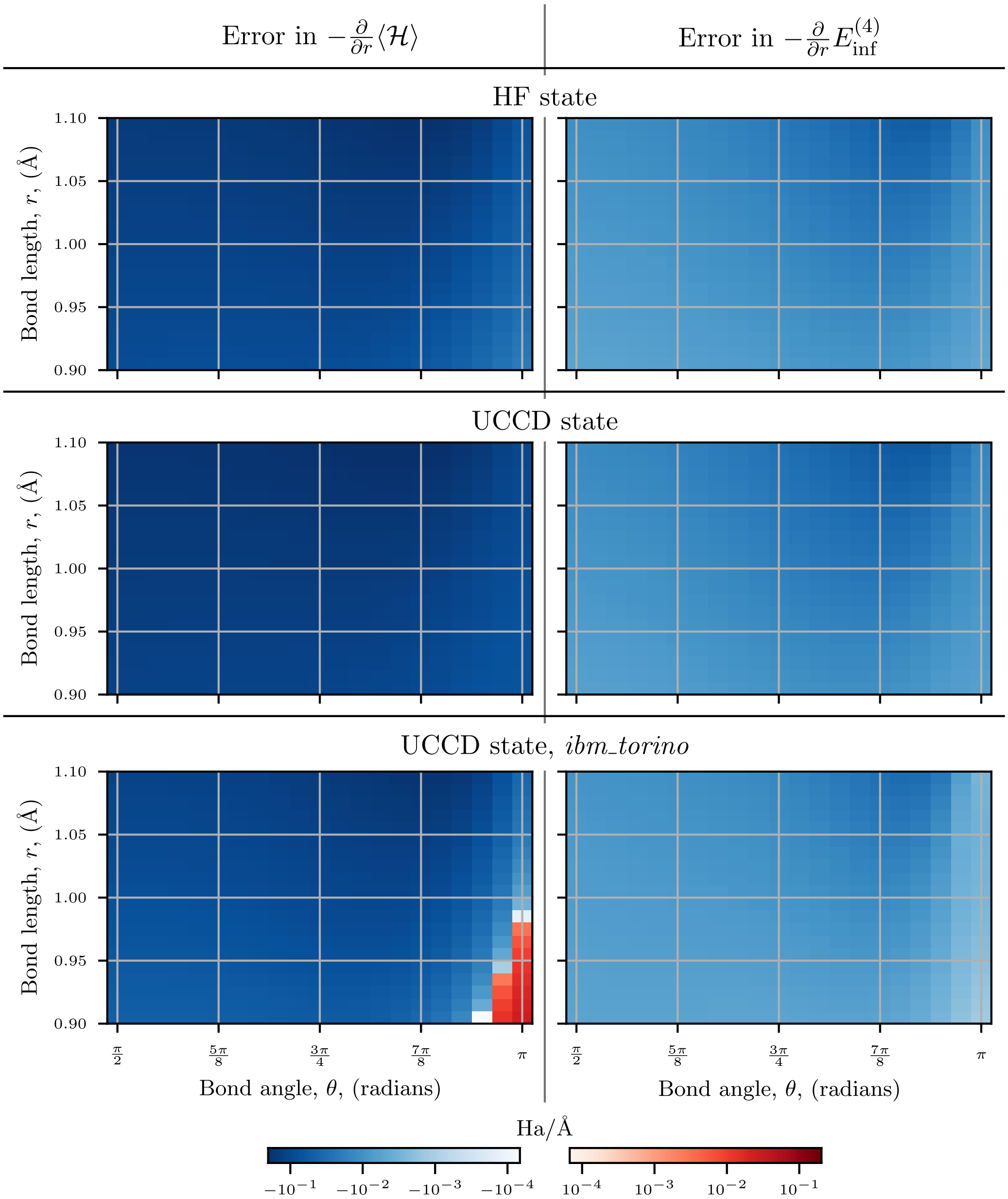


FIGURE 5.2: The error from FCI in the radial energy gradients computed using the Hartree-Fock state (top row), the optimised UCCD trial-state from Chapter 3 (middle row) and the noisy UCCD trial-state from *ibm_torino* (bottom row). Plots in the left-hand column use the variational energy, $\langle\mathcal{H}\rangle$, while plots in the right-hand column use the moments-corrected energy $E^{(4)}_{\text{inf}}$. Red indicates an overestimated force, blue indicates an underestimation.

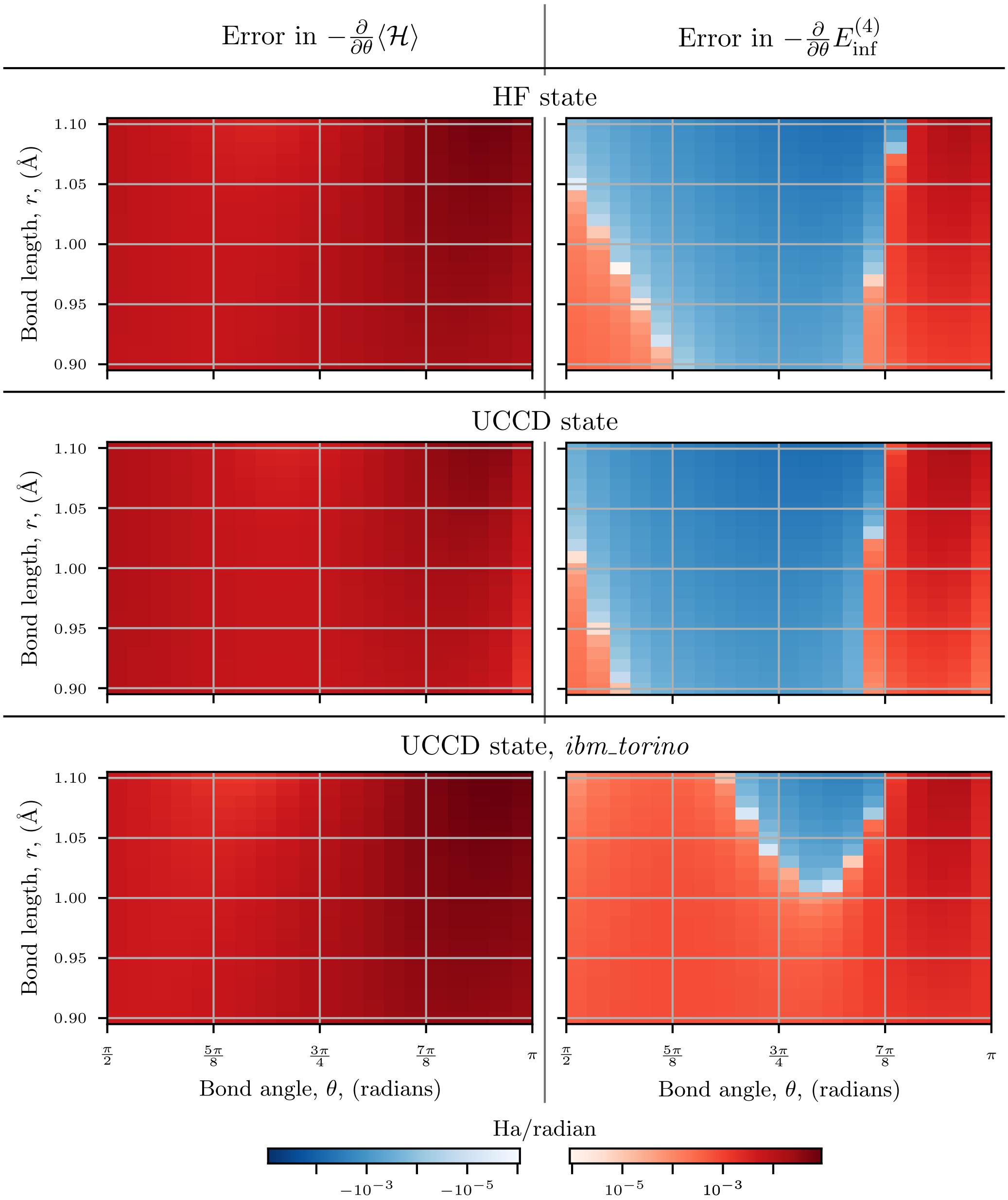


FIGURE 5.3: The error from FCI in the angular energy gradients computed using the Hartree-Fock state (top row), the optimised UCCD trial-state from Chapter 3 (middle row) and the noisy UCCD trial-state from *ibm_torino* (bottom row). Plots in the left-hand column use the variational energy, $\langle\mathcal{H}\rangle$, while plots in the right-hand column use the moments-corrected energy $E^{(4)}_{\text{inf}}$. Red indicates an overestimated force, blue indicates an underestimation.

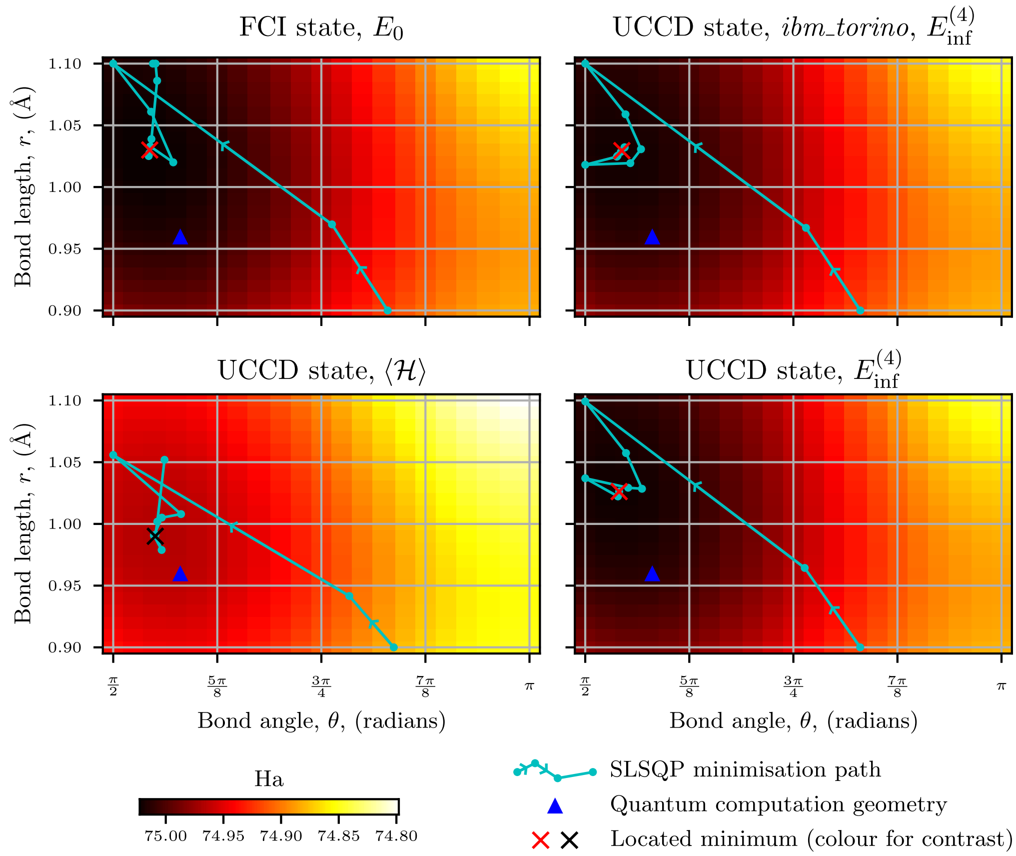


FIGURE 5.4: Example geometry optimisation results, using the FCI results (upper left), the UCCD-state expectation values (lower left), UCCD-state moments-corrected ground-state energy estimates (lower right) and the moments-corrected results from *ibm_torino* (upper right). The optimisation paths begin on the right and follow the cyan lines, with each dot representing one step of the SLSQP optimiser. The located minimum geometry is marked by a red or black cross (the choice of colour is purely for contrast purposes) and the blue triangle indicates the geometry at which the quantum computation was performed.

wavefunction response exactly. Though the FCI results here still include finite difference error, this could be removed through analytic treatment of the derivatives.

As shown in Figures 5.1-5.3, the moments-based correction can reduce the error in the energy and force estimation by an order of magnitude or more. While the variation in the error over the parameter space appears to be magnified by the moments-correction, which would make geometry optimisation less accurate, this magnification is primarily due to the log-scaled colour map and not the method.

To demonstrate the utility of the moments-based correction, Figure 5.4 shows the results of geometry optimisation using the UCCD trial-state, both with (lower right) and without (lower left) the moments-based correction. These are compared to minimisation results using the FCI ground-state (upper left) and results from the quantum hardware

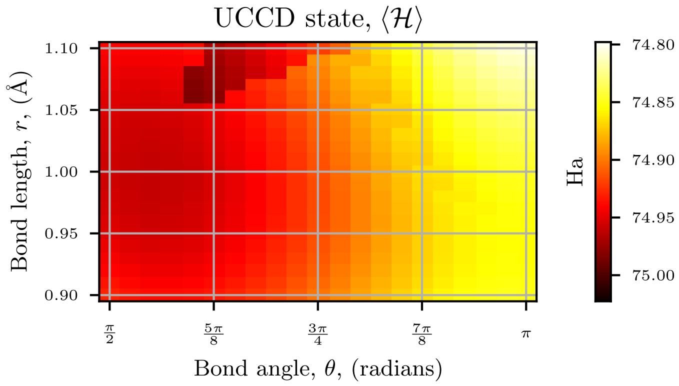


FIGURE 5.5: The impact of molecular orbital discontinuities. Due to degrees of freedom in the mixing of molecular orbitals, directly computing molecular energies can result in discontinuities in the energy as shown here (compare to Figure 5.4 lower left). While the discontinuities are dealt with in an ad hoc way here, a more systematic method is required.

(upper right, using the moments-based correction). We find that, due to the choice of minimal basis, the FCI calculations give a longer bond-length, ∼1.03Å, and tighter angle, ∼95° (0.52π radians), than expected from higher quality calculations [179], 0.96Å and 104.5° (0.58π radians) respectively. As such we compare our results to the FCI values rather than those from the literature, since agreement with larger bases is meaningless when results have not converged to the basis limit. Additionally, this means that the geometry for which the quantum computation was performed in Chapter 3 (blue triangle in Figure 5.4) is not at the targeted FCI equilibrium, which is likely to be the case in any practical application of the method to geometry optimisation since the equilibrium geometry is not known a priori. This means the optimisations performed here are more similar to how a practical geometry optimisation might proceed; by performing a variational minimisation at an approximate equilibrium geometry, evaluating the reduced density matrix from the quantum device, computing an approximate energy landscape, corrected by the moments method, from the single-point result and minimising over this landscape.

We find that the molecular geometry that minimises the UCCD trial-state energy, $\langle \mathcal{H} \rangle$, does not match the equilibrium geometry expected from the FCI calculation. On the other hand, application of the moments-based correction is enough to capture the expected location of the minimum to a good approximation.

However, in the presence of noise, there are artefacts present in the results using the UCCD circuit as seen in Figure 5.5. Notably these discontinuities occur even with the statevector simulation results i.e. they are not caused by noise in the quantum hardware. The boundary seen in this plot is due to discontinuity in the Hartree-Fock molecular orbitals with respect to variations in the molecular geometry. In short, this occurs because the Hartree-Fock state is invariant to mixing among the occupied spin orbitals; mixing among the unoccupied spin-orbitals; or the introduction of a phase on any orbital. This results in degrees of freedom that cannot be fixed by the Hartree-Fock procedure and therefore need not vary smoothly as the geometry is changed. However, the UCCD trial-state is not invariant to these effects and discontinuities in the mixing or phases of orbitals can lead to corresponding discontinuities in the UCCD energy.

This presents a problem, as to solve this we either need to enforce continuity of these degrees of freedom or reoptimise the UCCD state for each geometry. The latter is straightforward but computationally intensive, the aim of the moments-based correction in the context of molecular forces is to avoid these additional quantum computations if possible. The former, on the other hand, is not straightforward. It is possible that either seeding the Hartree-Fock solver with the molecular orbitals of a nearby nuclear configuration and/or classical orbital optimisation can be used to remove the discontinuities. However, this will likely require a deeper level of understanding and control over the classical computational tools than is relevant for the scope of this thesis.

As such, in Figures 5.1-5.3, the effects of these energy discontinuities have been minimised by manually identifying orbital discontinuities based on observing the molecular orbital coefficients. When these discontinuities are caused by discrete transformations, namely swapping molecular orbitals or applying a phase to particular orbitals, generating corrections is trivial. While these corrections eliminate all visible discontinuities, identifying discontinuities by eye is not practical and additionally fails to account for continuous transformations, i.e. orbital mixing. Thus, further investigation into seeding the Hartree-Fock solver and/or orbital optimisation is likely to be necessary. Alternatively, using the atomic orbitals in place of the molecular orbitals would remove the degree of freedom, though would likely require a deeper trial-state to effectively implement the Hartree-Fock protocol on the quantum hardware, which could lead to complications in transferring the trial-state between molecular geometries, as parameters that were allowed to vary previously must now be either frozen or reoptimised on quantum hardware.

5.3 Conclusion

Despite issues related to the consistency of Hartree-Fock computed orbitals, the possibility of using moments-based corrections to simplify quantum computation of molecular forces and geometry optimisation seems promising. Further investigation into the potential for orbital optimisation and/or seeding the Hartree-Fock solver is needed to determine if these techniques are sufficient to combat the orbital inconsistency, but if so it would allow energies and forces over regions in parameter space to be computed with reasonable accuracy from far fewer quantum computations. Additionally, for geometry optimisation it may be possible to perform approximate optimisation based on a single point in the vicinity of the expected minimum, performing another quantum computation at the calculated minimum and repeating. Such a scheme would offset the error incurred through the use of suboptimal trial-states while still allowing a decreased number of quantum evaluations compared to the naive technique.

Chapter 6

Scaling of circuit-based methods for moment estimation

Up to this stage, measurement of the Hamiltonian moments has been performed in a VQE-type framework, where the operator is decomposed into Pauli- or fermionic strings and measured term-by-term. However, despite various techniques designed to measure more operators with fewer state preparations, such as classical shadows [120, 121], or reduce the number of terms in the decomposition such as double factorisation [60, 61, 180], we would also like to investigate 'circuit-based' methods, i.e. methods where the Hamiltonian is encoded into the quantum circuit, that allow more direct access to the moments, preferably while keeping circuit depths below those required for QPE. Here we investigate the behaviour of two such techniques based on work by Duan [83] and later, by Seki and Yunoki [92] and compare these to VQE-style operator averaging in Section 6.1. We term these methods the Hadamard test method (or incoherent quantum power method) and the (coherent) quantum power method. While the resource requirements derived may seem prohibitive, we then indicate methods to improve this scaling in Section 6.2, both through previously suggested methods [92] (Section 6.2.1), and through new observations (Sections 6.2.2, 6.2.3 and 6.2.4). Finally, we perform numerical simulations for a quantum chemical system, the tri-hydrogen cation, H_3^+, to verify our analytic derivations in Section 6.3 and briefly discuss the process of converting moment estimates to ground-state energy estimates in Section 6.4.

6.1 Moment estimation techniques

In this section, we introduce the techniques to be compared and quantify the errors affecting each of them. By bounding these errors by a target precision, we state resource scaling relations for the number of ancilla qubits, circuit depth and number of trial-state copies. The scaling of errors for the Hadamard test (Section 6.1.2) and quantum power method (Section 6.1.3) were already given in [92], however, we give a more complete derivation, translate error scaling to resource scaling and consider the effects of finite sampling in more detail. Additionally, we extend the QPM method to allow computation of odd moments using the coherent differencing technique.

Formally, we define the moment estimation problem: Given a Hamiltonian, $\mathcal{H}$, that can be decomposed into F fragments, $\mathcal{H}_j$;

$$\mathcal{H} = \sum_{j=1}^{F} \mathcal{H}_j, \tag{6.1}$$

and copies of a trial-state, $|\psi\rangle$, we wish to estimate the Hamiltonian moments, $\langle\psi|\mathcal{H}^p|\psi\rangle$, to given additive precision, ε. We do not consider the cost of preparing the state $|\psi\rangle$ but we do consider the number of copies required.

A convenient device for encoding the Hamiltonian into a quantum circuit is the time-evolution operator,

$$U(t) = e^{-i\mathcal{H}t}. \tag{6.2}$$

Where possible, we will treat implementation of the time evolution operator as a black-box, however to obtain the final resource scaling we need to consider the error in implementation of the time-evolution which necessitates specifying a concrete form. For simplicity we use symmetric, s step Trotterisation (Figure 6.1) in this section while recognising that higher order Suzuki-Trotter formulae will likely provide better scaling [92]. As such the fragmentation of the Hamiltonian is necessary for both term-by-term operator averaging, and for implementation of the time-evolution operator by Trotterisation. The scaling of quantum resources for the various methods is summarised in Table 6.1.

6.1.1 Operator averaging

Method

The technique of Hamiltonian averaging was first introduced with the VQE algorithm [8] and allows energies of quantum states to be estimated on present-day devices, where further manipulation of the state, such as time evolution, is infeasible. Here, since we are interested in operators other than the Hamiltonian, i.e. the moments, we consider operator averaging. While this is essentially the same technique used in previous chapters,

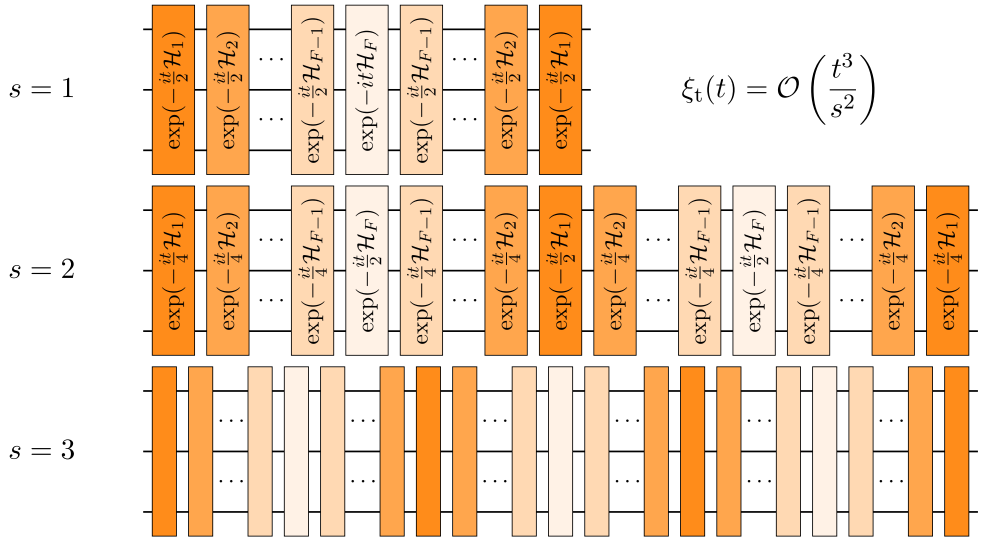


FIGURE 6.1: Example s-step symmetric Trotterisation. Gates of the same shade correspond to evolution under the same Hamiltonian fragment. The corresponding time-evolution approximation error is given in the upper right.

we provide a more complete discussion here. Given the fragmentation of the Hamiltonian, it is straightforward, if computationally intensive, to generate fragmentations for the first few Hamiltonian powers, which can then be evaluated term-by-term.

For example, using the simple Hamiltonian

$$\mathcal{H} = IX + XI + 2ZZ, \tag{6.3}$$

the Hamiltonian powers are then

$$\begin{aligned}
\mathcal{H}^2 &= \sum_{jk=1}^{F} \mathcal{H}_j \mathcal{H}_k \\
&= II + XX - 2iZY + XX + II - 2iYZ + 2iZY + 2iYZ + 4II \\
&= 2XX + 6II, \\
\mathcal{H}^3 &= 2XI + 2IX - 4YY + 4IX + 4XI + 8ZZ \\
&= 6XI + 6IX - 4YY + 8ZZ.
\end{aligned}$$

By simplifying these products through the Pauli operator identities, we obtain a decomposition of the Hamiltonian moments in terms of these ‘consolidated fragments’, $\mathcal{G}_j^{(p)}$,

$$\mathcal{H}^p = \sum_{j=1}^{G_p} \mathcal{G}_j^{(p)}, \tag{6.4}$$

where the number of consolidated fragments is $G_p \leq F^p$. Note that the consolidated fragments corresponding to $p = 1$ are trivially the Hamiltonian fragments themselves, $\mathcal{G}_j^{(1)} = \mathcal{H}_j$. Due to the (worst case) exponential growth in the number of consolidated fragments with the moment order, this technique is limited in the order of moment that can be estimated.

Derivation

The scaling analysis of the operator averaging method is fairly straightforward. First, consider the problem of sampling eigenvalues of the consolidated moments, $\mathcal{G}_j^{(p)}$, by measuring the trial-state in the eigenbasis and computing the mean, $\langle \mathcal{G}_j^{(p)} \rangle$. The standard error, which measures how close the sample mean is to the population mean for a given number of samples r, is

$$\begin{aligned} \xi_{\mathrm{r}}(r) &= \frac{\sigma_{\mathrm{sample}}}{\sqrt{r}} \\ &= \sqrt{\frac{\sum_k^r (x_k - \mu_{\mathrm{sample}})^2}{r(r-1)}} \end{aligned} \tag{6.5}$$

where σ_{sample} is the sample standard deviation, x_k is the result of the kth sample and μ_{sample} is the sample mean. The sample values are bounded by the spectral norm of $\mathcal{G}_j^{(p)}$, denoted

$$\omega_j^{(p)} \equiv ||\mathcal{G}_j^{(p)}||, \tag{6.6}$$

$$x_k - \mu_{\mathrm{sample}} \leq \mathcal{O}(\omega_j^{(p)}), \tag{6.7}$$

so the standard error becomes

$$\begin{aligned} \xi_{\mathrm{r}}(r) &= \mathcal{O}\left(\sqrt{\frac{\sum_k^r (\omega_j^{(p)})^2}{r(r-1)}}\right) \\ &= \mathcal{O}\left(\frac{\omega_j^{(p)}}{\sqrt{r}}\right). \end{aligned} \tag{6.8}$$

Therefore, the finite sampling error transforms the expectation value of a single consolidated fragment as:

$$\langle \mathcal{G}_j^{(p)} \rangle \to \langle \mathcal{G}_j^{(p)} \rangle_{\mathrm{approx}} = \langle \mathcal{G}_j^{(p)} \rangle + \xi_{\mathrm{r}}(r_j^{(p)}) = \langle \mathcal{G}_j^{(p)} \rangle + \mathcal{O}\left(\frac{\omega_j^{(p)}}{\sqrt{r_j^{(p)}}}\right), \tag{6.9}$$

where we now account for the fact that different consolidated fragments can be allocated different numbers of shots, $r_j^{(p)}$, and the total number of required shots is R. Then summing over j gives the error in the estimated moment due to finite sampling errors as

$$\langle \mathcal{H}^p \rangle \to \langle \mathcal{H}^p \rangle + \Xi_{\mathrm{r}}(R) = \langle \mathcal{H}^p \rangle + \sum_{j=1}^{G_p} \mathcal{O}\left(\frac{\omega_j^{(p)}}{\sqrt{r_j^{(p)}}} \right).$$

Where $\Xi_{\mathrm{r}}(R)$ is the finite sampling error contribution to the moment. Strictly this is also a function of the way the total R measurements are distributed between consolidated fragments. However, given a target error, ε, and since there are G_p terms in the summation, we can bound each error term for each consolidated fragment by $\mathcal{O}(\varepsilon/G_p)$ to obtain the corresponding number of measurements for each fragment required to bound the overall error by ε:

$$r_j^{(p)} = \mathcal{O}\left(\frac{G_p^2 (\omega_j^{(p)})^2}{\varepsilon^2} \right).$$

The total number of trial-state copies required is then,

$$R = \mathcal{O}\left(\frac{G_p^2}{\varepsilon^2} \sum_{j=1}^{G_p} (\omega_j^{(p)})^2 \right). \tag{6.10}$$

Note that this description implicitly includes grouping fragments into mutually commuting bases, since the weighted sum of the commuting fragments can itself be considered as a larger fragment, e.g. the Hamiltonian,

$$\mathcal{H} = (IX) + (XI) + (2ZZ),$$

has three fragments (each enclosed in parentheses) but is the same as the two-fragment Hamiltonian

$$\mathcal{H} = (IX + XI) + (2ZZ).$$

To express the required circuit depth, we need to consider the depth required to measure each consolidated fragment in its corresponding eigenbasis. We term this depth the 'diagonalisation depth' of consolidated fragment $\mathcal{G}_j^{(p)}$ and denote it by $d_j^{(p)}$. This is the depth required to implement the circuit which serves to rotate the trial-state into the eigenbasis of the fragment, or equivalently to rotate the fragment into the computational basis, i.e. to diagonalise it. The numerical value of the diagonalisation depth will evidently depend on the choice of fragment but also on the hardware details (e.g. qubit connectivity, base gate set etc.). For Pauli strings the diagonalisation depth is $\mathcal{O}(1)$, while for the single-excitations ($\mathrm{Re}(a_k^\dagger a_l)$ and $\mathrm{Im}(a_k^\dagger a_l)$) used in Chapter 3, it is $\mathcal{O}(|k-l|)$.

For the operator averaging method, the number of ancilla qubits, maximal circuit depth and required trial-state copies, respectively, are

$$\begin{aligned} Q_{\mathrm{OA}} &= 0, \\ D_{\mathrm{OA}} &= \max_j d_j^{(p)}, \\ R_{\mathrm{OA}} &= \mathcal{O}\left(\frac{G_p^2}{\varepsilon^2}\sum_{j=1}^{G_p}(\omega_j^{(p)})^2\right). \end{aligned} \tag{6.11}$$

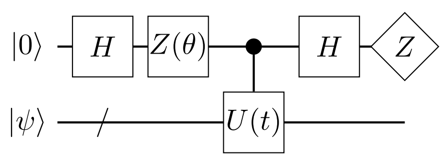
6.1.2 Hadamard test

Method

The computation of moments via the Hadamard test [83, 92] can be understood as estimating the gradient of the time evolution operator, $U(t) = \exp(-i\mathcal{H}t)$, which acts as a generating function for the moments,

$$(-i)^p\langle\mathcal{H}^p\rangle = \left.\frac{d^p\langle U(t)\rangle}{dt^p}\right|_{t=0}, \tag{6.12}$$

by running the circuit



for different values of θ and t and obtaining expectation values by sampling. The lower register is the system register, initially containing the trial-state $|\psi\rangle$ and the method requires one ancilla qubit. The parameter θ allows for selection of the real ($\theta = 0$) and imaginary ($\theta = \pi/2$) components of the expectation value. The moments can be estimated using the finite difference method [83, 92] as,

$$\begin{aligned} \langle\mathcal{H}^{2q}\rangle &\approx -\frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\mathrm{Pr}(0,kh), \\ \langle\mathcal{H}^{2q+1}\rangle &\approx -\frac{4}{h^{2q+1}}\left[\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\mathrm{Pr}(\pi/2,(k+1/2)h) - \frac{1}{2}\binom{2q}{q}\right], \end{aligned} \tag{6.13}$$

where $\mathrm{Pr}(\theta, t)$ is the probability of measuring 1 on the ancilla for the specified values of θ and t; and h is the finite difference step-size. Note that here, q is both half the target moment order (rounded up) and the number of circuits to be executed for that moment order. It is not the number of qubits in the system (which we do not define) nor the number of ancilla qubits (which is always 1)[1].

Derivation

In addition to finite sampling error, which will take a form analogous to Equation 6.9, the Hadamard test method also incurs error due to approximate implementation of the time-evolution operator, Ξ_{t}, and the finite difference method, Ξ_{h}. We first wish to express the output of the circuit in terms of its inputs, $|\psi\rangle$, $U_{\mathrm{approx}}(t)$ and θ, where

$$U_{\mathrm{approx}}(t) = e^{-i\mathcal{H}t} + \xi_{\mathrm{t}}(t) \approx e^{-i\mathcal{H}t}, \tag{6.14}$$

is the approximate time evolution which is implemented in practice. Immediately before measurement, it can be shown that the state of the combined ancilla and system registers has evolved to

$$\begin{aligned} |\Psi(\theta,t)\rangle &= \frac{1}{2}|0\rangle(1+e^{i\theta}[e^{-i\mathcal{H}t}+\xi_{\mathrm{t}}(t)])|\psi\rangle + \frac{1}{2}|1\rangle(1-e^{i\theta}[e^{-i\mathcal{H}t}+\xi_{\mathrm{t}}(t)])|\psi\rangle \\ &\equiv |0\rangle|\psi_0(\theta,t)\rangle + |1\rangle|\psi_1(\theta,t)\rangle, \end{aligned} \tag{6.15}$$

and the probability of the ancilla collapsing to the $|1\rangle$ state, can be obtained from the normalisation of the $|\psi_1\rangle$ state with which it is entangled[2]. For relevant choices of θ this gives,

$$\begin{aligned} \langle\psi_1(\theta,t)|\psi_1(\theta,t)\rangle &= \frac{1}{2} - \frac{1}{4}e^{i\theta}\langle\psi|(e^{-i\mathcal{H}t}+e^{-i2\theta}e^{i\mathcal{H}t})|\psi\rangle \\ &\quad - \frac{1}{4}e^{i\theta}\langle\psi|(\xi_{\mathrm{t}}(t)+e^{-i2\theta}\xi_{\mathrm{t}}^{\dagger}(t))|\psi\rangle, \\ \langle\psi_1(0,t)|\psi_1(0,t)\rangle &= \frac{1}{2} - \frac{1}{2}\sum_{j=0}^{\infty}\frac{(-1)^j t^{2j}}{(2j)!}\langle\mathcal{H}^{2j}\rangle - \frac{1}{2}\mathrm{Re}\langle\xi_{\mathrm{t}}(t)\rangle, \\ \langle\psi_1(\pi/2,t)|\psi_1(\pi/2,t)\rangle &= \frac{1}{2} - \frac{1}{2}\sum_{j=0}^{\infty}\frac{(-1)^j t^{2j+1}}{(2j+1)!}\langle\mathcal{H}^{2j+1}\rangle + \frac{1}{2}\mathrm{Im}\langle\xi_{\mathrm{t}}(t)\rangle. \end{aligned} \tag{6.16}$$

[1]The choice of q is to maintain consistency with the coherent quantum power method, where q is both half the moment order *and* the number of ancilla measurements.

[2]This can be shown by expressing the state as a density matrix and taking the expectation value of the projector $|1\rangle\langle 1|$ on the ancilla.

Since estimation will be performed using a finite number of samples, the calculated probabilites will be

$$\begin{aligned}
\Pr(0,t) &= \frac{1}{2} - \frac{1}{2}\sum_{j=0}^{\infty}\frac{(-1)^j t^{2j}}{(2j)!}\langle\mathcal{H}^{2j}\rangle - \frac{1}{2}\mathrm{Re}\langle\xi_{\mathrm{t}}(t)\rangle + \xi_{\mathrm{r}}(r,t),\\
\Pr(\pi/2,t) &= \frac{1}{2} + \frac{1}{2}\sum_{j=0}^{\infty}\frac{(-1)^j t^{2j+1}}{(2j+1)!}\langle\mathcal{H}^{2j+1}\rangle + \frac{1}{2}\mathrm{Im}\langle\xi_{\mathrm{t}}(t)\rangle + \xi_{\mathrm{r}}(r,t),
\end{aligned} \tag{6.17}$$

where ξ_{r} is a stochastic variable corresponding to shot noise. While ξ_{r} does not depend explicitly on t, it will vary each time a probability is measured and will be written as a function of t as a reminder that different evaluations (which occur at different choices of t) will have different numerical values. Given these expressions for the probabilities, we can now determine the error in Equation 6.13 by substitution.

Focusing first on the even moments, the expression obtained is

$$\begin{aligned}
\langle\mathcal{H}^{2q}\rangle &\approx -\frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\Pr(0,t)\\
&= 2\sum_{j=1}^{\infty}h^{2(j-q)}\left[\frac{(-1)^j}{(2j)!}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}k^{2j}\right]\langle\mathcal{H}^{2j}\rangle\\
&\quad+\frac{2}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\mathrm{Re}\langle\xi_{\mathrm{t}}(kh)\rangle + \frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\xi_{\mathrm{r}}(r,kh).
\end{aligned}$$

Where the summation in square brackets can be related to the generalised hypergeometric function. For any given value of j, this can be re-expressed as a rational polynomial of order $j-q$, with the first few values of j giving

$$\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}k^{2j} = 0, \qquad\qquad j-q<0,$$

and

$$\frac{(-1)^q(2(j-q+1))!}{(2j)!}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}k^{2j} \tag{6.18}$$

$$= \begin{cases} 1, & j-q=0, \\ q, & j-q=1, \\ \frac{1}{4}\left(5q^2-q\right), & j-q=2, \\ \frac{1}{18}\left(35q^3-21q^2+4q\right), & j-q=3, \\ \frac{1}{48}\left(175q^4-210q^3+101q^2-18q\right), & j-q=4, \\ \frac{1}{48}\left(385q^5-770q^4+671q^3-286q^2+48q\right), & j-q=5, \\ \vdots & \vdots \end{cases}$$

Defining the polynomials on the right hand side of equation Equation 6.18 as $R_{j-q}(q) = \mathcal{O}(q^{j-q})$ we can rewrite the expression for $\langle\mathcal{H}^{2q}\rangle$ as

$$\langle\mathcal{H}^{2q}\rangle \approx 2\sum_{l=0}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2q+2l}\rangle R_l(q)$$
$$+\frac{2}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\mathrm{Re}\langle\xi_{\mathrm{t}}(kh)\rangle+\frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\xi_{\mathrm{r}}(r,kh). \tag{6.19}$$

Noting the $l=0$ term is exactly $\langle\mathcal{H}^{2q}\rangle$, the error in the estimation of the even moments is

$$\delta\langle\mathcal{H}^{2q}\rangle = 2\sum_{l=1}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2q+2l}\rangle R_l(q)+\frac{2}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\mathrm{Re}\langle\xi_{\mathrm{t}}(kh)\rangle$$
$$+\frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\xi_{\mathrm{r}}(r,kh) \tag{6.20}$$

where the three error terms account for finite difference, approximate time evolution and finite sampling, respectively.

For the odd moments, substitution of the derived probability into Equation 6.13 gives

$$\begin{aligned}
\langle \mathcal{H}^{2q+1} \rangle &\approx -\frac{4}{h^{2q+1}} \left[\sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \mathrm{Pr}(\pi/2, (k+1/2)h) - \frac{1}{2}\binom{2q}{q} \right] \\
&= 2\sum_{j=0}^{\infty} h^{2j-2q} \left[\frac{(-1)^j}{(2j+1)!} \sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \left(k+\frac{1}{2}\right)^{2j+1} \right] \langle \mathcal{H}^{2j+1} \rangle \\
&\quad - \frac{2}{h^{2q+1}} \sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \mathrm{Im}\langle \xi_{\mathrm{t}}((k+1/2)h) \rangle \\
&\quad - \frac{4}{h^{2q+1}} \sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \xi_{\mathrm{r}}(r, (k+1/2)h),
\end{aligned}$$

where we have used the identity

$$\sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-j} = \binom{2q}{q}, \tag{6.21}$$

to cancel the constant terms. As with the even case, the expression in square brackets can be expressed as a rational polynomial

$$\sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \left(k+\frac{1}{2}\right)^{2j+1} = 0, \qquad j-q<0,$$

$$\frac{(-1)^q(2(j-q+1))!}{(2j+1)!} \sum_{k=0}^{q} (-1)^k \binom{2q+1}{q-k} \left(k+\frac{1}{2}\right)^{2j+1} \tag{6.22}$$

$$= \begin{cases} 1, & j-q=0 \\ 2q+1, & j-q=1 \\ 20q^2+16q+3, & j-q=2 \\ 280q^3+252q^2+74q+9, & j-q=3 \\ 2800q^4+2240q^3+776q^2+208q+15, & j-q=4 \\ 12320q^5+6160q^4+2992q^3+1496q^2+18q+9, & j-q=5 \\ \vdots & \vdots \end{cases}$$

Defining the right hand side as $S_{j-q}(q) = \mathcal{O}(q^{j-q})$, the expression can be simplified to

$$\begin{aligned}\langle\mathcal{H}^{2q+1}\rangle \approx 2\sum_{l=0}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2l+2q+1}\rangle S_l(q) \\ -\frac{2}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\mathrm{Im}\langle\xi_{\mathrm{t}}((k+1/2)h)\rangle \\ -\frac{4}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\xi_{\mathrm{r}}(r,(k+1/2)h),\end{aligned} \tag{6.23}$$

and recognising that the $l = 0$ term is $\langle\mathcal{H}^{2q+1}\rangle$, the error in the estimation of the odd moments can be written as

$$\begin{aligned}\delta\langle\mathcal{H}^{2q+1}\rangle = 2\sum_{l=1}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2l+2q+1}\rangle S_l(q) \\ -\frac{2}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\mathrm{Im}\langle\xi_{\mathrm{t}}((k+1/2)h) \\ -\frac{4}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\xi_{\mathrm{r}}(r,(k+1/2)h),\end{aligned} \tag{6.24}$$

where the terms, again, represent the finite difference, time-evolution and finite sampling errors respectively.

Given the expressions for the error in the even and odd moments, reproduced here for convenience:

$$\begin{aligned}\delta\langle\mathcal{H}^{2q}\rangle &= 2\sum_{l=1}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2q+2l}\rangle R_l(q) \\ &\quad+\frac{2}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\mathrm{Re}\langle\xi_{\mathrm{t}}(kh)\rangle \\ &\quad+\frac{4}{h^{2q}}\sum_{k=1}^{q}(-1)^k\binom{2q}{q-k}\xi_{\mathrm{r}}(r,kh), \\ \delta\langle\mathcal{H}^{2q+1}\rangle &= 2\sum_{l=1}^{\infty}\frac{(-1)^l}{(2(l+1))!}h^{2l}\langle\mathcal{H}^{2l+2q+1}\rangle S_l(q) \\ &\quad-\frac{2}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\mathrm{Im}\langle\xi_{\mathrm{t}}((k+1/2)h)\rangle \\ &\quad-\frac{4}{h^{2q+1}}\sum_{k=0}^{q}(-1)^k\binom{2q+1}{q-k}\xi_{\mathrm{r}}(r,(k+1/2)h),\end{aligned}$$

and the following asymptotic scaling relations:

$$\begin{aligned} R_l(q) &= \mathcal{O}(q^l), \\ S_l(q) &= \mathcal{O}(q^l), \\ \xi_{\mathrm{r}}(r,t) &= \mathcal{O}\left(\frac{1}{\sqrt{r}}\right), \end{aligned} \tag{6.25}$$

the contribution to the error in the moment estimate due to finite difference, time-evolution implementation and finite sampling can be written as

$$\Xi_{\mathrm{h}}(h) = \mathcal{O}(ph^2), \qquad \Xi_{\mathrm{t}}(h) = \mathcal{O}\left(\frac{\xi_{\mathrm{t}}(ph)}{h^p}\right), \qquad \Xi_{\mathrm{r}}(r,h) = \mathcal{O}\left(\frac{1}{h^p\sqrt{r}}\right). \tag{6.26}$$

To obtain scaling relations for the circuit depth, we need to decide on an implementation for the time-evolution. Using the s-step symmetric Trotter formula from Figure 6.1,

$$e^{-i\mathcal{H}t} = \left[\prod_{j=1}^{F} e^{-i\mathcal{H}_j t/2s} \prod_{j=1}^{F} e^{-i\mathcal{H}_{F-1-j} t/2s}\right]^s + \mathcal{O}\left(\frac{t^3}{s^2}\right), \tag{6.27}$$

allows the error to be expressed as

$$\delta\langle\mathcal{H}^p\rangle = \mathcal{O}(ph^2) + \mathcal{O}\left(\frac{p^3}{s^2h^{p-3}}\right) + \mathcal{O}\left(\frac{1}{h^p\sqrt{r}}\right), \tag{6.28}$$

and bounding each term by the target error ε and rearranging gives

$$h = \mathcal{O}\left(\sqrt{\frac{\varepsilon}{p}}\right), \qquad s = \mathcal{O}\left(\frac{p^{(p+3)/4}}{\varepsilon^{(p-1)/4}}\right), \qquad r = \mathcal{O}\left(\frac{p^p}{\varepsilon^{p+2}}\right). \tag{6.29}$$

The scaling of the quantum resources (number of ancillas, maximum circuit depth and trial-state copies, respectively) are then

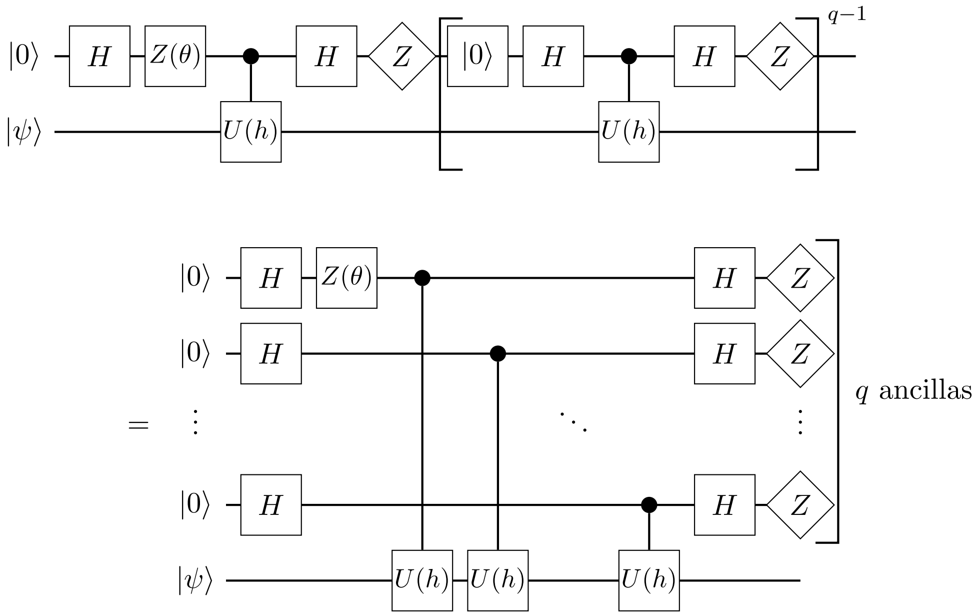
$$\begin{aligned} Q_{\mathrm{HT}} &= 1, \\ D_{\mathrm{HT}} &= \mathcal{O}\left(\frac{Fdp^{(p+3)/4}}{\varepsilon^{(p-1)/4}}\right), \\ R_{\mathrm{HT}} &= \mathcal{O}\left(\frac{p^{p+1}}{\varepsilon^{p+2}}\right), \end{aligned} \tag{6.30}$$

where the number of copies is increased by a factor of p to account for the q different times for which the circuit must be run and the F and d factors in the circuit depth correspond to the number of Hamiltonian fragments and the depth of implementing time-evolution by a single fragment. Note that while the scaling of the number of trial-state copies seems prohibitive, this can be improved through Richardson extrapolation techniques [92] (see Section 6.2.1).

6.1.3 Quantum power method

Method

While the Hadamard test method described above estimates the gradient of the time-evolution by computing expectation values and taking the finite difference classically, the quantum power method [92], implemented by either of the equivalent circuits below, can be viewed as computing the finite difference coherently, i.e. on the quantum device



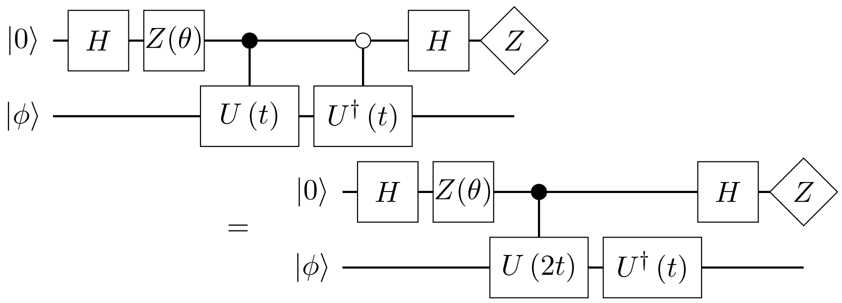
In the first circuit, the $|0\rangle$ gate is a reset operation and the bracketed circuit segment is repeated $q-1 = \lceil p/2 \rceil - 1$ times. In the second circuit, the system is instead interacted with $q = \lceil p/2 \rceil$ different ancilla qubits. It is trivial to show that the two circuits are identical, simply by replacing the measure-and-reset sequence with swapping in a fresh qubit. As with the Hadamard test, q is equal to half the moment order rounded up, however, in this case it is also equal to the number of ancilla qubits (or the number of measurments to be made on the single ancilla qubit) hence the choice of symbol. Note that while the second circuit appears to be significantly shallower, this will only be true in practice when the measure-and-reset sequence is expensive, since the single qubit gate times will be negligible compared to the time-evolution operator. The choice to use a single ancilla or multiple ancillas will be determined by hardware constraints, e.g. qubit connectivity, time required to perform measurement-and-reset relative to the time required to perform controlled time-evolution etc. In any case, the moments can be estimated as

$$\langle \mathcal{H}^{2q} \rangle \approx \left(\frac{2}{h}\right)^{2q} \mathrm{Pr}_q\left(0, h\right), \tag{6.31}$$
$$\langle \mathcal{H}^{2q+1} \rangle \approx -\left(\frac{2}{h}\right)^{2q+1} \left[\mathrm{Pr}_{q+1}\left(\pi/2, h\right) - \frac{1}{2}\mathrm{Pr}_q\left(0, h\right)\right],$$

where $\mathrm{Pr}_q\left(\theta, t\right)$ is the probability of measuring 1 on all q ancillas.

Derivation

The quantum power method will be affected by the same error sources as the Hadamard test – namely finite sampling, Ξ_{r}, finite difference, Ξ_{h}, and time-evolution approximation, Ξ_{t} – however, due to the differences in the way that information is extracted from the circuit, the manner in which the error sources propagate through the quantum and classical computations differs and leads to different overall scaling. For ease of computation, we will consider the directional controlled time evolution circuits



In terms of the ancilla measurement probabilities these circuits are identical to the standard controlled time evolution circuit used for the Hadamard Test, up to the substitution $t = h/2$, however, they are easier to analyse in terms of the quantum power method and may also be easier to implement in practice depending on the Hamiltonian of interest and native gateset of the quantum device in use. The quantum power method circuit can be written as the repeated application of the above circuits. To begin with, the state generated by one of these circuits is

$$\begin{aligned}|\Psi\rangle = \frac{1}{2}\Big(&|0\rangle \left[e^{i\mathcal{H}t} + e^{i\theta}e^{-i\mathcal{H}t} + \xi_{\mathrm{t}}^{\dagger}(t) + e^{i\theta}\xi_{\mathrm{t}}(t)\right]|\phi\rangle \\ &+ |1\rangle \left[e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t} + \xi_{\mathrm{t}}^{\dagger}(t) - e^{i\theta}\xi_{\mathrm{t}}(t)\right]|\phi\rangle\Big).\end{aligned} \tag{6.32}$$

If the ancilla is measured in the $|1\rangle$ state then the operation applied to the system register will be

$$V_1(\theta, t) = \frac{1}{2}\left[e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t} + \xi_{\mathrm{t}}^{\dagger}(t) - e^{i\theta}\xi_{\mathrm{t}}(t)\right]. \tag{6.33}$$

We wish to obtain the probability of measuring all q ancillas in the $|1\rangle$ state when the circuit is applied q successive times to the trial-state, $|\psi\rangle$, as required by Equation 6.31 to evaluate the even moments. The probability will be given by the magnitude of the state produced by q applications of V_1, i.e.

$$\mathrm{Pr}_q(0,t) = \langle\psi|[V_1^\dagger(0,t/2)]^q[V_1(0,t/2)]^q|\psi\rangle + \xi_\mathrm{r}(r,0), \tag{6.34}$$

where $\xi_\mathrm{r}(r,\theta)$ is the error due to the finite number of samples. The θ parameter simply serves as a reminder that the variable is stochastic and cannot be cancelled, which will be important when we come to evaluate the odd moments. To express this quantity analytically, it is useful to write

$$\begin{aligned}
V_1^\dagger(\theta,t)V_1(\theta,t) &= -\frac{e^{-i\theta}}{4}\left[e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t} + \xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right]^2 \\
&= -\frac{e^{-i\theta}}{4}\left(e^{2i\mathcal{H}t} + e^{2i\theta}e^{-2i\mathcal{H}t}\right) + \frac{1}{2} \\
&\quad - \frac{e^{-i\theta}}{4}\Bigg[\left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right)^2 + \left(e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t}\right)\left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right) \\
&\qquad + \left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right)\left(e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t}\right)\Bigg].
\end{aligned}$$

Using the series expansion of the time evolution operator and taking the special case $\theta = 0$ gives,

$$\begin{aligned}
V_1^\dagger(\theta,t)V_1(\theta,t) &= -\frac{e^{-i\theta}}{4}\sum_{j=0}^{\infty}\frac{(1+e^{2i\theta}(-1)^j)(2i\mathcal{H}t)^j}{j!} + \frac{1}{2} \\
&\quad - \frac{e^{-i\theta}}{4}\Bigg[\left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right)^2 + \left(e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t}\right)\left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right) \\
&\qquad + \left(\xi_\mathrm{t}^\dagger(t) - e^{i\theta}\xi_\mathrm{t}(t)\right)\left(e^{i\mathcal{H}t} - e^{i\theta}e^{-i\mathcal{H}t}\right)\Bigg], \\
V_1^\dagger(0,t)V_1(0,t) &= -\frac{1}{2}\sum_{j=1}^{\infty}\frac{(-1)^j}{(2j)!}\mathcal{H}^{2j}(2t)^{2j} \\
&\quad - \frac{1}{4}\Bigg[\left(\xi_\mathrm{t}^\dagger(t) - \xi_\mathrm{t}(t)\right)^2 + \left(e^{i\mathcal{H}t} - e^{-i\mathcal{H}t}\right)\left(\xi_\mathrm{t}^\dagger(t) - \xi_\mathrm{t}(t)\right) \\
&\qquad + \left(\xi_\mathrm{t}^\dagger(t) - \xi_\mathrm{t}(t)\right)\left(e^{i\mathcal{H}t} - e^{-i\mathcal{H}t}\right)\Bigg],
\end{aligned} \tag{6.35}$$

and so the product of q of these operators gives

$$\begin{aligned}
\left[V_1^\dagger(0,t)V_1(0,t)\right]^q &= \left[V_1^\dagger(0,t)\right]^q[V_1(0,t)]^q \\
&= \left[-\frac{1}{2}\sum_{j=1}^{\infty}\frac{(-1)^j}{(2j)!}\mathcal{H}^{2j}(2t)^{2j}\right]^q + \sum_{k=1}^{q}A_{kq}(t),
\end{aligned} \tag{6.36}$$

where we use the fact that $V_1(0,t)$ commutes with its adjoint, which can be seen since the two are related through multiplication by a phase. Here $A_{kq}(t)$ is defined to be the symmetric product over $q-k$ copies of the first term in Equation 6.35 and k copies of the second term. In the special case that ξ_{t} commutes with $\mathcal{H}$, then

$$\begin{aligned} A_{kq}(t) = \binom{q}{k} &\left[-\frac{1}{2} \sum_{j=1}^{\infty} \frac{(-1)^j}{(2j)!} \mathcal{H}^{2j} (2t)^{2j} \right]^{q-k} \\ &\cdot \left[-\frac{1}{4} \left[\left(\xi_{\mathrm{t}}^{\dagger}(t) - \xi_{\mathrm{t}}(t) \right)^2 + 2 \left(e^{i\mathcal{H}t} - e^{-i\mathcal{H}t} \right) \left(\xi_{\mathrm{t}}^{\dagger}(t) - \xi_{\mathrm{t}}(t) \right) \right] \right]^k, \end{aligned} \tag{6.37}$$

and Equation 6.36 would be the standard binomial formula. In any case, we can expand the first term in Equation 6.36 using the multinomial formula and extract the first term in the resulting summation to obtain

$$\begin{aligned} \left[V_1^{\dagger}(0,t) \right]^q \left[V_1(0,t) \right]^q &= \sum_{k=0}^{q} \binom{q}{k} \left[\mathcal{H}^2 t^2 \right]^{q-k} \left[-\frac{1}{2} \sum_{j=2}^{\infty} \frac{(-1)^j}{(2j)!} \mathcal{H}^{2j} (2t)^{2j} \right]^k + \sum_{k=1}^{q} A_{kq}(t) \\ &= \mathcal{H}^{2q} t^{2q} + \sum_{k=1}^{q} \binom{q}{k} \left[\mathcal{H}^2 t^2 \right]^{q-k} \left[-\frac{1}{2} \sum_{j=2}^{\infty} \frac{(-1)^j}{(2j)!} \mathcal{H}^{2j} (2t)^{2j} \right]^k \\ &\quad + \sum_{k=1}^{q} A_{kq}(t). \end{aligned} \tag{6.38}$$

Defining $B_{kq}(t)$ as follows, and taking expectation values with respect to the trial-state $|\psi\rangle$ gives

$$\begin{aligned} B_{kq}(t) &= \binom{q}{k} \left[\mathcal{H}^2 t^2 \right]^{q-k} \left[-\frac{1}{2} \sum_{j=2}^{\infty} \frac{(-1)^j}{(2j)!} \mathcal{H}^{2j} (2t)^{2j} \right]^k, \\ \Pr_q(0,h) &= \left(\frac{h}{2} \right)^{2q} \langle \mathcal{H}^{2q} \rangle + \sum_{k=1}^{q} \left\langle B_{kq}\left(\frac{h}{2} \right) + A_{kq}\left(\frac{h}{2} \right) \right\rangle + \xi_{\mathrm{r}}(r,0). \end{aligned} \tag{6.39}$$

where the sum of A_{kq} and the sum of B_{kq} are the time-evolution approximation and finite difference error respectively. Substituting this expression into Equation 6.31 gives

$$\begin{aligned} \langle \mathcal{H}^{2q} \rangle &\approx \left(\frac{2}{h} \right)^{2q} \Pr\left(0^{\otimes q}, h \right) \\ &= \langle \mathcal{H}^{2q} \rangle + \left(\frac{2}{h} \right)^{2q} \sum_{k=1}^{q} \left\langle B_{kq}\left(\frac{h}{2} \right) + A_{kq}\left(\frac{h}{2} \right) \right\rangle + \left(\frac{2}{h} \right)^{2q} \xi_{\mathrm{r}}(r,0), \end{aligned} \tag{6.40}$$

and the error in the evaluation of the even moments is

$$\delta\langle\mathcal{H}^{2q}\rangle = \left(\frac{2}{h}\right)^{2q} \sum_{k=1}^{q} \left\langle B_{kq}\left(\frac{h}{2}\right) + A_{kq}\left(\frac{h}{2}\right)\right\rangle + \left(\frac{2}{h}\right)^{2q} \xi_{\mathrm{r}}(r,0). \tag{6.41}$$

To compute the odd moments, we also need to evaluate the probability for $\theta = \pi/2$, given by

$$\begin{aligned}
\mathrm{Pr}_{q+1}(\pi/2,h) &= \langle\psi|V_1^\dagger(\pi/2,h/2)[V_1^\dagger(0,h/2)]^q[V_1(0,h/2)]^q V_1(\pi/2,h/2)|\psi\rangle + \xi_{\mathrm{r}}(r,\pi/2) \\
&= \langle\psi|\left([V_1^\dagger(0,h/2)]^q[V_1(0,h/2)]^q\right)\left(V_1^\dagger(\pi/2,h/2)V_1(\pi/2,h/2)\right)|\psi\rangle \\
&\quad + \xi_{\mathrm{r}}(r,\pi/2).
\end{aligned} \tag{6.42}$$

The assumption that $V_1(0,h/2)$ commutes with $V_1^\dagger(\pi/2,h/2)$ can be shown provided the approximate time-evolution is symmetric. Therefore, we obtain

$$\begin{aligned}
V_1^\dagger(\pi/2,t)V_1(\pi/2,t) &= \frac{i}{4}\sum_{j=0}^{\infty}\frac{(1-(-1)^j)(2i\mathcal{H}t)^j}{j!} + \frac{1}{2} \\
&\quad + \frac{i}{4}\Bigg[\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)^2 + \left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right) \\
&\qquad + \left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)\left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\Bigg] \\
&= -\mathcal{H}t + \frac{1}{2} - \frac{1}{2}\sum_{j=1}^{\infty}\frac{(-1)^j\mathcal{H}^{2j+1}(2t)^{2j+1}}{(2j+1)!} \\
&\quad + \frac{i}{4}\Bigg[\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)^2 + \left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right) \\
&\qquad + \left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)\left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\Bigg].
\end{aligned} \tag{6.43}$$

Recognising the final term as the error due to the time-evolution approximation and the sum over j as the error due to the finite difference method and denoting these by A' and B' respectively

$$\begin{aligned}
A'(t) &= \frac{i}{4}\Bigg[\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)^2 + \left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right) \\
&\qquad + \left(\xi_{\mathrm{t}}^\dagger(t) - i\xi_{\mathrm{t}}(t)\right)\left(e^{i\mathcal{H}t} - ie^{-i\mathcal{H}t}\right)\Bigg], \\
B'(t) &= -\frac{1}{2}\sum_{j=1}^{\infty}\frac{(-1)^j\mathcal{H}^{2j+1}(2t)^{2j+1}}{(2j+1)!},
\end{aligned} \tag{6.44}$$

and suppressing the argument to A_{kq}, B_{kq}, A', and B' (all arguments are $h/2$), the measured probability can then be written as

$$\begin{aligned}
&\mathrm{Pr}_{q+1}(\pi/2, h) \\
&\quad = \left\langle \left[-\frac{h}{2}\mathcal{H} + \frac{1}{2} + B' + A' \right] \cdot \left[\left(\frac{h}{2}\right)^{2q} \mathcal{H}^{2q} + \sum_{k=1}^{q} (B_{kq} + A_{kq}) \right] \right\rangle + \xi_{\mathrm{r}}(r, \pi/2) \\
&\quad = -\left(\frac{h}{2}\right)^{2q+1} \langle \mathcal{H}^{2q+1} \rangle - \frac{h}{2} \left\langle \mathcal{H} \sum_{k=1}^{q} (B_{kq} + A_{kq}) \right\rangle + \left(\frac{h}{2}\right)^{2q} \langle \mathcal{H}^{2q}(B' + A') \rangle \\
&\qquad + \left\langle (B' + A') \sum_{k=1}^{q} (B_{kq} + A_{kq}) \right\rangle + \xi_{\mathrm{r}}(r, \pi/2) \\
&\qquad + \frac{1}{2}\left(\frac{h}{2}\right)^{2q} \langle \mathcal{H}^{2q} \rangle + \frac{1}{2} \sum_{k=1}^{q} \langle B_{kq} + A_{kq} \rangle
\end{aligned} \tag{6.45}$$

Substituting this along with the probability for $\theta = 0$ into the equation for the moments (Equation 6.31) gives

$$\begin{aligned}
\langle \mathcal{H}^{2q+1} \rangle \approx& -\left(\frac{2}{h}\right)^{2q+1} \left[\mathrm{Pr}_{q+1}(\pi/2, h) - \frac{1}{2}\mathrm{Pr}_{q}(0, h) \right] \\
&= \langle \mathcal{H}^{2q+1} \rangle + \left(\frac{2}{h}\right)^{2q} \left\langle \mathcal{H} \sum_{k=1}^{q} (B_{kq} + A_{kq}) \right\rangle - \frac{2}{h} \langle \mathcal{H}^{2q}(B' + A') \rangle \\
&\quad - \left(\frac{2}{h}\right)^{2q+1} \xi_{\mathrm{r}}(r, \pi/2) - \left(\frac{2}{h}\right)^{2q+1} \xi_{\mathrm{r}}(r, 0) \\
&\quad - \left(\frac{2}{h}\right)^{2q+1} \left\langle (B' + A') \sum_{k=1}^{q} (B_{kq} + A_{kq}) \right\rangle
\end{aligned} \tag{6.46}$$

The error in the estimation of the odd moments is therefore

$$\begin{aligned}
\delta \langle \mathcal{H}^{2q+1} \rangle =& \left(\frac{2}{h}\right)^{2q} \left\langle \mathcal{H} \sum_{k=1}^{q} [B_{kq} + A_{kq}] \right\rangle - \left(\frac{2}{h}\right)^{2q+1} \xi_{\mathrm{r}}(r, 0) \\
&- \frac{2}{h} \langle \mathcal{H}^{2q} [B' + A'] \rangle - \left(\frac{2}{h}\right)^{2q+1} \xi_{\mathrm{r}}(r, \pi/2) \\
&- \left(\frac{2}{h}\right)^{2q+1} \left\langle [B' + A'] \sum_{k=1}^{q} [B_{kq} + A_{kq}] \right\rangle
\end{aligned} \tag{6.47}$$

To evaluate the scaling of the error in moment estimation, we will first need to consider the scaling of A, A', B and B'. Noting that

$$e^{i\mathcal{H}t} - e^{-i\mathcal{H}t} = \mathcal{O}(t), \tag{6.48}$$

and assuming that, as $t \to 0$, $\xi_{\mathrm{t}}(t)$ decays at least as fast as $\mathcal{O}(t^2)$, then

$$\begin{aligned}
\sum_{k=1}^{q} A_{kq}(t) &= \mathcal{O}\left(\sum_{k=1}^{q} \binom{q}{k} t^{2q-2k} \left(\xi_{\mathrm{t}}(t)^2 + t\xi_{\mathrm{t}}(t)\right)^k\right) = \mathcal{O}\left(qt^{2q-1}\xi_{\mathrm{t}}(t)\right), \\
A'(t) &= \mathcal{O}\left(\xi_{\mathrm{t}}(t)^2 + \xi_{\mathrm{t}}(t)\right) = \mathcal{O}\left(\xi_{\mathrm{t}}(t)\right), \\
\sum_{k=1}^{q} B_{kq}(t) &= \mathcal{O}\left(\sum_{k=1}^{q} \binom{q}{k} t^{2q-2k} t^{4k}\right) = \mathcal{O}\left(qt^{2q+2}\right), \\
B'(t) &= \mathcal{O}\left(t^3\right). \qquad (6.49)
\end{aligned}$$

Substituting these into the relevant expressions gives

$$\begin{aligned}
\delta\langle\mathcal{H}^{2q}\rangle &= \mathcal{O}\left(qh^2\right) + \mathcal{O}\left(\frac{q}{h}\xi_{\mathrm{t}}(h)\right) + \mathcal{O}\left(\frac{1}{h^{2q}\sqrt{r}}\right), \\
\delta\langle\mathcal{H}^{2q+1}\rangle &= \mathcal{O}\left(qh^2\right) + \mathcal{O}\left(\frac{q}{h}\xi_{\mathrm{t}}(h)\right) + \mathcal{O}\left(\frac{1}{h^{2q+1}\sqrt{r}}\right). \qquad (6.50)
\end{aligned}$$

Combining these together, we get

$$\begin{aligned}
\delta\langle\mathcal{H}^{p}\rangle &= \mathcal{O}\left(ph^2\right) + \mathcal{O}\left(\frac{p}{h}\xi_{\mathrm{t}}(h)\right) + \mathcal{O}\left(\frac{1}{h^{p}\sqrt{r}}\right) \\
&\equiv \Xi_{\mathrm{h}}(h) + \Xi_{\mathrm{t}}(h) + \Xi_{\mathrm{r}}(r,h). \qquad (6.51)
\end{aligned}$$

By bounding each term by ε, and assuming the s-step symmetric Trotterisation, $\xi_{\mathrm{t}}(t) = \mathcal{O}(t^3/s^2)$, we can obtain

$$\begin{aligned}
h &= \mathcal{O}\left(\sqrt{\frac{\varepsilon}{p}}\right), \\
s &= \mathcal{O}(1), \qquad (6.52) \\
r &= \mathcal{O}\left(\frac{p^p}{\varepsilon^{p+2}}\right),
\end{aligned}$$

and converting these to quantum circuit resource costs (number of ancillas, maximum circuit depth and circuit repetitions respectively) gives

$$\begin{aligned}
Q_{\mathrm{QPM}} &= 1 \text{ or } \lfloor p/2 \rfloor, \\
D_{\mathrm{QPM}} &= \mathcal{O}\left(Fdp\right), \qquad (6.53) \\
R_{\mathrm{QPM}} &= \mathcal{O}\left(\frac{p^p}{\varepsilon^{p+2}}\right).
\end{aligned}$$

where the circuit depth includes factors of F, d and p to account for the number of Hamiltonian fragments, depth of time-evolving by a single fragment and the fact that there are q controlled time evolution operations. As with the Hadamard test, the number

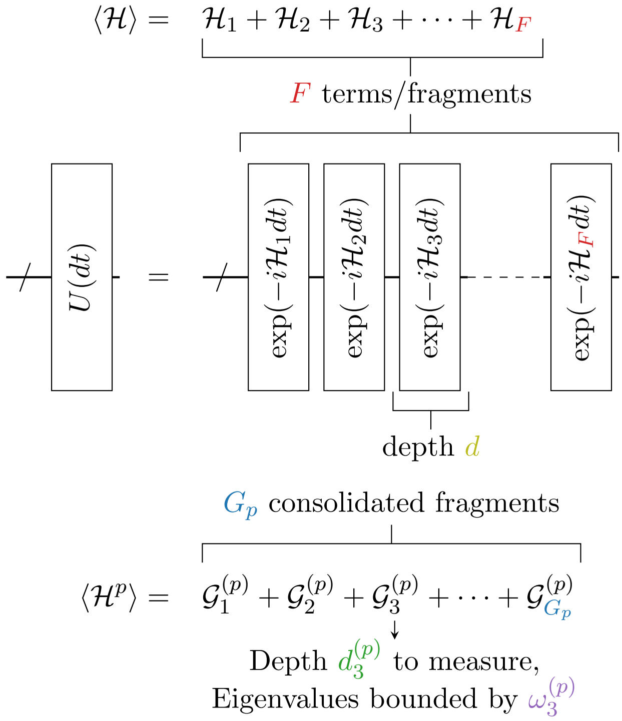


FIGURE 6.2: Summary of properties relevant to analytical scaling expressions in Table 6.1. F is the number of fragments in the Hamiltonian; d is the depth to implement time-evolution of a single fragment (on average); $G_p \leq F^p$ is the number of consolidated fragments in the pth moment; $d_j^{(p)}$ is the depth required to diagonalise the consolidated fragment $\mathcal{G}_j^{(p)}$; and $\omega_j^{(p)}$ is a bound on the eigenvalues of the consolidated fragment $\mathcal{G}_j^{(p)}$.

of trial-state copies required seems large, but this can be improved through Richardson extrapolation [92] (and Section 6.2.1) and also through a measurement recycling scheme introduced in Section 6.2.3.

6.2 Improved moment estimation

In this section we consider a number of adaptations that may reduce the cost of the Hadamard test and quantum power method techniques.

6.2.1 Richardson extrapolation

The Richardson extrapolation technique was suggested alongside the original formulation of the quantum power method [92] and reduces the number of trial-state copies required for moment estimation. The method involves evaluating the finite difference at large differences (large h) and combining the estimates in such a way that the finite difference and Trotterisation errors cancel to order h^{2m}, where m is the order of the Richardson

Resource	Ancilla qubits	Circuit Depth	Circuit Repetitions[a]
OA	0	$\max\limits_{j\in[1,G_p]} d_j^{(p)}$	$\mathcal{O}\left(\frac{{G_p}^2}{\varepsilon^2}\sum_{j=1}^{G_p}(\omega_j^{(p)})^2\right)$
HT	1	$\mathcal{O}\left(\frac{Fdp^{(p+3)/4}}{\varepsilon^{(p-1)/4}}\right)$	$\mathcal{O}\left(\frac{p^{p+1}}{\varepsilon^{p+2}}\right)$
QPM No qubit reset	$\left\lfloor\frac{p}{2}\right\rfloor$	$\mathcal{O}\left(Fdp\right)$	$\mathcal{O}\left(\frac{p^{p}}{\varepsilon^{p+2}}\right)$
QPM With qubit reset	1	As above	As above

TABLE 6.1: Summary of analytic scaling for estimation of the moment, $\langle\mathcal{H}^p\rangle$, to additive precision ε using Operator Averaging (OA), the Hadamard Test (HT) and the Quantum Power Method (QPM), assuming multi-step symmetric Trotterisation (Figure 6.1). Other quantities are defined in Figure 6.2. [a]Note that Richardson extrapolation is not considered here, since scaling estimates are complicated by sub-leading order terms in the time-evolution approximation error due to cancellation of leading orders.

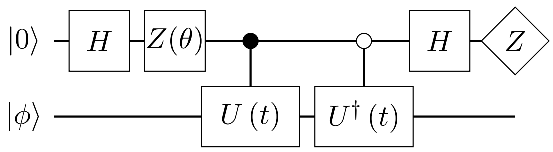
extrapolation. Note that this is not the same as zero-noise extrapolation [139] which mitigates hardware noise. Here we are cancelling algorithmic noise due to the finite difference and Trotter errors and using this reduced error to indirectly reduce the number of trial-state copies required. This reduction comes about because sampling noise is amplified at small evolution times, so it is more efficient to perform multiple estimates for long evolution times than it is to perform a single estimate at a shorter time.

Time reversal invariance

We will begin by showing that the moment estimates obtained via the Hadamard test and quantum power method are invariant under time inversion $t \to -t$ (or equivalently $h \to -h$). This is important as this symmetry leads to error terms with an odd power in t vanishing. Denoting the moment estimates derived using time t as $\langle\mathcal{H}^p\rangle_t$, then we can express a general algorithmic error term as a polynomial (of infinite order) in t, seperate the even and odd terms, and show the odd ones vanishing:

$$\begin{aligned}
\langle \mathcal{H}^p \rangle_t &= \langle \mathcal{H}^p \rangle_{-t} \\
\langle \mathcal{H}^p \rangle + \sum_{j=1}^{\infty} b_j t^j &= \langle \mathcal{H}^p \rangle + \sum_{j=1}^{\infty} b_j (-t)^j \\
\sum_{j=1}^{\infty} b_{2j} t^{2j} + \sum_{j=0}^{\infty} b_{2j+1} t^{2j+1} &= \sum_{j=1}^{\infty} b_{2j} t^{2j} - \sum_{j=0}^{\infty} b_{2j+1} t^{2j+1} \\
2 \sum_{j=0}^{\infty} b_{2j+1} t^{2j+1} &= 0, \quad \forall t \\
\Rightarrow b_{2j+1} &= 0. \qquad (6.54)
\end{aligned}$$

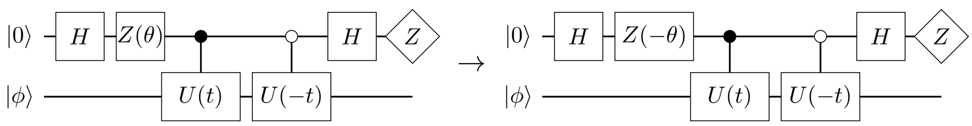
It remains to show that the estimates are symmetric under the time inversion. To do this, we examine the circuit element



which (as discussed in Section 6.1.3) is the building block for the QPM and HT circuits. Assuming symmetric Trotterisation, we have $U^\dagger(t) = U(-t)$ so

$$t \to -t$$

$$\Rightarrow$$



which can be shown by commuting the time-evolution gates, introducing X gates on the ancilla to reverse the controls, and pushing these gates through the Hadamards to obtain a Z gate at the start of the circuit (which has no effect on the $|0\rangle$ state) and another before the measurement (which introduces an unobserved phase). The effect of time-inversion is therefore to take $\theta \to -\theta$. This indicates that if $\theta = 0$, the measurement probabilities will be invariant under time inversion. However, if $\theta = \pi/2$, then the time inversion effectively introduces a Z gate on the ancilla following the time evolution, when moved to the end of the circuit, this becomes an X gate and swaps the probabilities of measuring 1 and 0. For the Hadamard test this means that time inversion causes the transformations

$$\begin{aligned} h &\to -h, \\ \Pr(0,h) &\to \Pr(0,h), \\ \Pr(\pi/2,h) &\to 1-\Pr(\pi/2,h). \end{aligned} \tag{6.55}$$

With minimal additional work it can be shown that Equation 6.13 is invariant under these transformations. Analysis of Equation 6.31 is slightly more involved. First we note that the analysis applied to the Hadamard test above also holds for the quantum power method,

$$h \to -h \qquad \Rightarrow \qquad \Pr_{q+1}(\pi/2,h) \to \Pr_{q+1}(-\pi/2,h), \tag{6.56}$$

however this swaps the probabilities related to the first qubit only (as this is the only qubit with a $Z(\theta)$ gate). So, with respect to the circuit with $(\theta, t) = (\pi/2, h)$, the right hand side of the above is the probability of measuring 0 on the first qubit and 1 on the remaining q qubits. Note that since there are only two outcomes of measurement on the first qubit, this can be written as

$$\Pr_{q+1}(-\pi/2,h) = \Pr_q(0,h) - \Pr_{q+1}(\pi/2,h), \tag{6.57}$$

where the terms on the right hand side correspond to the probability of measuring 1 on the later q qubits (and anything on the first, the $\theta = 0$ parameter then refers to a $Z(\theta)$ gate on the second qubit) and the probability of measuring 1 on all $q+1$ qubits. Substitution into Equation 6.31 shows that the quantum power method is also invariant under time reversal symmetry, and therefore error terms with odd powers of t must vanish.

Cancellation of error terms

Having shown time reversal invariance of the moment estimates and hence the inherent cancellation of odd order error terms in t, we now wish to use Richardson extrapolation to cancel the even order terms. This can be accomplished by the formula [92]

$$\langle \mathcal{H}^p \rangle_h^{(1)} = \frac{\langle \mathcal{H}^p \rangle_{h\lambda} - \lambda^2 \langle \mathcal{H}^p \rangle_h}{1-\lambda^2} = \langle \mathcal{H}^p \rangle + \mathcal{O}\left(h^4\right), \tag{6.58}$$

where $\langle \mathcal{H}^p \rangle_h^{(1)}$ is the first order Richardson corrected estimate and $\lambda > 1$ is a real scalar. Higher order extrapolations can be defined recursively as [92]

$$\langle \mathcal{H}^p \rangle_h^{(m+1)} = \frac{\langle \mathcal{H}^p \rangle_{h\lambda}^{(m)} - \lambda^{2(m+1)} \langle \mathcal{H}^p \rangle_h^{(m)}}{1 - \lambda^{2(m+1)}} = \langle \mathcal{H}^p \rangle + \mathcal{O}\left(h^{2m+2}\right). \tag{6.59}$$

By reusing moment estimations, the mth order Richardson-corrected estimate can be obtained using estimation at only $m+1$ different times [92] as demonstrated in Figure 6.3. To show that the error following Richardson extrapolation behaves as stated, suppose the mth Richardson estimate is given by

$$\langle \mathcal{H}^p \rangle_h^{(m)} = \langle \mathcal{H}^p \rangle + \sum_{k=m+1} b_{2k}^{(m)} h^{2k} + \frac{\xi_{\mathrm{r}}^{(m)}(h)}{h^p}. \tag{6.60}$$

For $m = 0$, i.e. no Richardson extrapolation, this is true for the quantum power method as given by the combination of Equations 6.41, 6.47 and 6.54 and it is also true for the Hadamard test given sufficiently accurate time-evolution implementation (recall that for higher order moments, we require correspondingly higher order time-evolution). From this initial observation it can be shown by induction that the above holds for any order Richardson extrapolation. Substituting Equation 6.60 (for $m - 1$) into 6.59 gives

$$\begin{aligned}
\langle \mathcal{H}^p \rangle_h^{(m)} &= \frac{1}{1-\lambda^{2m}} \langle \mathcal{H}^p \rangle + \frac{1}{1-\lambda^{2m}} \sum_{k=m} b_{2k} \lambda^{2k} h^{2k} + \frac{1}{1-\lambda^{2m}} \frac{\xi_{\mathrm{r}}(\lambda h)}{\lambda^p h^p} \\
&\quad - \frac{\lambda^{2m}}{1-\lambda^{2m}} \langle \mathcal{H}^p \rangle - \frac{\lambda^{2m}}{1-\lambda^{2m}} \sum_{k=m} b_{2k} h^{2k} - \frac{\lambda^{2m}}{1-\lambda^{2m}} \frac{\xi_{\mathrm{r}}(h)}{h^p} \\
&= \langle \mathcal{H}^p \rangle + \sum_{k=m+1} \frac{\lambda^{2k} - \lambda^{2m}}{1-\lambda^{2m}} b_{2k} h^{2k} + \frac{1}{1-\lambda^{2m}} \left(\frac{\xi_{\mathrm{r}}(\lambda h)}{\lambda^p h^p} - \lambda^{2m} \frac{\xi_{\mathrm{r}}(h)}{h^p} \right).
\end{aligned} \tag{6.61}$$

Replacing $\xi_{\mathrm{r}}(t)$ by the asymptotic scaling $\mathcal{O}(1/\sqrt{r})$ leads to

$$\langle \mathcal{H}^p \rangle_h^{(m)} = \langle \mathcal{H}^p \rangle + \sum_{k=m+1} \frac{\lambda^{2k} - \lambda^{2m}}{1-\lambda^{2m}} b_{2k} h^{2k} + \frac{1+\lambda^{2m+p}}{(1-\lambda^{2m})\lambda^p} \frac{1}{h^p} \mathcal{O}\left(\frac{1}{\sqrt{r}}\right). \tag{6.62}$$

This matches the general form of Equation 6.60, so defining $b_{2k}^{(m)}$ (where $k > m$) and $\xi_{\mathrm{r}}^{(m)}(h)$ as follows

$$
\begin{aligned}
b_{2k}^{(m)} &\equiv \begin{cases} \dfrac{\lambda^{2k} - \lambda^{2m}}{1 - \lambda^{2m}} b_{2k}^{(m-1)}, & m > 0, \\ b_{2k}, & m = 0 \end{cases} \\
&= \left[\prod_{j=1}^{m} \frac{\lambda^{2k} - \lambda^{2j}}{1 - \lambda^{2j}} \right] b_{2k}, \quad k > m \geq 0, \\
\xi_{\mathrm{r}}^{(m)}(h) &\equiv \left[\prod_{j=1}^{m} \frac{1 + \lambda^{2j+p}}{(1 - \lambda^{2j})\lambda^{p}} \right] \mathcal{O}\left(\frac{1}{\sqrt{r}} \right), \quad m \geq 0,
\end{aligned} \tag{6.63}
$$

shows that Equation 6.60 holding for $m - 1$ implies it holds for m, and therefore for all $m \geq 0$. To simplify the expressions for $b_{2k}^{(m)}$ and $\xi_{\mathrm{r}}^{(m)}$, we can bound the numerators and denominators in the products, taking an upper bound on the numerators and a lower bound on the denominator gives an upper bound on the products as follows

$$
\begin{aligned}
b_{2k}^{(m)} &\leq \left[\prod_{j=1}^{m} \frac{\lambda^{2k} - \lambda^{2m}}{1 - \lambda^{2}} \right] b_{2k} = \frac{(\lambda^{2k} - \lambda^{2m})^{m}}{(1 - \lambda^{2})^{m}} b_{2k} \\
\xi_{\mathrm{r}}^{(m)}(h) &\leq \left[\prod_{j=1}^{m} \frac{1 + \lambda^{2m+p}}{(1 - \lambda^{2})\lambda^{p}} \right] \mathcal{O}\left(\frac{1}{\sqrt{r}} \right) = \frac{(1 + \lambda^{2m+p})^{m}}{(1 - \lambda^{2})^{m} \lambda^{pm}} \mathcal{O}\left(\frac{1}{\sqrt{r}} \right)
\end{aligned} \tag{6.64}
$$

Bounding the error terms in Equation 6.60 by ε then gives

$$
\begin{aligned}
h &= \mathcal{O}\left(\frac{(1 - \lambda^{2})^{m/(2(m+1))}}{\lambda^{m}} \frac{\varepsilon^{1/(2(m+1))}}{(b_{2(m+1)})^{1/(2(m+1))}} \right) \\
r &= \mathcal{O}\left(\frac{(1 + \lambda^{2m+p})^{2m} (b_{2(m+1)})^{p/(m+1)}}{([1 - \lambda^{2}]^{m} \varepsilon)^{2+p/(m+1)}} \right) \approx \mathcal{O}\left(\frac{1}{\varepsilon^{2+p/(m+1)}} \right),
\end{aligned} \tag{6.65}
$$

where our focus is on the scaling of the number of trial-state copies, r, using the Richardson extrapolation as compared to without. Notably, with the extrapolation it is possible to make the number of copies arbitrarily close to the standard quantum limit, $\mathcal{O}(1/\varepsilon^{2})$, (for a fixed p) at the expense of running circuits at m different times. While further analysis of the scaling of the prefactor, including the tightness of the bounds used to arrive at this point, is required, this is left to future work. We do note that since the moments can theoretically grow exponentially with p, it seems likely that increasing the moment order and demanding constant additive error will greatly increase the quantum resources required. However, it is not clear yet whether constant additive error is required or whether, for example, constant multiplicative error is sufficient to obtain reasonable ground-state energy estimates.

$$
\begin{tikzcd}[column sep=small]
\text{Error: } \mathcal{O}(pt^2) + \mathcal{O}\left(\frac{1}{t^p\sqrt{r}}\right) & \langle \mathcal{H}^p \rangle_h \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda h} \arrow[dl] \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda^2 h} \arrow[dl] \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda^3 h} \arrow[dl] \\
\text{Error: } \mathcal{O}(p^2t^4) + \mathcal{O}\left(a_{1,p}(\lambda)\frac{1}{t^p\sqrt{r}}\right) & \langle \mathcal{H}^p \rangle_h^{(1)} \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda h}^{(1)} \arrow[dl] \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda^2 h}^{(1)} \arrow[dl] & \\
\text{Error: } \mathcal{O}(p^3t^6) + \mathcal{O}\left(a_{2,p}(\lambda)\frac{1}{t^p\sqrt{r}}\right) & \langle \mathcal{H}^p \rangle_h^{(2)} \arrow[d] & \langle \mathcal{H}^p \rangle_{\lambda h}^{(2)} \arrow[dl] & & \\
\text{Error: } \mathcal{O}(p^4t^8) + \mathcal{O}\left(a_{3,p}(\lambda)\frac{1}{t^p\sqrt{r}}\right) & \langle \mathcal{H}^p \rangle_h^{(3)} & & a_{m,p}(\lambda) = \frac{(1+\lambda^{p+2m})^m}{(1-\lambda^2)^m\lambda^{mp}} &
\end{tikzcd}
$$

FIGURE 6.3: Reusing $m + 1 = 4$ moment estimations to evaluate the mth order Richardson extrapolation. The remaining error scaling is displayed on the left (where $t \in \{h, \lambda h, \lambda^2 h, \lambda^3 h\}$) assuming r shots per moment estimation.

6.2.2 Scale invariance and additive precision

Motivated by the suggestion that constant additive precision may not be required, we consider scaling the Hamiltonian by a factor $\kappa > 0$, to obtain a new Hamiltonian

$$\mathcal{H}_\kappa = \kappa \mathcal{H}. \tag{6.66}$$

This should not be confused with the modified Hamiltonian introduced in Chapter 4. It is straightforward to show the action on the moments and cumulants of this transformation is

$$\begin{aligned} \langle \mathcal{H}_\kappa^p \rangle &= \kappa^p \langle \mathcal{H}^p \rangle \\ c_{\kappa,p} &= \kappa^p c_p. \end{aligned} \tag{6.67}$$

Indeed, this scaling of the Hamiltonian can be considered as a unit conversion, for example if $\kappa = 1000$, this would convert a Hamiltonian expressed in hartree into one expressed in millihartree. Therefore, the pth moment and cumulant, which have units of energy to the pth power, will be scaled as above and it can be shown that the infimum energy estimate scales as

$$E_{\text{inf},\kappa}^{(p)} = \kappa E_{\text{inf}}^{(p)}, \quad \forall p. \tag{6.68}$$

Consider performing moment estimation using the Hadamard test or quantum power method with Hamiltonian, $\mathcal{H}$, and evolution time, h, to obtain the relevant probabilities of measuring 1. The same circuit can equally be considered as performing the method with Hamiltonian, $\mathcal{H}_\kappa$, and evolution time, h/κ. Since these two perspectives relate to the same circuit they will therefore give the same output following post-processing and amount only to different ways of viewing the same calculation. Nevertheless, we hope to gain insight into the effects of errors by comparing these viewpoints.

In each of these cases, we will ignore the algorithmic error terms Ξ_{h} and Ξ_{t} and focus on the stochastic error in the moments due to the finite number of measurements

$$\Xi_{\mathrm{r}}(r,h) = \mathcal{O}\left(\frac{1}{h^p\sqrt{r}}\right), \qquad \Xi_{\mathrm{r}}(r,h/\kappa) = \mathcal{O}\left(\frac{\kappa^p}{h^p\sqrt{r}}\right). \tag{6.69}$$

The second case has an exponentially larger stochastic error, which cancels out when the scaled moment is converted back to the original moment, $\langle\mathcal{H}^p\rangle = 1/\kappa^p\langle\mathcal{H}_\kappa^p\rangle$. However, if we instead use the scaled moments in the infimum theorem, Equation 1.29, before converting back to the original Hamiltonian, we should obtain the same results as if we had used the unscaled moments directly, i.e. we expect

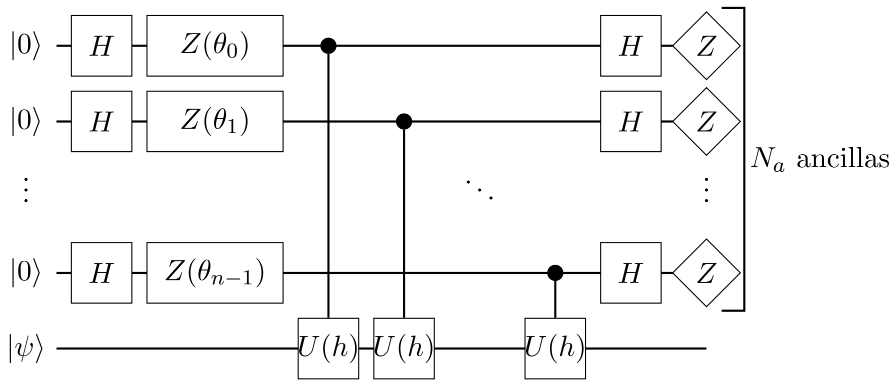
$$E_{\mathrm{inf}}^{(p)}(\mathcal{H},h) = \kappa E_{\mathrm{inf}}^{(p)}(\mathcal{H}_\kappa,h/\kappa), \tag{6.70}$$

due to the scale-invariance of the infimum formula[3]. This means that while the error in the moments being used on the right-hand side are exponentially larger than the errors on the left, the error in the final outcome is larger only by a factor of κ. This suggests that additive error may not be a meaningful way of analysing the infimum theorem and another approach, such as considering relative errors may be more informative. Further investigation will be the topic of future work.

6.2.3 Measurement recycling

When running the quantum power method circuit as given in Section 6.1.3, it is possible to compute all moments up to p by simply discarding the measurement results of unneeded ancillas, i.e. to measure the the probability of a pair of qubits both being measured in the 1 state, we can simply ignore the third, fourth etc. qubits. Interestingly, there is an ambiguity as to which ancillas to discard and it may be possible to obtain an increased number of effective circuit repetitions by considering every possible partition. Figure 6.4 illustrates this concept for the simple case of measuring qubit correlations, but to formalise this, for the quantum power method, we consider a different form of the circuit:

[3]Even if scale-invariance breaks down and Equation 6.70 does not hold, this would indicate that tuning κ could increase or decrease the error in the final output. However, we do not expect this to be the case.



where we have the freedom to assign Z-rotation angles, $\theta_j \in \{0, \pi/2\}$, to each qubit individually, in order to adjust the number of possible partitions for different moments. Note that we now have N_a ancillas instead of q. q ancillas is the minimum number required to measure the $2q$th moment, but we now want to consider the potential benefits of using more ancillas than strictly necessary.

While the number of possible partitions grows combinatorially, which naively presents a computational bottleneck to considering them all, it is possible to consider all partitions in an efficient manner as follows:

Let $a\&b$ be the bitwise product of integers a and b, let $|a|_{\mathrm{w}}$ be the hamming weight of the integer a (where the subscript w is used to avoid confusion with the absolute value of a), and let the set of all integers output by the quantum circuit be B, where each integer $b \in B$ has a corresponding frequency f_b. Let D be the integer such that the jth bit of D is 1 if and only if the jth ancilla has $\theta_j = 0$. It immediately follows that for the integer $D' = 2^n - 1 - D$, the jth bit is 1 if and only if the jth ancilla has $\theta_j = \pi/2$, where n is the number of ancillas. Then the probability, P_{2q}, required to evaluate the $2q$th moment can be computed from all possible subsets of qubits as,

$$\mathrm{P}_{2q} = \binom{|D|_w}{q}^{-1} \sum_{b\in B} f_b \binom{|b\&D|_w}{q}, \tag{6.71}$$

and the probability for the $(2q+1)$th moment is,

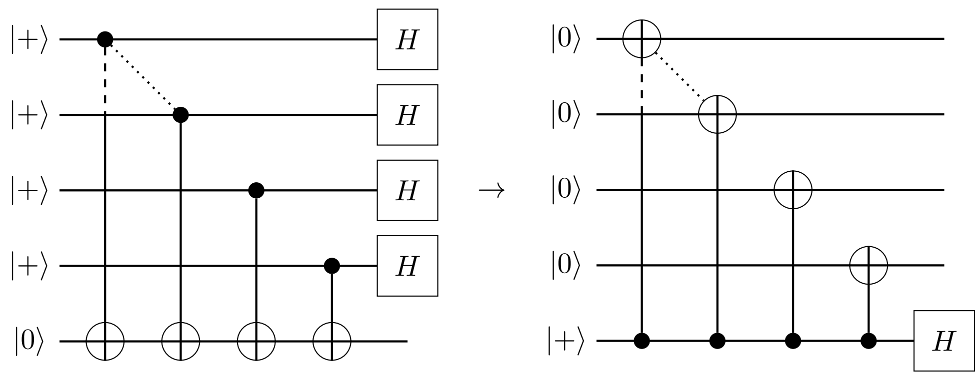
$$\mathrm{P}_{2q+1} = \frac{1}{|D'|_w} \binom{|D|_w}{q}^{-1} \sum_{b\in B} f_b |b\&D'|_w \binom{|b\&D|_w}{q}. \tag{6.72}$$

This can be seen since the term in the summation is the number of partitions for which all qubits in the partition were measured in the $|1\rangle$ state for the given bitstring, weighted by the probability of the bitstring being measured, and the prefactor is the inverse of the number of possible partitions. In other words, the probability of success is the number of successful trials divided by the total number of trials. The procedure is efficient since it does not explicitly sum over partitions but only over measured bitstrings which is upper bounded by r. To express this in another way, by performing the explicit summation over bitstrings instead of over partitions it suffices to be able to count how many possible partitions there are and we avoid needing to list out all of them.

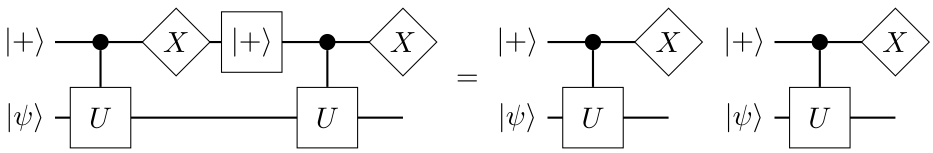
However, it is not completely clear how much additional information measurement recycling provides. Taking a simple problem, it can be shown that in the worst case, no additional information is obtained, setting the trial-state, and time-evolution as

$$\begin{aligned}|\psi\rangle &= |0\rangle,\\ U(t) &= X,\end{aligned}$$

gives the circuit:



Since the circuit produces a GHZ state, where the state of a single ancilla is maximally correlated with each other ancilla, measuring additional ancillas can never provide additional information. It is, however, not clear if this is due to violation of the small time-step limit or a situation which can occur more generally. In contrast to this, taking the case that the trial-state is an eigenstate of the Hamiltonian (and therefore also of the time evolution operator), we can write the circuit as



where resetting the trial-state has no effect since the only operation to be performed on it was the application of a global phase[4]. Therefore using measurement recycling on the left hand circuit is identical to running twice as many copies of the single-measurement circuit, and serves to increase the effective number of shots. In practice, we would hope the trial-state is close to the energy eigenstate and so this scenario would be more likely

[4]The phase applied will depend on the measurement outcome, and will not be the same as if U were applied directly with no control, but it will be global as long as $|\psi\rangle$ is an eigenstate of U.

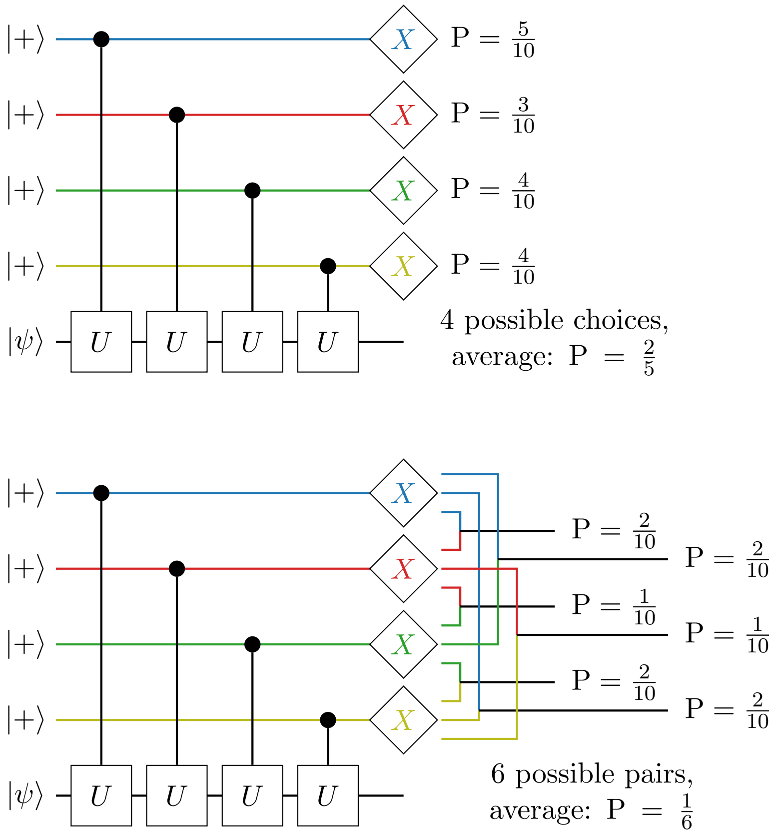


FIGURE 6.4: Measurement recycling examples. Top: Each ancilla has the same measurement probabilities as each other qubit (the listed probabilities, P, are estimated from the results of ten trials), which can be seen through symmetry by commuting the controlled operations. Therefore, the probability of an ancilla being in the 1 state might be estimated more efficiently by considering the outcomes of the other qubits. Since there are 4 qubits, the effective number of shots is increased by a factor of 4. Bottom: When measuring correlations between pairs of qubits, there are 6 possible ways to choose a pair of qubits, so the effective number of shots is increased by a factor of 6.

than the GHZ state, where the trial-state is as far from an energy eigenstate as possible. Indeed, we find in numerical simulation in Section 6.3 that measurement recycling can reduce the variance of probability estimates.

6.2.4 Classically tractable trial-states

Another improvement we can make to circuit-based moment estimation comes not in decreasing the algorithmic cost, but in increasing the resilience to hardware noise with minimal additional quantum resources through the use of classically tractable trial-states.

Essentially, to perform classical calculations, we require that both the trial-state and the operator are tractable i.e. have a polynomial sized classical representation. In the operator averaging scheme, we consider the case where the operator is tractable (since the decomposition into Pauli/fermionic strings must contain only polynomially many terms) but the trial-state is not (e.g. UCCSD states). On the other hand, with circuit based methods, we can aim to estimate higher order moments which are not necessarily tractable (since we do not need to explicitly write out the Pauli/fermionic decomposition) with respect to a state that may or may not be tractable. Note that in deciding whether an object is tractable, we are considering the asymptotic scaling of representing such an object, not the actual amount of memory required to store it.

If the trial-state used for the Hadamard test or quantum power method is tractable, we note that an error mitigation technique similar to the reference-state error mitigation described in Section 3.4.7 can be employed. We will focus on the quantum power method, noting that the extension to the Hadamard test is straightforward, and assume that finite difference, time-evolution approximation and finite sampling errors are well controlled, but that error in the implementation of the quantum circuit, e.g. qubit decoherence, imperfect gates etc, is limiting the performance of the moment estimation. Then, given a trial-state for which the energy can be computed efficiently, it is possible to determine the probability of a single ancilla qubit being measured in the $|1\rangle$ state by inverting Equation 6.31, for the specific case $2q+1=1$, as

$$\mathrm{Pr}_1(\pi/2, h) = -\frac{h}{2}\langle\mathcal{H}\rangle - \frac{1}{2}. \tag{6.73}$$

With this expected probability, $\mathrm{Pr}_{1,\mathrm{noiseless}}(\pi/2, h)$, and the noisy probability from the noisy quantum device, $\mathrm{Pr}_{1,\mathrm{noisy}}(\pi/2, h)$, we can calibrate a simple depolarising error model

$$\begin{aligned}\mathrm{Pr}_{q,\mathrm{noisy}}(\theta, h) &= (1-x)\mathrm{Pr}_{q,\mathrm{noiseless}}(\theta, h) + x\mathrm{Pr}_{q,\mathrm{mixed}}(\theta, h) \\ &= (1-x)\mathrm{Pr}_{q,\mathrm{noiseless}}(\theta, h) + \frac{x}{2^q}\end{aligned} \tag{6.74}$$

where $\mathrm{Pr}_q(\theta)$ is the probability of measuring 1 on q qubits with the corresponding rotation angle, θ, on the first qubit and x is the circuit error level (previously denoted q in Equation 3.21). Since we have both the noisy and noiseless probabilities when $q=1$, $\theta=\pi/2$ we can solve for x and use this to correct other probabilities:

$$\begin{aligned}\mathrm{Pr}_{q,\mathrm{noiseless}}(\theta, h) = &\frac{2\mathrm{Pr}_{1,\mathrm{noiseless}}(\pi/2, h) - 1}{2\mathrm{Pr}_{1,\mathrm{noisy}}(\pi/2, h) - 1}\mathrm{Pr}_{q,\mathrm{noisy}}(\theta, h) \\ &+ \frac{1}{2^q}\frac{2\mathrm{Pr}_{1,\mathrm{noisy}}(\pi/2, h) - 2\mathrm{Pr}_{1,\mathrm{noiseless}}(\pi/2, h)}{2\mathrm{Pr}_{1,\mathrm{noisy}}(\pi/2, h) - 1}.\end{aligned} \tag{6.75}$$

Note that if we are measuring all moments up to p with a single circuit, then we can use the energy computation to correct the higher-order moment estimations and the only

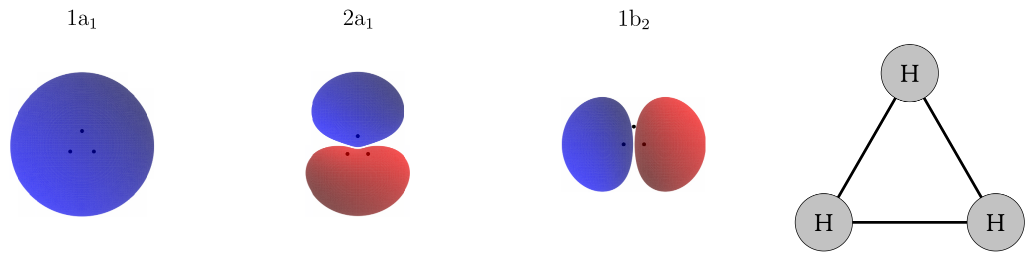


FIGURE 6.5: The three molecular orbitals of the trihydrogen cation, H_3^+. The label above each orbital refers to its symmetry – a_1 is symmetric with respect to rotation about the vertical axis, b_1 is antisymmetric. The first number in the label simply distinguishes the two a_1 orbitals. On the right is an enlarged version of the nuclear geometry.

additional quantum computational cost corresponds to an increase in variance due to the extrapolation, which can be accounted for by running additional shots. Since the model is calibrated at the same time as the computation, there will be reduced vulnerability to fluctuations in error rates. This is especially true if the ancilla qubit is measured and reset rather than using multiple ancillas, since different qubits may experience different error rates.

6.3 Numerical simulations

To validate the scaling presented above, we perform numerical simulations based on a 3-qubit representation of the tri-hydrogen cation obtained using the *QuantumSymmetry* package [181]. *QuantumSymmetry* uses symmetries of the molecule, such as rotation, reflection and spin symmetries to efficiently reduce the number of qubits required to represent a trial-state in a fixed symmetry sector. The spatial orbitals of the trihydrogen cation are plotted in Figure 6.5. In this case the six spin-orbitals are encoded into 3 qubits with the first and second qubits directly representing the occupation of the $2a_1$ spin-up and spin-down orbitals respectively, while the third qubit represents the occupation of the $1b_2$ spatial-orbital which must be either unoccupied or doubly occupied. The occupation of the $1a_1$ spin-orbitals can then be inferred from the requirement that the ground-state has a single spin-up and a single spin-down electron.

We choose our trial-state to be

$$|\psi\rangle = |00\rangle \otimes R_y(0.1)|0\rangle = \cos(0.05)|000\rangle + \sin(0.05)|001\rangle, \tag{6.76}$$

as this gives a better demonstration of the behaviour of the infimum energy-estimate than the $|000\rangle$ state (see Section 6.4).

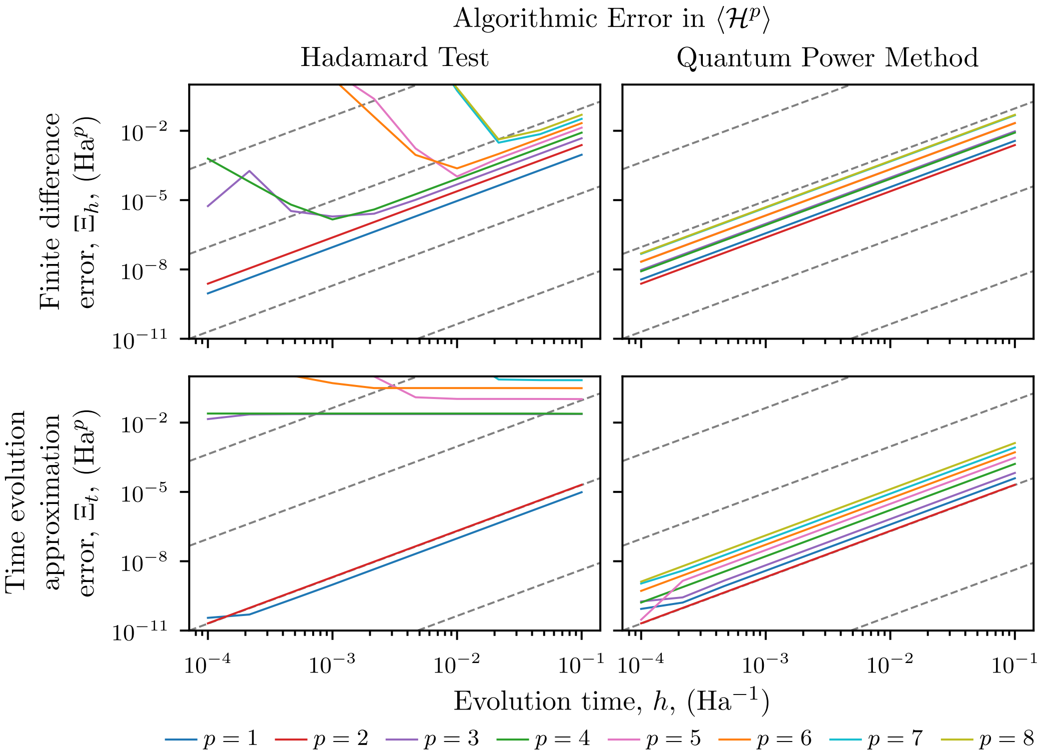


FIGURE 6.6: Algorithmic error scaling for the H_3^+ test case (top: finite difference, bottom: Trotterisation) in Hadamard test (left) and QPM (right). Each colored line indicates a different order moment as indicated in the legend, while the dotted lines represent $O(h^2)$ scaling which is the expected scaling except for the time-evolution approximation error for the Hadamard test with $p > 1$.

6.3.1 Algorithmic error

Using a single symmetric Trotter step, i.e. setting $s = 1$ in Figure 6.1, we evaluate the finite difference and time-evolution approximation errors numerically for both the Hadamard test and quantum power method circuits and compare to the analytic scaling in Figure 6.6, finding that the finite difference error initially scales as expected. However, for the Hadamard test, due to the summation over the different probabilities with alternating sign, the method begins to show the effects of floating-point precision. This is particularly noticeable for higher moments, which involve summation of more terms, with larger magnitude. This floating point error is the cause of the error diverging as the evolution time is reduced in the upper-left plot of Figure 6.6. QPM on the other hand only ever takes the difference of at most two probabilities for any moment order and so does not suffer as badly from floating point error. While the floating point error accumulation seems to render the Hadamard test method unusable, we note that this method was only proposed for evaluating low order moments [92] for which the divergence only occurs once the error is small.

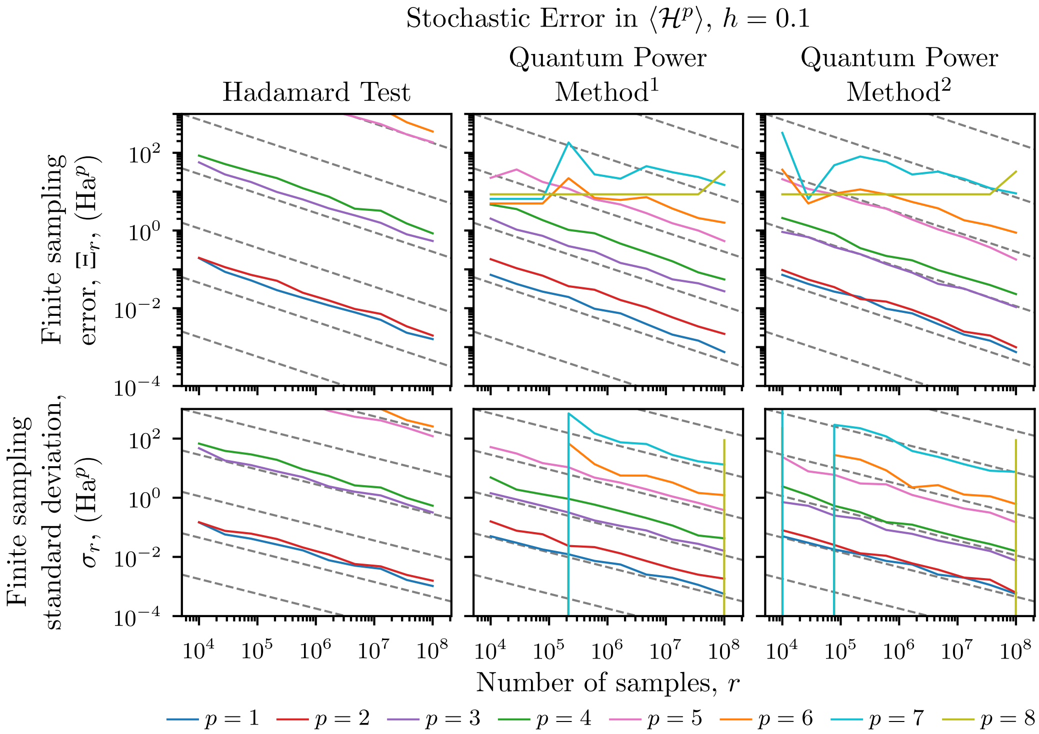


FIGURE 6.7: Stochastic error scaling for the H_3^+ test case (top: mean unsigned error, bottom: standard deviation) in Hadamard test (left) and QPM with and without measurement recycling (center and right respectively). Each coloured line indicates a different order moment as indicated in the legend, while the dashed lines represent the expected $O(1/\sqrt{r})$ scaling.

Since only first-order Trotterisation is used in the Hadamard test, we do not expect convergence of moments with $p > 2$, though convergence can be achieved using more sophisticated time-evolution approximation methods, such as higher order Suzuki-Trotter decompositions [92] or increasing the number of steps in the Trotterisation. For $p = 2$, we observe scaling with $\mathcal{O}(h^2)$ where the analytical calculation in Section 6.1.3 suggests $\mathcal{O}(h)$. In fact, this is not unexpected, since we showed in Section 6.2.1 that error terms with an odd power of h must vanish. For $p = 1$ using the Hadamard test, and for all moments using QPM, the scaling is as expected and in agreement with numerical simulations from [92].

6.3.2 Stochastic error

To investigate the scaling of stochastic error, Ξ_r, i.e. error resulting from the finite number of measurements, we sample from the statevector outputs from the Hadamard test and QPM circuits and calculate the difference between exact treatment of the probabilities and the sampled probabilities. The results are displayed in Figure 6.7 for fixed evolution time and varying number of samples.

We generally see the expected $\mathcal{O}(1/\sqrt{r})$ scaling for both the mean unsigned error (upper panels) and standard deviation (lower panels). However, when the probability approaches 0 and the number of samples is not sufficient to resolve it, the error in QPM reaches a ceiling. This occurred only for the higher order moments ($p \geq 6$), and is essentially due to the fact that the estimated probability has reached 0 and the corresponding moment estimate is therefore also 0. Then reducing the number of samples does not change the probability and also leaves the moment estimate unchanged. We can also observe the effect of measurement recycling by comparing the center and right-hand columns, finding a modest decrease in error. Note that measurement recycling is expected to have no effect on moments with $p \in \{1, 8\}$ since there is only one valid partition using 5 ancillas (with angles $\theta_0 = \pi/2$ and $\theta_j = 0$ for $j \geq 1$), while the effect is strongest for $p \in \{4, 5\}$ as these have the most valid partitions. The number of partitions can be increased through the use of more ancilla qubits (except for $p = 1$). Since the measurement recycling method is given the same samples as the non-recycling method, there should be no statistical effects between the two results, indicating the recycling method is able to extract additional information from the same set of samples. The results for higher order moments using the Hadamard test are not shown, while the data shows the stochastic errors scaling as expected, the finite precision effects render these results unimportant.

We also do not plot results for the scaling of the stochastic error as a function of h, since working with smaller h values causes the error to reach the upper bound making it difficult to observe scaling behaviour without using more samples. Whether this required increase in the number of samples will limit the utility of the method remains to be seen, but improvements such as Richardson extrapolation (Section 6.2.1) and measurement recycling (Section 6.2.3) may help offset this.

6.4 Converting moments to ground-state energy estimates

While it is highly non-trivial to show an analytical error scaling of the infimum theorem, Equation 1.29, that converts moments to corrected ground-state energy estimates, it is possible to derive a scaling relation for the error in the ground-state energy estimate from the related Lanczos diagonalisation technique [76], a simplified version of which was discussed in Section 1.3. The error can be bounded by [78]

$$\xi_{\text{Lanczos}} \leq \frac{1-\eta^2}{\eta^2} \frac{E_{N-1} - E_0}{T_p\left(1 + 2\frac{E_1 - E_0}{E_{N-1} - E_1}\right)}, \tag{6.77}$$

where $\eta = |\langle\psi|\phi_0\rangle|$ is the magnitude of the overlap between the trial-state, $|\psi\rangle$, and the true ground-state, $|\phi_0\rangle$, E_j is the energy of the jth excited state, $N = 2^n$ (where n is the number of qubits in the system) is the size of the Hilbert space, p is the number of Lanczos iterations (and will be related to the number of moments we use) and T_p is the pth Chebyshev polynomial of the first kind.

Under the assumption that the number of Lanczos iterations is large, $p \gg 0$. (i.e. in the asymptotic limit) and that the first excitation energy is small compared to the extent of the energy spectrum

$$E_{0,1} \equiv E_1 - E_0 \qquad \ll \qquad E_{0,N-1} \equiv E_{N-1} - E_0 \qquad \approx \qquad E_{N-1} - E_1, \tag{6.78}$$

then the error can be simplified to

$$\xi_{\text{Lanczos}} = \mathcal{O}\left(\frac{1-\eta^2}{\eta^2} E_{0,N-1}\left(1 + 2\sqrt{\frac{E_{0,1}}{E_{0,N-1}}}\right)^{-p}\right). \tag{6.79}$$

Numerically, it has been seen that the infimum theorem obeys a similar error scaling [182]. Bounding this by a target error, ε, and assuming the overlap with the ground-state is small gives

$$\begin{aligned} p &= \mathcal{O}\left(\frac{\log\left(\frac{1-\eta^2}{\eta^2}\frac{E_{0,N-1}}{\varepsilon}\right)}{\log\left(1 + 2\sqrt{\frac{E_{0,1}}{E_{0,N-1}}}\right)}\right) \\ &= \mathcal{O}\left(\sqrt{\frac{E_{0,N-1}}{E_{0,1}}}\log\frac{E_{0,N-1}}{\varepsilon\eta^2}\right), \end{aligned} \tag{6.80}$$

indicating that to achieve convergence the required number of moments is logarithmic in the inverse precision and inverse overlap. However, this only applies empirically and does not consider the error in the moments.

To investigate the use of approximate moments, under the effects of both finite difference and Trotter error, we use the moments from Figure 6.6 to obtain ground-state energy estimates from the infimum theorem and plot the error relative to the true ground-state energy. The error plotted in Figure 6.8 is therefore the combination of error due to finite difference and Trotterisation which decrease as the finite-difference step-size is reduced, leading to convergence towards the exact infimum theorem result, with remaining error

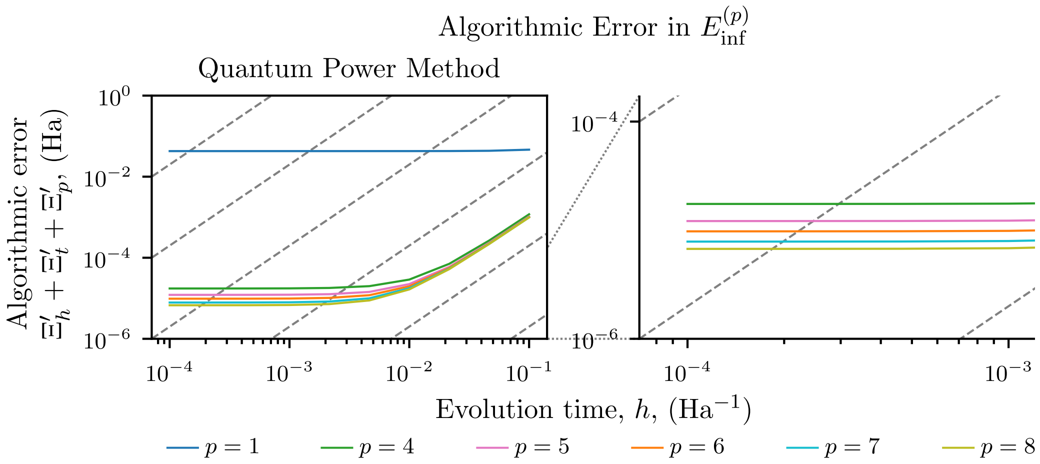


FIGURE 6.8: Algorithmic error scaling in QPM for the H_3^+ test case, i.e. finite difference (Ξ'_{h}), time-evolution approximation (Ξ'_{t}) and ground-state overlap (Ξ'_{p}) errors. Note that the prime indicates that these are errors in $E_{\mathrm{inf}}^{(p)}$, not in $\langle \mathcal{H}^p \rangle$. Each coloured line indicates a different maximum order moment in the cluster expansion as indicated in the legend, the dotted lines represent $O(1/h^2)$, which is the error scaling in the moments. The right hand plot is an enlarged view of the lower-left section of the data, showing the improvement due to increasing moment order. Note that the infimum theorem does not converge for $p = 2$ and $p = 3$ and so is not shown.

due to the limited ground-state overlap. Only results for the quantum power method are shown, since the low level Trotterisation prevents the Hadamard test from converging. Note that the trial-state (Equation 6.76) was specifically chosen to have high overlap with the ground-state, $|\langle \psi_{\mathrm{trial}} | \psi_{\mathrm{ground}} \rangle| \approx 0.9916$. Without the small rotation on the third qubit the state has better overlap, $|\langle \psi_{\mathrm{trial}} | \psi_{\mathrm{ground}} \rangle| \approx 0.9929$, but the 6th order infimum estimate does not exhibit a stationary point or inflection point. The rotation is included to better demonstrate convergence as the moment order is increased.

At large evolution times for the higher order ($p \geq 4$) estimates we observe scaling as $\mathcal{O}(1/h^2)$, while for shorter evolution times we see that different order estimates converge to different levels of error based on how much of the missing ground-state overlap they account for. Note that the first order estimate, $p = 1$ (which also corresponds to the trial-state energy), is already dominated by the missing trial-state overlap even at $h = 0.1$.

The right-hand plot of Figure 6.8 is a magnified view of the convergence of the higher order moments, showing a roughly exponential convergence as expected [78, 182], however higher order infimum estimates are required to draw any concrete conclusions.

6.5 A brief comparison to quantum phase estimation

Given the preceding analysis of the quantum power method, it seems obvious to ask how this compares to quantum phase estimation. While a full scaling analysis is beyond the scope of this thesis, we can consider the toy problem introduced in Section 6.3, for which we already have numerical results. Given an evolution time of 0.1, the fourth order infimum estimate reaches an algorithmic error of 10^{-3} hartree (from Figure 6.8), the precise question we would like to answer is then; what are the resource requirements to reach this precision with quantum phase estimation? Firstly, we note that the conversion from the binary output, k, of QPE using N_a ancillas, to an energy value, E_k, is given by

$$E_k = \frac{2\pi}{2^{N_a} t} k, \tag{6.81}$$

when the unitary from Section 1.2.4 is replaced by

$$U \rightarrow e^{-i\mathcal{H}t}. \tag{6.82}$$

The parameter t is a scaling factor, from a physics perspective it has units of time (inverse energy). A large value of t decreases the distance between E_k and E_{k+1} effectively increasing the resolution of QPE by wrapping the accumulated phase around more of the unit circle. On the other hand, if t is too large, the phases can loop around the entire unit circle and become indistinguishable. To avoid this, an upper bound based on the width of the energy spectrum must be placed on t. Here, since we wish to compare resource estimates against the quantum power method, we set

$$\begin{aligned} t &= \frac{2\pi}{\Delta}, \\ \Delta &= E_{\text{max}} - E_{\text{min}}, \end{aligned} \tag{6.83}$$

where E_{min} (E_{max}) is the minimum (maximum) eigenvalue for which the trial-state has a non-zero overlap with the corresponding eigenstate. In this case the overlaps, computed by diagonalising the Hamiltonian matrix, are 98.6% and 1.4% respectively and $\Delta = 1.72$. We note that this choice of t is not practical for two reasons: Firstly, the exact eigenvalues and eigenstates would not normally be known a priori and an approximate bound would be required instead. Secondly, the chosen value causes the spectrum to wrap exactly once around the unit circle and therefore the endpoints (i.e. E_{min} and E_{max}) overlap. With reference to Equation 6.81, this means that increasing the energy by Δ requires increasing the measurement output by 2^{N_a} which cannot be recorded on the N_a ancillas. However, as we wish to lower bound the resource cost of

QPE we can accept both these concessions and simply note that the analysis favours QPE over QPM.

The number of ancillas required to achieve milliHartree resolution can then be computed from

$$\begin{aligned}
(E_{k+1} - E_k) &= \frac{2\pi}{2^{N_a} t} \\
10^{-3} &= \frac{\Delta}{2^{N_a}} \\
N_a &= \log_2\left(\frac{\Delta}{10^{-3}}\right) \\
&= 10.
\end{aligned} \tag{6.84}$$

In practice, since the ground-state energy will not have a finite binary expansion, the resulting bitstrings will be distributed about the string corresponding to the true eigenvalue. To account for this we could use additional qubits so that nearby strings will also give energies within the allowed tolerance. Here, to be favourable to QPE, we simply assume the eigenenergy has an exact binary expansion.

Already we find that the Quantum Power Method results in shorter circuits than Quantum Phase Estimation, since for QPM we require 3 (symmetric) Trotter steps to estimate the first five moments (one with an ancilla angle $\theta = \pi/2$ and two with angles $\theta = 0$) while QPE requires at least one Trotter step for each ancilla, for a lower bound of at least 10 steps. Additionally, once Trotter error and the significantly longer evolution times for QPE[5] are considered, the QPM circuits will be much shorter than the QPE circuits with the same target accuracy. Assuming one Trotter step per time t the total number of steps in QPE is 2047. However, heuristics suggest that to bound the accumulated error by 10^{-3} requires at least three steps, resulting in a total of 6141 steps. This suggests the QPE circuit required to reach millihartree accuracy for this problem would be between 600 and 2000 times deeper than the QPM circuit.

Of course, the reduced circuit depth does not come free. To bound statistical sampling error in the first moment by 10^{-3} would require around 10^8 to 10^9 shots, based on extrapolating the plots in Figure 6.7. While Chapter 2 and Chapter 3 have shown that the infimum theorem is much more robust to shot noise than the corresponding first moment, further investigation is required to confirm that this holds for circuit-based methods as it does for operator averaging. As such, developing techniques to reduce the required number of measurements is crucial to the viability of these methods. Conversely, since there is a large overlap between the trial- and ground-states, we expect QPE to succeed with a probability of 98% with only a single shot.

[5]recall that the jth ancilla requires application of U^{2^j}, which is equivalent to evolution for time $2^j t$.

6.6 Summary

While the operator averaging method employed in previous chapters provides a compelling proof-of-principle, for the estimation of ground-state energies via Hamiltonian moments, due to the large number of measurements required and the classical pre-computational cost, circuit based methods for evaluation of Hamiltonian moments seem a promising alternative. Here, we analysed the Hadamard test [83, 92] and quantum power method techniques [92] to obtain expected resource scaling as summarised in Table 6.1. We then validated this scaling through numerical simulation using the trihydrogen cation represented using three qubits. Since our simulations used only first order Trotter, the Hadamard test did not display convergence for $p > 2$, though this could be achieved by increasing the number of Trotter steps as the target accuracy decreases, or with higher order Suziki-Trotter formulae. While no analytic expression is known for the scaling of the error in the infimum theorem, as the evolution time is decreased the estimate appears to converge to its zero-error value as $\mathcal{O}(h^2)$ with the convergence value being determined by the trial-state and the order at which the cluster expansion is truncated.

Further investigation is needed into the behaviour of the moments and infimum estimate in the presence of sampling error, in particular accounting for Richardson extrapolation and measurement recycling schemes in order to offset the high cost of evaluating expectation values at small evolution times. Once the scaling of the infimum estimate is better understood it may be possible to make comparisons to other methods from the literature such as quantum phase estimation [7] or other ground-state energy estimation techniques such as [183–185], among others, though initial comparisons suggest a significant circuit depth saving compared to textbook QPE, offset by a much higher number of circuit repetitions.

Chapter 7

Conclusion

While traditional computing methods struggle to simulate complex quantum systems, it is expected that fault-tolerant quantum computation will provide an algorithmic speed-up in this domain. What is less clear is whether near-term quantum devices can provide any such advantage. Initial implementations of variational quantum algorithms were limited by the trade-off between trial-circuit complexity and noise in the quantum device, leading to the development of error mitigation techniques. However, even with error mitigation, the effectiveness of the variational quantum eigensolver was still limited by the expressivity of the trial-state. Of particular note was the density matrix purification technique which provided a high degree of resilience to noise, but restricted the trial-state to single Slater-determinant states. However, in Chapter 2 we saw that computation of Hamiltonian moments from the quantum processor can extend the reach of variational quantum algorithms beyond the limit imposed by the trial-state. Additionally, we showed empirically that the quantum-computed moments method already possesses a high degree of noise robustness which complements error mitigation techniques. In Chapter 3, we extended the application of the moments-based correction to more expressive trial-states over more qubits leading to additional sources of noise and necessitating more general error mitigation schemes than density matrix purification. In particular, the reference-state error mitigation technique was found to be highly effective. These results were the largest quantum chemical calculation (in number of qubits) performed on quantum hardware to reach millihartree precision of the FCI result, though they have since been exceeded by sample-based quantum diagonalisation methods [44, 45].

Though ground-state energies are important for quantum chemistry, they are not the sole quantity of interest. As such Chapter 4 considered how to compute other ground-state properties, using the permanent electric dipole moment of water as an example. Unlike previous quantum computations of molecular properties, our technique is not limited by the trial-state expressivity due to the Hamiltonian moments effectively generating a subspace around the trial-state. While we are limited by the minimal basis we employ, our result is much closer to this limit than the direct evaluation of the dipole moment. Likewise, Chapter 5 considered the computation of molecular forces and was able to generate a potential energy surface and corresponding forces based on results computed at a single nuclear geometry. While further work is needed to account for inconsistencies in the identification of molecular orbitals, this could be a promising technique for performing geometry optimisation on a quantum computer.

Finally, we examined the analytical scaling of moments estimation methods in Chapter 6 for the term-by-term computation used in Chapters 2 and 3 and for proposed circuit-based techniques. Notably, Richardson extrapolation seems to be necessary to obtain convergence of the moments to additive precision, though there are further questions to be answered, particularly relating to the measurement recycling scheme and the actual precision to which we require the moments for the infimum theorem. The analytical scaling of the algorithmic (finite difference and time-evolution approximation) errors and stochastic (finite sampling) errors was verified numerically using a simple three-qubit electronic-structure Hamiltonian. Further investigation is needed into the convergence of the infimum estimate, computed with noisy moments, to the zero-noise limit, following which it may be possible to compare resource estimates to other ground-state energy estimation techniques in the asymptotic case. On the other hand, if this analysis proves difficult, it would still be possible to benchmark methods numerically on small-scale systems as was performed for the three qubit representation of the tri-hydrogen cation, indicating a significant circuit depth reduction at the cost of a large increase to the number of circuit repetitions and overall runtime in this specific case.

Bibliography

[1] Stefan Heinen, Max Schwilk, Guido Falk von Rudorff, and O Anatole von Lilienfeld. Machine learning the computational cost of quantum chemistry. *Machine Learning: Science and Technology*, 1(2):025002, 2020. URL https://dx.doi.org/10.1088/2632-2153/ab6ac4.

[2] Richard P. Feynman. Simulating physics with computers. *International Journal of Theoretical Physics*, 21(6):467–488, 1982. URL https://doi.org/10.1007/BF02650179.

[3] Michael A. Nielsen and Isaac L. Chuang. *Quantum Computation and Quantum Information: 10th Anniversary Edition*. Cambridge University Press, 2010.

[4] P. Jordan and E. Wigner. Über das Paulische Äquivalenzverbot. *Zeitschrift für Physik*, 47(9):631–651, 1928. URL https://doi.org/10.1007/BF01331938.

[5] Sergey B. Bravyi and Alexei Yu. Kitaev. Fermionic quantum computation. *Annals of Physics*, 298(1):210–226, 2002. ISSN 0003-4916. URL https://www.sciencedirect.com/science/article/pii/S0003491602962548.

[6] Jacob T. Seeley, Martin J. Richard, and Peter J. Love. The Bravyi-Kitaev transformation for quantum computation of electronic structure. *The Journal of Chemical Physics*, 137(22):224109, 2012. ISSN 0021-9606. URL https://doi.org/10.1063/1.4768229.

[7] Alexei Y. Kitaev. Quantum measurements and the abelian stabilizer problem. *Electron. Colloquium Comput. Complex.*, TR96, 1995. URL https://eccc.weizmann.ac.il/eccc-reports/1996/TR96-003/index.html.

[8] Alberto Peruzzo, Jarrod McClean, Peter Shadbolt, Man-Hong Yung, Xiao-Qi Zhou, Peter J. Love, Alán Aspuru-Guzik, and Jeremy L. O'Brien. A variational eigenvalue solver on a photonic quantum processor. *Nature Communications*, 5 (1):4213, 2014. URL https://doi.org/10.1038/ncomms5213.

[9] Mario Motta and Julia E. Rice. Emerging quantum computing algorithms for quantum chemistry. *WIREs Computational Molecular Science*, 12(3):e1580, 2022. doi: https://doi.org/10.1002/wcms.1580. URL https://wires.onlinelibrary.wiley.com/doi/abs/10.1002/wcms.1580.

[10] Austin G. Fowler, Ashley M. Stephens, and Peter Groszkowski. High-threshold universal quantum computation on the surface code. *Phys. Rev. A*, 80:052312, 2009. URL https://link.aps.org/doi/10.1103/PhysRevA.80.052312.

[11] David S. Wang, Austin G. Fowler, and Lloyd C. L. Hollenberg. Surface code quantum computing with error rates over 1%. *Phys. Rev. A*, 83:020302, 2011. URL https://link.aps.org/doi/10.1103/PhysRevA.83.020302.

[12] Seth Lloyd. Universal quantum simulators. *Science*, 273(5278):1073–1078, 1996. URL https://www.science.org/doi/abs/10.1126/science.273.5278.1073.

[13] Markus Reiher, Nathan Wiebe, Krysta M. Svore, Dave Wecker, and Matthias Troyer. Elucidating reaction mechanisms on quantum computers. *PNAS*, 114(29): 7555–7560, 2017. ISSN 0027-8424. URL https://www.pnas.org/content/114/29/7555.

[14] John Preskill. Quantum Computing in the NISQ era and beyond. *Quantum*, 2:79, 2018. ISSN 2521-327X. URL https://doi.org/10.22331/q-2018-08-06-79.

[15] Jarrod R. McClean, Sergio Boixo, Vadim N. Smelyanskiy, Ryan Babbush, and Hartmut Neven. Barren plateaus in quantum neural network training landscapes. *Nat. Commun.*, 9(1):4812, 2018. URL https://doi.org/10.1038/s41467-018-07090-4.

[16] P. J. J. O'Malley, R. Babbush, I. D. Kivlichan, J. Romero, J. R. McClean, R. Barends, J. Kelly, P. Roushan, A. Tranter, N. Ding, B. Campbell, Y. Chen, Z. Chen, B. Chiaro, A. Dunsworth, A. G. Fowler, E. Jeffrey, E. Lucero, A. Megrant, J. Y. Mutus, M. Neeley, C. Neill, C. Quintana, D. Sank, A. Vainsencher, J. Wenner, T. C. White, P. V. Coveney, P. J. Love, H. Neven, A. Aspuru-Guzik, and J. M. Martinis. Scalable quantum simulation of molecular energies. *Phys. Rev. X*, 6: 031007, 2016. URL https://link.aps.org/doi/10.1103/PhysRevX.6.031007.

[17] Abhinav Kandala, Antonio Mezzacapo, Kristan Temme, Maika Takita, Markus Brink, Jerry M. Chow, and Jay M. Gambetta. Hardware-efficient variational quantum eigensolver for small molecules and quantum magnets. *Nature*, 549(7671): 242–246, 2017. URL https://doi.org/10.1038/nature23879.

[18] Yangchao Shen, Xiang Zhang, Shuaining Zhang, Jing-Ning Zhang, Man-Hong Yung, and Kihwan Kim. Quantum implementation of the unitary coupled cluster

for simulating molecular electronic structure. *Phys. Rev. A*, 95:020501, 2017. URL https://link.aps.org/doi/10.1103/PhysRevA.95.020501.

[19] Alexander J. McCaskey, Zachary P. Parks, Jacek Jakowski, Shirley V. Moore, Titus D. Morris, Travis S. Humble, and Raphael C. Pooser. Quantum chemistry as a benchmark for near-term quantum computers. *npj Quantum Information*, 5 (1):99, 2019. URL https://doi.org/10.1038/s41534-019-0209-0.

[20] Google AI Quantum and Collaborators. Hartree-Fock on a superconducting qubit quantum computer. *Science*, 369(6507):1084–1089, 2020. ISSN 0036-8075. URL https://science.sciencemag.org/content/369/6507/1084.

[21] Philippe Suchsland, Francesco Tacchino, Mark H. Fischer, Titus Neupert, Panagiotis Kl. Barkoutsos, and Ivano Tavernelli. Algorithmic error mitigation scheme for current quantum processors. *Quantum*, 5:492, 2021. URL https://doi.org/10.22331/q-2021-07-01-492.

[22] Shaojun Guo, Jinzhao Sun, Haoran Qian, Ming Gong, Yukun Zhang, Fusheng Chen, Yangsen Ye, Yulin Wu, Sirui Cao, Kun Liu, Chen Zha, Chong Ying, Qingling Zhu, He-Liang Huang, Youwei Zhao, Shaowei Li, Shiyu Wang, Jiale Yu, Daojin Fan, Dachao Wu, Hong Su, Hui Deng, Hao Rong, Yuan Li, Kaili Zhang, Tung-Hsun Chung, Futian Liang, Jin Lin, Yu Xu, Lihua Sun, Cheng Guo, Na Li, Yong-Heng Huo, Cheng-Zhi Peng, Chao-Yang Lu, Xiao Yuan, Xiaobo Zhu, and Jian-Wei Pan. Experimental quantum computational chemistry with optimized unitary coupled cluster ansatz. *Nature Physics*, 20(8):1240–1246, 2024. URL https://doi.org/10.1038/s41567-024-02530-z.

[23] Yunseong Nam, Jwo-Sy Chen, Neal C. Pisenti, Kenneth Wright, Conor Delaney, Dmitri Maslov, Kenneth R. Brown, Stewart Allen, Jason M. Amini, Joel Apisdorf, Kristin M. Beck, Aleksey Blinov, Vandiver Chaplin, Mika Chmielewski, Coleman Collins, Shantanu Debnath, Kai M. Hudek, Andrew M. Ducore, Matthew Keesan, Sarah M. Kreikemeier, Jonathan Mizrahi, Phil Solomon, Mike Williams, Jaime David Wong-Campos, David Moehring, Christopher Monroe, and Jungsang Kim. Ground-state energy estimation of the water molecule on a trapped-ion quantum computer. *npj Quantum Information*, 6(1):33, 2020. URL https://doi.org/10.1038/s41534-020-0259-3.

[24] Andrew Eddins, Mario Motta, Tanvi P. Gujarati, Sergey Bravyi, Antonio Mezzacapo, Charles Hadfield, and Sarah Sheldon. Doubling the size of quantum simulators by entanglement forging. *PRX Quantum*, 3:010309, 2022. URL https://link.aps.org/doi/10.1103/PRXQuantum.3.010309.

[25] Mario Motta, Gavin O. Jones, Julia E. Rice, Tanvi P. Gujarati, Rei Sakuma, Ieva Liepuoniute, Jeannette M. Garcia, and Yu-ya Ohnishi. Quantum chemistry simulation of ground- and excited-state properties of the sulfonium cation on a superconducting quantum processor. *Chem. Sci.*, 14:2915–2927, 2023. doi: 10.1039/D2SC06019A. URL http://dx.doi.org/10.1039/D2SC06019A.

[26] Ieva Liepuoniute, Mario Motta, Thaddeus Pellegrini, Julia E. Rice, Tanvi P. Gujarati, Sofia Gil, and Gavin O. Jones. Simulation of a diels–alder reaction on a quantum computer. *Phys. Chem. Chem. Phys.*, 26:25181–25191, 2024. doi: 10.1039/D4CP01314J. URL http://dx.doi.org/10.1039/D4CP01314J.

[27] T. E. O'Brien, G. Anselmetti, F. Gkritsis, V. E. Elfving, S. Polla, W. J. Huggins, O. Oumarou, K. Kechedzhi, D. Abanin, R. Acharya, et al. Purification-based quantum error mitigation of pair-correlated electron simulations. *Nat. Phys.*, 19 (12):1787–1792, 2023. URL https://doi.org/10.1038/s41567-023-02240-y.

[28] Luning Zhao, Joshua Goings, Kyujin Shin, Woomin Kyoung, Johanna I. Fuks, June-Koo Kevin Rhee, Young Min Rhee, Kenneth Wright, Jason Nguyen, Jungsang Kim, and Sonika Johri. Orbital-optimized pair-correlated electron simulations on trapped-ion quantum computers. *Npj Quantum Inf.*, 9(1):60, 2023. URL https://doi.org/10.1038/s41534-023-00730-8.

[29] Tim Weaving, Alexis Ralli, Peter J. Love, Sauro Succi, and Peter V. Coveney. Contextual subspace variational quantum eigensolver calculation of the dissociation curve of molecular nitrogen on a superconducting quantum computer. *npj Quantum Information*, 11(1):25, Feb 2025. ISSN 2056-6387. doi: 10.1038/s41534-024-00952-4. URL https://doi.org/10.1038/s41534-024-00952-4.

[30] Werner Dobrautz, Igor O. Sokolov, Ke Liao, Pablo López Ríos, Martin Rahm, Ali Alavi, and Ivano Tavernelli. Toward real chemical accuracy on current quantum hardware through the transcorrelated method. *Journal of Chemical Theory and Computation*, 20(10):4146–4160, May 2024. ISSN 1549-9618. doi: 10.1021/acs.jctc.4c00070. URL https://doi.org/10.1021/acs.jctc.4c00070.

[31] Tanvi P. Gujarati, Mario Motta, Triet Nguyen Friedhoff, Julia E. Rice, Nam Nguyen, Panagiotis Kl. Barkoutsos, Richard J. Thompson, Tyler Smith, Marna Kagele, Mark Brei, Barbara A. Jones, and Kristen Williams. Quantum computation of reactions on surfaces using local embedding. *npj Quantum Information*, 9(1), September 2023. ISSN 2056-6387. doi: 10.1038/s41534-023-00753-1. URL http://dx.doi.org/10.1038/s41534-023-00753-1.

[32] Róbert Izsák, Christoph Riplinger, Nick S. Blunt, Bernardo de Souza, Nicole Holzmann, Ophelia Crawford, Joan Camps, Frank Neese, and Patrick Schopf. Quantum computing in pharma: A multilayer embedding approach for near future applications. *Journal of Computational Chemistry*, 44(3):406–421, 2023. doi: https://doi.org/10.1002/jcc.26958. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/jcc.26958.

[33] Huan Ma, Jie Liu, Honghui Shang, Yi Fan, Zhenyu Li, and Jinlong Yang. Multiscale quantum algorithms for quantum chemistry. *Chem. Sci.*, 14:3190–3205, 2023. doi: 10.1039/D2SC06875C. URL http://dx.doi.org/10.1039/D2SC06875C.

[34] Max Rossmannek, Fabijan Pavošević, Angel Rubio, and Ivano Tavernelli. Quantum embedding method for the simulation of strongly correlated systems on quantum computers. *J. Phys. Chem. Lett.*, 14(14):3491–3497, 2023. URL https://doi.org/10.1021/acs.jpclett.3c00330.

[35] Raffaele Santagati, Alan Aspuru-Guzik, Ryan Babbush, Matthias Degroote, Leticia González, Elica Kyoseva, Nikolaj Moll, Markus Oppel, Robert M. Parrish, Nicholas C. Rubin, Michael Streif, Christofer S. Tautermann, Horst Weiss, Nathan Wiebe, and Clemens Utschig-Utschig. Drug design on quantum computers. *Nature Physics*, 20(4):549–557, Apr 2024. ISSN 1745-2481. doi: 10.1038/s41567-024-02411-5. URL https://doi.org/10.1038/s41567-024-02411-5.

[36] Akhil Shajan, Danil Kaliakin, Abhishek Mitra, Javier Robledo Moreno, Zhen Li, Mario Motta, Caleb Johnson, Abdullah Ash Saki, Susanta Das, Iskandar Sitdikov, Antonio Mezzacapo, and Kenneth M. Merz Jr. Towards quantum-centric simulations of extended molecules: sample-based quantum diagonalization enhanced with density matrix embedding theory, 2024. URL https://arxiv.org/abs/2411.09861.

[37] Patrick Ettenhuber, Mads Bøttger Hansen, Pier Paolo Poier, Irfansha Shaik, Stig Elkjaer Rasmussen, Niels Kristian Madsen, Marco Majland, Frank Jensen, Lars Olsen, and Nikolaj Thomas Zinner. Calculating the energy profile of an enzymatic reaction on a quantum computer. *Journal of Chemical Theory and Computation*, Mar 2025. ISSN 1549-9618. doi: 10.1021/acs.jctc.5c00022. URL https://doi.org/10.1021/acs.jctc.5c00022.

[38] Renke Huang, Chenyang Li, and Francesco A. Evangelista. Leveraging small-scale quantum computers with unitarily downfolded hamiltonians. *PRX Quantum*, 4:020313, Apr 2023. doi: 10.1103/PRXQuantum.4.020313. URL https://link.aps.org/doi/10.1103/PRXQuantum.4.020313.

[39] Alessandro Tammaro, Davide E. Galli, Julia E. Rice, and Mario Motta. N-electron valence perturbation theory with reference wave functions from quantum computing: Application to the relative stability of hydroxide anion and hydroxyl radical. *The Journal of Physical Chemistry A*, 127(3):817–827, 2023. doi: 10.1021/acs.jpca.2c07653. URL https://doi.org/10.1021/acs.jpca.2c07653. PMID: 36638358.

[40] Daniel Yoffe, Noga Entin, Amir Natan, and Adi Makmal. A qubit-efficient variational selected configuration-interaction method. *Quantum Science and Technology*, 10(1):015020, 2024. URL https://dx.doi.org/10.1088/2058-9565/ad7d32.

[41] Keita Kanno, Masaya Kohda, Ryosuke Imai, Sho Koh, Kosuke Mitarai, Wataru Mizukami, and Yuya O. Nakagawa. Quantum-selected configuration interaction: classical diagonalization of hamiltonians in subspaces selected by quantum computers, 2023. URL https://arxiv.org/abs/2302.11320.

[42] Javier Robledo-Moreno, Mario Motta, Holger Haas, Ali Javadi-Abhari, Petar Jurcevic, William Kirby, Simon Martiel, Kunal Sharma, Sandeep Sharma, Tomonori Shirakawa, Iskandar Sitdikov, Rong-Yang Sun, Kevin J. Sung, Maika Takita, Minh C. Tran, Seiji Yunoki, and Antonio Mezzacapo. Chemistry beyond exact solutions on a quantum-centric supercomputer, 2024. URL https://arxiv.org/abs/2405.05068.

[43] Danil Kaliakin, Akhil Shajan, Javier Robledo Moreno, Zhen Li, Abhishek Mitra, Mario Motta, Caleb Johnson, Abdullah Ash Saki, Susanta Das, Iskandar Sitdikov, Antonio Mezzacapo, and Kenneth M. Merz Jr. Accurate quantum-centric simulations of supramolecular interactions, 2024. URL https://arxiv.org/abs/2410.09209.

[44] Ieva Liepuoniute, Kirstin D. Doney, Javier Robledo-Moreno, Joshua A. Job, Will S. Friend, and Gavin O. Jones. Quantum-centric study of methylene singlet and triplet states, 2024. URL https://arxiv.org/abs/2411.04827.

[45] Alan Duriez, Pamela C. Carvalho, Marco Antonio Barroca, Federico Zipoli, Ben Jaderberg, Rodrigo Neumann Barros Ferreira, Kunal Sharma, Antonio Mezzacapo, Benjamin Wunsch, and Mathias Steiner. Computing band gaps of periodic materials via sample-based quantum diagonalization, 2025. URL https://arxiv.org/abs/2503.10901.

[46] Aidan Pellow-Jarman, Shane McFarthing, Doo Hyung Kang, Pilsun Yoo, Eyuel Eshetu Elala, Rowan Pellow-Jarman, P. Mai Nakliang, Jaewan Kim, and

June-Koo Kevin Rhee. Hivqe: Handover iterative variational quantum eigensolver for efficient quantum chemistry calculations, 2025. URL https://arxiv.org/abs/2503.06292.

[47] Nobuyuki Yoshioka, Mirko Amico, William Kirby, Petar Jurcevic, Arkopal Dutt, Bryce Fuller, Shelly Garion, Holger Haas, Ikko Hamamura, Alexander Ivrii, Ritajit Majumdar, Zlatko Minev, Mario Motta, Bibek Pokharel, Pedro Rivero, Kunal Sharma, Christopher J. Wood, Ali Javadi-Abhari, and Antonio Mezzacapo. Diagonalization of large many-body hamiltonians on a quantum processor, 2024. URL https://arxiv.org/abs/2407.14431.

[48] Manolo C. Per, Nathan Rhodes, Maiyuren Srikumar, and Joshua W. Dai. Chemically-accurate prediction of the ionisation potential of helium using a quantum processor, 2025. URL https://arxiv.org/abs/2502.02023.

[49] B. P. Lanyon, J. D. Whitfield, G. G. Gillett, M. E. Goggin, M. P. Almeida, I. Kassal, J. D. Biamonte, M. Mohseni, B. J. Powell, M. Barbieri, A. Aspuru-Guzik, and A. G. White. Towards quantum chemistry on a quantum computer. *Nature Chemistry*, 2(2):106–111, 2010. URL https://doi.org/10.1038/nchem.483.

[50] Jiangfeng Du, Nanyang Xu, Xinhua Peng, Pengfei Wang, Sanfeng Wu, and Dawei Lu. Nmr implementation of a molecular hydrogen quantum simulation with adiabatic state preparation. *Phys. Rev. Lett.*, 104:030502, 2010. URL https://link.aps.org/doi/10.1103/PhysRevLett.104.030502.

[51] S. Paesani, A. A. Gentile, R. Santagati, J. Wang, N. Wiebe, D. P. Tew, J. L. O'Brien, and M. G. Thompson. Experimental bayesian quantum phase estimation on a silicon photonic chip. *Phys. Rev. Lett.*, 118:100503, 2017. URL https://link.aps.org/doi/10.1103/PhysRevLett.118.100503.

[52] Zhaokai Li, Xiaomei Liu, Hefeng Wang, Sahel Ashhab, Jiangyu Cui, Hongwei Chen, Xinhua Peng, and Jiangfeng Du. Quantum simulation of resonant transitions for solving the eigenproblem of an effective water hamiltonian. *Phys. Rev. Lett.*, 122:090504, 2019. URL https://link.aps.org/doi/10.1103/PhysRevLett.122.090504.

[53] Nick S. Blunt, Laura Caune, Róbert Izsák, Earl T. Campbell, and Nicole Holzmann. Statistical phase estimation and error mitigation on a superconducting quantum processor. *PRX Quantum*, 4:040341, 2023. URL https://link.aps.org/doi/10.1103/PRXQuantum.4.040341.

[54] Kentaro Yamamoto, Samuel Duffield, Yuta Kikuchi, and David Muñoz Ramo. Demonstrating bayesian quantum phase estimation with quantum error detection.

Phys. Rev. Res., 6:013221, 2024. URL https://link.aps.org/doi/10.1103/PhysRevResearch.6.013221.

[55] Wim van Dam, Hongbin Liu, Guang Hao Low, Adam Paetznick, Andres Paz, Marcus Silva, Aarthi Sundaram, Krysta Svore, and Matthias Troyer. End-to-end quantum simulation of a chemical system, 2024. URL https://arxiv.org/abs/2409.05835.

[56] Sam McArdle and David P. Tew. Improving the accuracy of quantum computational chemistry using the transcorrelated method, 2020. URL https://arxiv.org/abs/2006.11181.

[57] Mario Motta, Tanvi P. Gujarati, Julia E. Rice, Ashutosh Kumar, Conner Masteran, Joseph A. Latone, Eunseok Lee, Edward F. Valeev, and Tyler Y. Takeshita. Quantum simulation of electronic structure with a transcorrelated hamiltonian: improved accuracy with a smaller footprint on the quantum computer. *Phys. Chem. Chem. Phys.*, 22:24270–24281, 2020. doi: 10.1039/D0CP04106H. URL http://dx.doi.org/10.1039/D0CP04106H.

[58] Philipp Schleich, Jakob S. Kottmann, and Alán Aspuru-Guzik. Improving the accuracy of the variational quantum eigensolver for molecular systems by the explicitly-correlated perturbative [2]r12-correction. *Phys. Chem. Chem. Phys.*, 24:13550–13564, 2022. doi: 10.1039/D2CP00247G. URL http://dx.doi.org/10.1039/D2CP00247G.

[59] Igor O. Sokolov, Werner Dobrautz, Hongjun Luo, Ali Alavi, and Ivano Tavernelli. Orders of magnitude increased accuracy for quantum many-body problems on quantum computers via an exact transcorrelated method. *Phys. Rev. Res.*, 5: 023174, Jun 2023. doi: 10.1103/PhysRevResearch.5.023174. URL https://link.aps.org/doi/10.1103/PhysRevResearch.5.023174.

[60] David Poulin, M. B. Hastings, Dave Wecker, Nathan Wiebe, Andrew C. Doberty, and Matthias Troyer. The Trotter step size required for accurate quantum simulation of quantum chemistry. *Quant. Inf. Comput.*, 15(5-6):0361–0384, 2015. URL https://doi.org/10.26421/QIC15.5-6-1.

[61] Oumarou Oumarou, Maximilian Scheurer, Robert M. Parrish, Edward G. Hohenstein, and Christian Gogolin. Accelerating Quantum Computations of Chemistry Through Regularized Compressed Double Factorization. *Quantum*, 8:1371, 2024. ISSN 2521-327X. URL https://doi.org/10.22331/q-2024-06-13-1371.

[62] Joonho Lee, Dominic W. Berry, Craig Gidney, William J. Huggins, Jarrod R. McClean, Nathan Wiebe, and Ryan Babbush. Even more efficient quantum computations of chemistry through tensor hypercontraction. *PRX Quantum*, 2:030305, Jul 2021. doi: 10.1103/PRXQuantum.2.030305. URL https://link.aps.org/doi/10.1103/PRXQuantum.2.030305.

[63] Mario Motta, Erika Ye, Jarrod R. McClean, Zhendong Li, Austin J. Minnich, Ryan Babbush, and Garnet Kin-Lic Chan. Low rank representations for quantum simulation of electronic structure. *npj Quantum Information*, 7(1):83, May 2021. ISSN 2056-6387. doi: 10.1038/s41534-021-00416-z. URL https://doi.org/10.1038/s41534-021-00416-z.

[64] Mekena Metcalf, Nicholas P. Bauman, Karol Kowalski, and Wibe A. de Jong. Resource-efficient chemistry on quantum computers with the variational quantum eigensolver and the double unitary coupled-cluster approach. *Journal of Chemical Theory and Computation*, 16(10):6165–6175, Oct 2020. ISSN 1549-9618. doi: 10.1021/acs.jctc.0c00421. URL https://doi.org/10.1021/acs.jctc.0c00421.

[65] Nicholas P. Bauman, Eric J. Bylaska, Sriram Krishnamoorthy, Guang Hao Low, Nathan Wiebe, Christopher E. Granade, Martin Roetteler, Matthias Troyer, and Karol Kowalski. Downfolding of many-body hamiltonians using active-space models: Extension of the sub-system embedding sub-algebras approach to unitary coupled cluster formalisms. *The Journal of Chemical Physics*, 151(1):014107, 07 2019. ISSN 0021-9606. doi: 10.1063/1.5094643. URL https://doi.org/10.1063/1.5094643.

[66] Nicholas P. Bauman, Guang Hao Low, and Karol Kowalski. Quantum simulations of excited states with active-space downfolded hamiltonians. *The Journal of Chemical Physics*, 151(23):234114, 12 2019. ISSN 0021-9606. doi: 10.1063/1.5128103. URL https://doi.org/10.1063/1.5128103.

[67] Bauman Nicholas P, Jaroslav Chládek, Libor Veis, Jiří Pittner, and Kowalski Karol. Variational quantum eigensolver for approximate diagonalization of downfolded hamiltonians using generalized unitary coupled cluster ansatz. *Quantum Science and Technology*, 6(3):034008, jun 2021. doi: 10.1088/2058-9565/abf602. URL https://dx.doi.org/10.1088/2058-9565/abf602.

[68] Nicholas P. Bauman and Karol Kowalski. Coupled cluster downfolding theory: towards universal many-body algorithms for dimensionality reduction of composite quantum systems in chemistry and materials science. *Materials Theory*, 6(1):17, May 2022. ISSN 2509-8012. doi: 10.1186/s41313-022-00046-8. URL https://doi.org/10.1186/s41313-022-00046-8.

[69] Ilya G Ryabinkin, Artur F Izmaylov, and Scott N Genin. A posteriori corrections to the iterative qubit coupled cluster method to minimize the use of quantum resources in large-scale calculations. *Quantum Science and Technology*, 6(2):024012, mar 2021. doi: 10.1088/2058-9565/abda8e. URL https://dx.doi.org/10.1088/2058-9565/abda8e.

[70] Ryan Babbush, Nathan Wiebe, Jarrod McClean, James McClain, Hartmut Neven, and Garnet Kin-Lic Chan. Low-depth quantum simulation of materials. *Phys. Rev. X*, 8:011044, 2018. URL https://link.aps.org/doi/10.1103/PhysRevX.8.011044.

[71] Cheng-Lin Hong, Ting Tsai, Jyh-Pin Chou, Peng-Jen Chen, Pei-Kai Tsai, Yu-Cheng Chen, En-Jui Kuo, David Srolovitz, Alice Hu, Yuan-Chung Cheng, and Hsi-Sheng Goan. Accurate and efficient quantum computations of molecular properties using daubechies wavelet molecular orbitals: A benchmark study against experimental data. *PRX Quantum*, 3:020360, Jun 2022. doi: 10.1103/PRXQuantum.3.020360. URL https://link.aps.org/doi/10.1103/PRXQuantum.3.020360.

[72] Jakob S. Kottmann et al. Reducing qubit requirements while maintaining numerical precision for the variational quantum eigensolver: A basis-set-free approach. *The Journal of Physical Chemistry Letters*, 12(1):663–673, 2021. URL https://doi.org/10.1021/acs.jpclett.0c03410.

[73] J. Cioslowski. Connected moments expansion: A new tool for quantum many-body theory. *Phys. Rev. Lett.*, 58:83–85, 1987. URL https://link.aps.org/doi/10.1103/PhysRevLett.58.83.

[74] Brad Ganoe and Martin Head-Gordon. Doubles connected moments expansion: A tractable approximate horn–weinstein approach for quantum chemistry. *Journal of Chemical Theory and Computation*, 19(24):9187–9201, 2023. doi: 10.1021/acs.jctc.3c00929. URL https://doi.org/10.1021/acs.jctc.3c00929. PMID: 38051773.

[75] Lloyd C. L. Hollenberg and N. S. Witte. Analytic solution for the ground-state energy of the extensive many-body problem. *Phys. Rev. B*, 54:16309–16312, 1996. URL https://link.aps.org/doi/10.1103/PhysRevB.54.16309.

[76] Cornelius Lanczos. An iteration method for the solution of the eigenvalue problem of linear differential and integral operators. *J. Res. Natl. Bur. Stand. B*, 45:255–282, 1950. URL https://doi.org/10.6028/jres.045.026.

[77] Pauli Virtanen et al. Scipy 1.0: fundamental algorithms for scientific computing in python. *Nature Methods*, 17(3):261–272, 2020. URL https://doi.org/10.1038/s41592-019-0686-2.

[78] Christopher Conway Paige. *The Computation of Eigenvalues and Eigenvectors of Very Large Sparse Matrices*. PhD thesis, University of London, UK, 1971. URL https://www.cs.mcgill.ca/~chris/pubClassic/PaigeThesis.pdf.

[79] Lloyd C. L. Hollenberg. Plaquette expansion in lattice Hamiltonian models. *Phys. Rev. D*, 47:1640–1644, 1993. URL https://link.aps.org/doi/10.1103/PhysRevD.47.1640.

[80] Erik A van Doorn. Representations and bounds for zeros of orthogonal polynomials and eigenvalues of sign-symmetric tri-diagonal matrices. *Journal of Approximation Theory*, 51(3):254–266, 1987. ISSN 0021-9045. URL https://www.sciencedirect.com/science/article/pii/0021904587900384.

[81] Mourad Ismail and Xin Li. Bound on the extreme zeros of orthogonal polynomials. *Proceedings of The American Mathematical Society - PROC AMER MATH SOC*, 115, 1992. URL http://www.jstor.org/stable/2159575.

[82] Lloyd C. L. Hollenberg, David C. Bardos, and N. S. Witte. Lanczos cluster expansion for non-extensive systems. *Z. Phys. D*, 38(3):249–252, 1996. URL https://doi.org/10.1007/s004600050089.

[83] Aochen Duan. Matrix product states in quantum information processing. Master's thesis, University of Melbourne, Australia, 2015. URL http://hdl.handle.net/11343/267657.

[84] Michael Alexander Jones. Moments-based corrections to variational quantum computation. Master's thesis, University of Melbourne, Australia, 2019. URL http://hdl.handle.net/11343/281917.

[85] Harish J. Vallury, Michael A. Jones, Charles D. Hill, and Lloyd C. L. Hollenberg. Quantum computed moments correction to variational estimates. *Quantum*, 4:373, 2020. ISSN 2521-327X. URL https://doi.org/10.22331/q-2020-12-15-373.

[86] Mario Motta, Chong Sun, Adrian T. K. Tan, Matthew J. O'Rourke, Erika Ye, Austin J. Minnich, Fernando G. S. L. Brandão, and Garnet Kin-Lic Chan. Determining eigenstates and thermal states on a quantum computer using quantum imaginary time evolution. *Nature Physics*, 16(2):205–210, 2020. URL https://doi.org/10.1038/s41567-019-0704-4.

[87] Kübra Yeter-Aydeniz, Raphael C. Pooser, and George Siopsis. Practical quantum computation of chemical and nuclear energy levels using quantum imaginary time evolution and Lanczos algorithms. *npj Quantum Information*, 6(1):63, 2020. URL https://doi.org/10.1038/s41534-020-00290-1.

[88] Thomas E. Baker. Lanczos recursion on a quantum computer for the green's function and ground state. *Phys. Rev. A*, 103:032404, Mar 2021. doi: 10.1103/PhysRevA.103.032404. URL https://link.aps.org/doi/10.1103/PhysRevA.103.032404.

[89] William Kirby, Mario Motta, and Antonio Mezzacapo. Exact and efficient Lanczos method on a quantum computer. *Quantum*, 7:1018, 2023. ISSN 2521-327X. URL https://doi.org/10.22331/q-2023-05-23-1018.

[90] Thomas E. Baker. Block lanczos method for excited states on a quantum computer. *Phys. Rev. A*, 110:012420, Jul 2024. doi: 10.1103/PhysRevA.110.012420. URL https://link.aps.org/doi/10.1103/PhysRevA.110.012420.

[91] Lloyd C.L. Hollenberg, Matthew P. Wilson, and N.S. Witte. General nonperturbative mass gap to first order in $1/V$. *Phys. Lett. B*, 361:81–86, 1995. URL https://doi.org/10.1016/0370-2693(95)01049-V.

[92] Kazuhiro Seki and Seiji Yunoki. Quantum power method by a superposition of time-evolved states. *PRX Quantum*, 2:010333, 2021. URL https://link.aps.org/doi/10.1103/PRXQuantum.2.010333.

[93] R. McWeeny. Some recent advances in density matrix theory. *Rev. Mod. Phys.*, 32:335–369, 1960. URL https://link.aps.org/doi/10.1103/RevModPhys.32.335.

[94] Qiming Sun, Timothy C. Berkelbach, Nick S. Blunt, George H. Booth, Sheng Guo, Zhendong Li, Junzi Liu, James D. McClain, Elvira R. Sayfutyarova, Sandeep Sharma, Sebastian Wouters, and Garnet Kin-Lic Chan. Pyscf: the python-based simulations of chemistry framework. *Wiley Interdisciplinary Reviews: Computational Molecular Science*, 8(1):e1340, 2017. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/wcms.1340.

[95] Dave Wecker, Matthew B. Hastings, Nathan Wiebe, Bryan K. Clark, Chetan Nayak, and Matthias Troyer. Solving strongly correlated electron models on a quantum computer. *Phys. Rev. A*, 92:062318, 2015. URL https://link.aps.org/doi/10.1103/PhysRevA.92.062318.

[96] Ian D. Kivlichan, Jarrod McClean, Nathan Wiebe, Craig Gidney, Alán Aspuru-Guzik, Garnet Kin-Lic Chan, and Ryan Babbush. Quantum simulation of electronic structure with linear depth and connectivity. *Phys. Rev. Lett.*, 120:110501, 2018. URL https://link.aps.org/doi/10.1103/PhysRevLett.120.110501.

[97] Juan Miguel Arrazola, Olivia Di Matteo, Nicolás Quesada, Soran Jahangiri, Alain Delgado, and Nathan Killoran. Universal quantum circuits for quantum chemistry. *Quantum*, 6:742, 2022. ISSN 2521-327X. URL https://doi.org/10.22331/q-2022-06-20-742.

[98] Quantum user interface tool. quispace.org, (2018-2020). URL quispace.org.

[99] Vincent E. Elfving, Benno W. Broer, Mark Webber, Jacob L. Gavartin, Mathew D. Halls, K. Patrick Lorton, and Arteum D. Bochevarov. How will quantum computers provide an industrially relevant computational advantage in quantum chemistry?, 2020. URL https://arxiv.org/abs/2009.12472.

[100] Yukio Kawashima, Erika Lloyd, Marc P. Coons, Yunseong Nam, Shunji Matsuura, Alejandro J. Garza, Sonika Johri, Lee Huntington, Valentin Senicourt, Andrii O. Maksymov, Jason H. V. Nguyen, Jungsang Kim, Nima Alidoust, Arman Zaribafiyan, and Takeshi Yamazaki. Optimizing electronic structure simulations on a trapped-ion quantum computer using problem decomposition. *Nature Communications Physics*, 4(1):245, 2021. URL https://doi.org/10.1038/s42005-021-00751-9.

[101] Ivan Kassal and Alán Aspuru-Guzik. Quantum algorithm for molecular properties and geometry optimization. *The Journal of Chemical Physics*, 131(22):224102, 2009. URL https://doi.org/10.1063/1.3266959.

[102] Sergey Bravyi, Jay M. Gambetta, Antonio Mezzacapo, and Kristan Temme. Tapering off qubits to simulate fermionic hamiltonians, 2017. URL https://arxiv.org/abs/1701.08213.

[103] P. A. M. Dirac. Note on the interpretation of the density matrix in the many-electron problem. *Proc. Cambridge Phil. Soc.*, 27(2):240–243, 1931. URL https://doi.org/10.1017/S0305004100010343.

[104] Ali Javadi-Abhari, Matthew Treinish, Kevin Krsulich, Christopher J. Wood, Jake Lishman, Julien Gacon, Simon Martiel, Paul D. Nation, Lev S. Bishop, Andrew W. Cross, Blake R. Johnson, and Jay M. Gambetta. Quantum computing with qiskit, 2024. URL https://arxiv.org/abs/2405.08810.

[105] Ryan Babbush, Dominic W. Berry, Jarrod R. McClean, and Hartmut Neven. Quantum simulation of chemistry with sublinear scaling in basis size. *npj Quantum Information*, 5(1):92, 2019. URL https://doi.org/10.1038/s41534-019-0199-y.

[106] Vladyslav Verteletskyi, Tzu-Ching Yen, and Artur F. Izmaylov. Measurement optimization in the variational quantum eigensolver using a minimum clique cover.

The Journal of Chemical Physics, 152(12):124114, 2020. URL https://doi.org/10.1063/1.5141458.

[107] J. I. Colless, V. V. Ramasesh, D. Dahlen, M. S. Blok, M. E. Kimchi-Schwartz, J. R. McClean, J. Carter, W. A. de Jong, and I. Siddiqi. Computation of molecular spectra on a quantum processor with an error-resilient algorithm. *Phys. Rev. X*, 8: 011021, 2018. URL https://link.aps.org/doi/10.1103/PhysRevX.8.011021.

[108] Jules Tilly, P. V. Sriluckshmy, Akashkumar Patel, Enrico Fontana, Ivan Rungger, Edward Grant, Robert Anderson, Jonathan Tennyson, and George H. Booth. Reduced density matrix sampling: Self-consistent embedding and multiscale electronic structure on current generation quantum computers. *Phys. Rev. Res.*, 3: 033230, 2021. URL https://link.aps.org/doi/10.1103/PhysRevResearch.3.033230.

[109] Michael A. Jones, Harish J. Vallury, Charles D. Hill, and Lloyd C. L. Hollenberg. Chemistry beyond the Hartree–Fock energy via quantum computed moments. *Sci. Rep.*, 12(1):8985, 2022. URL https://doi.org/10.1038/s41598-022-12324-z.

[110] Soichi Shirai, Hokuto Iwakiri, Keita Kanno, Takahiro Horiba, Keita Omiya, Hirotoshi Hirai, and Sho Koh. Computational analysis of chemical reactions using a variational quantum eigensolver algorithm without specifying spin multiplicity. *ACS Omega*, 8(22):19917–19925, 2023. URL https://doi.org/10.1021/acsomega.3c01875.

[111] Harish J. Vallury, Michael A. Jones, Gregory A. L. White, Floyd M. Creevey, Charles D. Hill, and Lloyd C. L. Hollenberg. Noise-robust ground state energy estimates from deep quantum circuits. *Quantum*, 7:1109, 2023. ISSN 2521-327X. URL https://doi.org/10.22331/q-2023-09-11-1109.

[112] Akel Hashim, Ravi K. Naik, Alexis Morvan, Jean-Loup Ville, Bradley Mitchell, John Mark Kreikebaum, Marc Davis, Ethan Smith, Costin Iancu, Kevin P. O'Brien, Ian Hincks, Joel J. Wallman, Joseph Emerson, and Irfan Siddiqi. Randomized compiling for scalable quantum computing on a noisy superconducting quantum processor. *Phys. Rev. X*, 11:041039, 2021. URL https://link.aps.org/doi/10.1103/PhysRevX.11.041039.

[113] J. Goings, L. Zhao, J. Jakowski, T. Morris, and R. Pooser. Molecular symmetry in vqe: A dual approach for trapped-ion simulations of benzene. In *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, pages 76–82, Los Alamitos, CA, USA, 2023. IEEE Computer Society. URL https://doi.ieeecomputersociety.org/10.1109/QCE57702.2023.10187.

[114] Andrew G. Taube and Rodney J. Bartlett. New perspectives on unitary coupled-cluster theory. *International Journal of Quantum Chemistry*, 106(15):3393–3401, 2006. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/qua.21198.

[115] Jia Chen, Hai-Ping Cheng, and J. K. Freericks. Flexibility of the factorized form of the unitary coupled cluster Ansatz. *The J. Chem. Phys.*, 156(4):044106, 2022. ISSN 0021-9606. URL https://doi.org/10.1063/5.0074311.

[116] D.J. Thouless. Stability conditions and nuclear rotations in the Hartree-Fock theory. *Nucl. Phys.*, 21:225 – 232, 1960. ISSN 0029-5582. URL http://www.sciencedirect.com/science/article/pii/0029558260900481.

[117] Igor O. Sokolov, Panagiotis Kl. Barkoutsos, Pauline J. Ollitrault, Donny Greenberg, Julia Rice, Marco Pistoia, and Ivano Tavernelli. Quantum orbital-optimized unitary coupled cluster methods in the strongly correlated regime: Can quantum algorithms outperform their classical equivalents? *J. Chem. Phys.*, 152(12):124107, 2020. ISSN 0021-9606. URL https://doi.org/10.1063/1.5141835.

[118] Harper R. Grimsley, Sophia E. Economou, Edwin Barnes, and Nicholas J. Mayhall. An adaptive variational algorithm for exact molecular simulations on a quantum computer. *Nature Communications*, 10(1):3007, 2019. URL https://doi.org/10.1038/s41467-019-10988-2.

[119] Ho Lun Tang, V.O. Shkolnikov, George S. Barron, Harper R. Grimsley, Nicholas J. Mayhall, Edwin Barnes, and Sophia E. Economou. Qubit-ADAPT-VQE: An adaptive algorithm for constructing hardware-efficient ansätze on a quantum processor. *PRX Quantum*, 2:020310, 2021. URL https://link.aps.org/doi/10.1103/PRXQuantum.2.020310.

[120] Hsin-Yuan Huang, Richard Kueng, and John Preskill. Predicting many properties of a quantum system from very few measurements. *Nat. Phys.*, 16(10):1050–1057, 2020. URL https://doi.org/10.1038/s41567-020-0932-7.

[121] Andrew Zhao, Nicholas C. Rubin, and Akimasa Miyake. Fermionic partial tomography via classical shadows. *Phys. Rev. Lett.*, 127:110504, 2021. URL https://link.aps.org/doi/10.1103/PhysRevLett.127.110504.

[122] Gary J Mooney. *Entanglement in superconducting quantum devices and improving quantum circuit compilation.* PhD thesis, University of Melbourne, Australia, 2022. URL https://minerva-access.unimelb.edu.au/items/a7ae6399-00a8-4082-a2b9-b4858bf246f7.

[123] Gurobi Optimization, LLC. Gurobi Optimizer Reference Manual, 2023. URL https://www.gurobi.com.

[124] Bradley Efron and Robert J. Tibishirani. *An Introduction to the Bootstrap*. CRC Press, 1994.

[125] Filip B. Maciejewski, Zoltán Zimborás, and Michał Oszmaniec. Mitigation of readout noise in near-term quantum devices by classical post-processing based on detector tomography. *Quantum*, 4:257, 2020. ISSN 2521-327X. URL https://doi.org/10.22331/q-2020-04-24-257.

[126] Paul D. Nation, Hwajung Kang, Neereja Sundaresan, and Jay M. Gambetta. Scalable mitigation of measurement errors on quantum computers. *PRX Quantum*, 2:040326, 2021. URL https://link.aps.org/doi/10.1103/PRXQuantum.2.040326.

[127] C. Michelot. A finite algorithm for finding the projection of a point onto the canonical simplex of $\mathcal{R}^n$. *J. Optim. Theor. Appl.*, 50(1):195–200, 1986. URL https://doi.org/10.1007/BF00938486.

[128] John A. Smolin, Jay M. Gambetta, and Graeme Smith. Efficient method for computing the maximum-likelihood quantum state from measurements with additive gaussian noise. *Phys. Rev. Lett.*, 108:070502, 2012. URL https://link.aps.org/doi/10.1103/PhysRevLett.108.070502.

[129] X. Bonet-Monroig, R. Sagastizabal, M. Singh, and T. E. O'Brien. Low-cost error mitigation by symmetry verification. *Phys. Rev. A*, 98:062339, 2018. URL https://link.aps.org/doi/10.1103/PhysRevA.98.062339.

[130] A. J. Coleman. Structure of fermion density matrices. *Rev. Mod. Phys.*, 35:668–686, 1963. URL https://link.aps.org/doi/10.1103/RevModPhys.35.668.

[131] Mitchell J Knight, Harry M Quiney, and Andy M Martin. Reduced density matrix approach to ultracold few-fermion systems in one dimension. *New J. Phys.*, 24(5):053004, 2022. URL https://dx.doi.org/10.1088/1367-2630/ac643d.

[132] David A. Mazziotti. Pure-N-representability conditions of two-fermion reduced density matrices. *Phys. Rev. A*, 94:032516, 2016. URL https://link.aps.org/doi/10.1103/PhysRevA.94.032516.

[133] Nicholas C Rubin, Ryan Babbush, and Jarrod McClean. Application of fermionic marginal constraints to hybrid quantum algorithms. *New J. Phys.*, 20(5):053020, 2018. URL https://dx.doi.org/10.1088/1367-2630/aab919.

[134] Zhenyu Cai, Ryan Babbush, Simon C. Benjamin, Suguru Endo, William J. Huggins, Ying Li, Jarrod R. McClean, and Thomas E. O'Brien. Quantum error mitigation. *Rev. Mod. Phys.*, 95:045005, 2023. URL https://link.aps.org/doi/10.1103/RevModPhys.95.045005.

[135] Scott E. Smart and David A. Mazziotti. Quantum-classical hybrid algorithm using an error-mitigating n-representability condition to compute the mott metal-insulator transition. *Phys. Rev. A*, 100:022517, 2019. URL https://link.aps.org/doi/10.1103/PhysRevA.100.022517.

[136] Phalgun Lolur, Mårten Skogh, Werner Dobrautz, Christopher Warren, Janka Biznárová, Amr Osman, Giovanna Tancredi, Göran Wendin, Jonas Bylander, and Martin Rahm. Reference-state error mitigation: A strategy for high accuracy quantum computation of chemistry. *J. Chem. Theory Comput.*, 19(3):783–789, 2023. URL https://doi.org/10.1021/acs.jctc.2c00807.

[137] Piotr Czarnik, Andrew Arrasmith, Patrick J. Coles, and Lukasz Cincio. Error mitigation with Clifford quantum-circuit data. *Quantum*, 5:592, 2021. ISSN 2521-327X. URL https://doi.org/10.22331/q-2021-11-26-592.

[138] Alexander M. Dalzell, Nicholas Hunter-Jones, and Fernando G. S. L. Brandão. Random quantum circuits transform local noise into global white noise. *Communications in Mathematical Physics*, 405(3):78, 2024. ISSN 1432-0916. doi: 10.1007/s00220-024-04958-z. URL https://doi.org/10.1007/s00220-024-04958-z.

[139] Kristan Temme, Sergey Bravyi, and Jay M. Gambetta. Error mitigation for short-depth quantum circuits. *Phys. Rev. Lett.*, 119:180509, 2017. URL https://link.aps.org/doi/10.1103/PhysRevLett.119.180509.

[140] Abhinav Kandala, Kristan Temme, Antonio D. Córcoles, Antonio Mezzacapo, Jerry M. Chow, and Jay M. Gambetta. Error mitigation extends the computational reach of a noisy quantum processor. *Nature*, 567(7749):491–495, 2019. URL https://doi.org/10.1038/s41586-019-1040-7.

[141] Qi Gao, Gavin O. Jones, Mario Motta, Michihiko Sugawara, Hiroshi C. Watanabe, Takao Kobayashi, Eriko Watanabe, Yu-ya Ohnishi, Hajime Nakamura, and Naoki Yamamoto. Applications of quantum computing for investigations of electronic transitions in phenylsulfonyl-carbazole tadf emitters. *npj Computational Materials*, 7(1):70, 2021. URL https://doi.org/10.1038/s41524-021-00540-6.

[142] Shigeki Gocho, Hajime Nakamura, Shu Kanno, Qi Gao, Takao Kobayashi, Taichi Inagaki, and Miho Hatanaka. Excited state calculations using variational quantum eigensolver with spin-restricted ansätze and automatically-adjusted constraints. *npj Computational Materials*, 9(1):13, 2023. URL https://doi.org/10.1038/s41524-023-00965-1.

[143] Maria A. Castellanos, Mario Motta, and Julia E. Rice. Quantum computation of $\pi \rightarrow \pi^*$ and $n \rightarrow \pi^*$ excited states of aromatic heterocycles. *Molecular Physics*,

122(7-8):e2282736, 2024. doi: 10.1080/00268976.2023.2282736. URL https://doi.org/10.1080/00268976.2023.2282736.

[144] Pauline J. Ollitrault, Abhinav Kandala, Chun-Fu Chen, Panagiotis Kl. Barkoutsos, Antonio Mezzacapo, Marco Pistoia, Sarah Sheldon, Stefan Woerner, Jay M. Gambetta, and Ivano Tavernelli. Quantum equation of motion for computing molecular excitation energies on a noisy quantum processor. *Phys. Rev. Res.*, 2:043140, 2020. URL https://link.aps.org/doi/10.1103/PhysRevResearch.2.043140.

[145] M. Ganzhorn, D.J. Egger, P. Barkoutsos, P. Ollitrault, G. Salis, N. Moll, M. Roth, A. Fuhrer, P. Mueller, S. Woerner, I. Tavernelli, and S. Filipp. Gate-efficient simulation of molecular eigenstates on a quantum computer. *Phys. Rev. Appl.*, 11: 044092, 2019. URL https://link.aps.org/doi/10.1103/PhysRevApplied.11.044092.

[146] Frank T. Cerasoli, Kyle Sherbert, Jagoda Sławińska, and Marco Buongiorno Nardelli. Quantum computation of silicon electronic band structure. *Phys. Chem. Chem. Phys.*, 22:21816–21822, 2020. doi: 10.1039/D0CP04008H. URL http://dx.doi.org/10.1039/D0CP04008H.

[147] Takahiro Ohgoe, Hokuto Iwakiri, Masaya Kohda, Kazuhide Ichikawa, Yuya O. Nakagawa, Hubert Okadome Valencia, and Sho Koh. Demonstrating quantum computation for quasiparticle band structures. *Phys. Rev. Res.*, 6:L022022, 2024. URL https://link.aps.org/doi/10.1103/PhysRevResearch.6.L022022.

[148] Dileep Singh, Shashank Mehendale, Arvind, and Kavita Dorai. Ground and excited state energy calculations of the h2 molecule using a variational quantum eigensolver algorithm on an nmr quantum simulator. arXiv:physics.quant-ph/2407.01000, 2024. URL https://arxiv.org/abs/2407.01000.

[149] Shree Hari Sureshbabu, Manas Sajjan, Sangchul Oh, and Sabre Kais. Implementation of quantum machine learning for electronic structure calculations of periodic systems on quantum computing devices. *Journal of Chemical Information and Modeling*, 61(6):2667–2674, 2021. URL https://doi.org/10.1021/acs.jcim.1c00294.

[150] Shaobo Zhang, Akib Karim, Harry M. Quiney, and Muhammad Usman. Full band structure calculation of semiconducting materials on a noisy quantum processor, 2024. URL http://dx.doi.org/10.1103/PhysRevA.110.062415.

[151] Spencer T. Stober, Stuart M. Harwood, Dimitar Trenev, Panagiotis Kl. Barkoutsos, Tanvi P. Gujarati, and Sarah Mostame. Considerations for evaluating thermodynamic properties with hybrid quantum-classical computing work flows.

Phys. Rev. A, 105:012425, 2022. URL https://link.aps.org/doi/10.1103/PhysRevA.105.012425.

[152] Julia E. Rice, Tanvi P. Gujarati, Mario Motta, Tyler Y. Takeshita, Eunseok Lee, Joseph A. Latone, and Jeannette M. Garcia. Quantum computation of dominant products in lithium–sulfur batteries. *J. Chem. Phys.*, 154(13):134115, 2021. ISSN 0021-9606. URL https://doi.org/10.1063/5.0044068.

[153] Palak Chawla, Shweta, K. R. Swain, Tushti Patel, Renu Bala, Disha Shetty, Kenji Sugisaki, Sudhindu Bikash Mandal, Jordi Riu, Jan Nogué, V. S. Prasannaa, and B. P. Das. Relativistic variational-quantum-eigensolver calculations of molecular electric dipole moments on quantum hardware. *Phys. Rev. A*, 111:022817, 2025. doi: 10.1103/PhysRevA.111.022817. URL https://link.aps.org/doi/10.1103/PhysRevA.111.022817.

[154] Zhiwen Zong, Sainan Huai, Tianqi Cai, Wenyan Jin, Ze Zhan, Zhenxing Zhang, Kunliang Bu, Liyang Sui, Ying Fei, Yicong Zheng, Shengyu Zhang, Jianlan Wu, and Yi Yin. Determination of molecular energies via quantum imaginary time evolution in a superconducting qubit system. *Sci. China Phys. Mech. Astron.*, 67 (4):240311, 2024. URL https://doi.org/10.1007/s11433-023-2315-0.

[155] Michael A. Jones, Harish J. Vallury, and Lloyd C.L. Hollenberg. Ground-state-energy calculation for the water molecule on a superconducting quantum processor. *Phys. Rev. Appl.*, 21:064017, 2024. URL https://link.aps.org/doi/10.1103/PhysRevApplied.21.064017.

[156] Mario Motta, William Kirby, Ieva Liepuoniute, Kevin J Sung, Jeffrey Cohn, Antonio Mezzacapo, Katherine Klymko, Nam Nguyen, Nobuyuki Yoshioka, and Julia E Rice. Subspace methods for electronic structure simulations on quantum computers. *Electronic Structure*, 6(1):013001, 2024. doi: 10.1088/2516-1075/ad3592. URL https://dx.doi.org/10.1088/2516-1075/ad3592.

[157] Jeffery Yu, Javier Robledo Moreno, Joseph T. Iosue, Luke Bertels, Daniel Claudino, Bryce Fuller, Peter Groszkowski, Travis S. Humble, Petar Jurcevic, William Kirby, Thomas A. Maier, Mario Motta, Bibek Pokharel, Alireza Seif, Amir Shehata, Kevin J. Sung, Minh C. Tran, Vinay Tripathi, Antonio Mezzacapo, and Kunal Sharma. Quantum-centric algorithm for sample-based krylov diagonalization, 2025. URL https://arxiv.org/abs/2501.09702.

[158] Libor Veis, Jakub Višňák, Hiroaki Nishizawa, Hiromi Nakai, and Jiří Pittner. Quantum chemistry beyond born–oppenheimer approximation on a quantum computer: A simulated phase estimation study. *International Journal of Quantum*

Chemistry, 116(18):1328–1336, 2016. doi: https://doi.org/10.1002/qua.25176. URL https://onlinelibrary.wiley.com/doi/abs/10.1002/qua.25176.

[159] Xiaoxia Cai, Wei-Hai Fang, Heng Fan, and Zhendong Li. Quantum computation of molecular response properties. *Phys. Rev. Res.*, 2:033324, Aug 2020. doi: 10.1103/PhysRevResearch.2.033324. URL https://link.aps.org/doi/10.1103/PhysRevResearch.2.033324.

[160] Renato Olarte Hernandez, Armand Soldera, and Benoît Champagne. Vibronic structure of the uv/visible absorption spectra of phenol and phenolate: A hybrid density functional theory-doktorov's quantum algorithm approach. *The Journal of Physical Chemistry A*, 129(1):396–406, 2025. doi: 10.1021/acs.jpca.4c06960. URL https://doi.org/10.1021/acs.jpca.4c06960. PMID: 39718998.

[161] Ashutosh Kumar, Ayush Asthana, Vibin Abraham, T. Daniel Crawford, Nicholas J. Mayhall, Yu Zhang, Lukasz Cincio, Sergei Tretiak, and Pavel A. Dub. Quantum simulation of molecular response properties in the nisq era. *Journal of Chemical Theory and Computation*, 19(24):9136–9150, 2023. doi: 10.1021/acs.jctc.3c00731. URL https://doi.org/10.1021/acs.jctc.3c00731. PMID: 38054645.

[162] Kaixuan Huang, Xiaoxia Cai, Hao Li, Zi-Yong Ge, Ruijuan Hou, Hekang Li, Tong Liu, Yunhao Shi, Chitong Chen, Dongning Zheng, Kai Xu, Zhi-Bo Liu, Zhendong Li, Heng Fan, and Wei-Hai Fang. Variational quantum computation of molecular linear response properties on a superconducting quantum processor. *The Journal of Physical Chemistry Letters*, 13(39):9114–9121, 2022. doi: 10.1021/acs.jpclett.2c02381. URL https://doi.org/10.1021/acs.jpclett.2c02381. PMID: 36154018.

[163] Shi-Ning Sun, Brian Marinelli, Jin Ming Koh, Yosep Kim, Long B. Nguyen, Larry Chen, John Mark Kreikebaum, David I. Santiago, Irfan Siddiqi, and Austin J. Minnich. Quantum computation of frequency-domain molecular response properties using a three-qubit itoffoli gate. *npj Quantum Information*, 10(1):55, May 2024. ISSN 2056-6387. doi: 10.1038/s41534-024-00850-9. URL https://doi.org/10.1038/s41534-024-00850-9.

[164] Lloyd C. L. Hollenberg and Michael J. Tomlinson. Staggered magnetisation in the heisenberg antiferromagnet. *Australian Journal of Physics*, 47(2):137–144, 1994. URL https://doi.org/10.1071/PH940137.

[165] N. S. Witte, Lloyd C. L. Hollenberg, and Zheng Weihong. Two-dimensional xxz model ground-state properties using an analytic lanczos expansion. *Phys. Rev.*

B, 55:10412–10419, 1997. URL https://link.aps.org/doi/10.1103/PhysRevB.55.10412.

[166] Harish J. Vallury and Lloyd C. L. Hollenberg. Arbitrary ground state observables from quantum computed moments. In *2023 IEEE International Conference on Quantum Computing and Engineering (QCE)*, volume 01, pages 295–299, 2023. doi: 10.1109/QCE57702.2023.00040. URL https://doi.ieeecomputersociety.org/10.1109/QCE57702.2023.00040.

[167] Saul T. Epstein, Andrew C. Hurley, Robert E. Wyatt, and Robert G. Parr. Integrated and integral hellmann—feynman formulas. *The Journal of Chemical Physics*, 47(4):1275–1286, 08 1967. ISSN 0021-9606. doi: 10.1063/1.1712080. URL https://doi.org/10.1063/1.1712080.

[168] R. McWeeny and B.T. Sutcliffe. *Methods of Molecular Quantum Mechanics*. Pure and Applied Mathematics. Academic Press, 1969. ISBN 9780124865501.

[169] VebjØSRN Bakken, Trygve Helgaker, Wim Klopper, and Kenneth Ruud and. The calculation of molecular geometrical properties in the hellmann—feynman approximation. *Molecular Physics*, 96(4):653–671, 1999. doi: 10.1080/00268979909483002. URL https://doi.org/10.1080/00268979909483002.

[170] P. Pulay. Ab initio calculation of force constants and equilibrium geometries in polyatomic molecules. *Molecular Physics*, 17(2):197–204, 1969. URL https://doi.org/10.1080/00268976900100941.

[171] Robert M. Parrish, Edward G. Hohenstein, Peter L. McMahon, and Todd J. Martinez. Hybrid quantum/classical derivative theory: Analytical gradients and excited-state dynamics for the multistate contracted variational quantum eigensolver, 2019. URL https://arxiv.org/abs/1906.08728.

[172] Kosuke Mitarai, Yuya O. Nakagawa, and Wataru Mizukami. Theory of analytical energy derivatives for the variational quantum eigensolver. *Phys. Rev. Res.*, 2:013129, 2020. URL https://link.aps.org/doi/10.1103/PhysRevResearch.2.013129.

[173] Thomas E. O'Brien, Michael Streif, Nicholas C. Rubin, Raffaele Santagati, Yuan Su, William J. Huggins, Joshua J. Goings, Nikolaj Moll, Elica Kyoseva, Matthias Degroote, Christofer S. Tautermann, Joonho Lee, Dominic W. Berry, Nathan Wiebe, and Ryan Babbush. Efficient quantum computation of molecular forces and other energy gradients. *Phys. Rev. Res.*, 4:043210, 2022. URL https://link.aps.org/doi/10.1103/PhysRevResearch.4.043210.

[174] Kenji Sugisaki, Hiroyuki Wakimoto, Kazuo Toyota, Kazunobu Sato, Daisuke Shiomi, and Takeji Takui. Quantum algorithm for numerical energy gradient calculations at the full configuration interaction level of theory. *The Journal of Physical Chemistry Letters*, 13(48):11105–11111, 2022. doi: 10.1021/acs.jpclett.2c02737. URL https://doi.org/10.1021/acs.jpclett.2c02737. PMID: 36444985.

[175] Juntao Lai, Yi Fan, Qiang Fu, Zhenyu Li, and Jinlong Yang. Accurate and efficient calculations of hellmann–feynman forces for quantum computation. *The Journal of Chemical Physics*, 159(11):114113, 09 2023. ISSN 0021-9606. doi: 10.1063/5.0167444. URL https://doi.org/10.1063/5.0167444.

[176] Thomas E. O'Brien, Bruno Senjean, Ramiro Sagastizabal, Xavier Bonet-Monroig, Alicja Dutkiewicz, Francesco Buda, Leonardo DiCarlo, and Lucas Visscher. Calculating energy derivatives for quantum chemistry on a quantum computer. *npj Quantum Information*, 5(1):113, 2019. URL https://doi.org/10.1038/s41534-019-0213-4.

[177] Dmitry A. Fedorov, Matthew J. Otten, Stephen K. Gray, and Yuri Alexeev. Ab initio molecular dynamics on quantum computers. *The Journal of Chemical Physics*, 154(16):164103, 2021. ISSN 0021-9606. URL https://doi.org/10.1063/5.0046930.

[178] Igor O. Sokolov, Panagiotis Kl. Barkoutsos, Lukas Moeller, Philippe Suchsland, Guglielmo Mazzola, and Ivano Tavernelli. Microcanonical and finite-temperature ab initio molecular dynamics simulations on quantum computers. *Phys. Rev. Res.*, 3:013125, 2021. URL https://link.aps.org/doi/10.1103/PhysRevResearch.3.013125.

[179] A.R. Hoy and P.R. Bunker. A precise solution of the rotation bending schrödinger equation for a triatomic molecule with application to the water molecule. *Journal of Molecular Spectroscopy*, 74(1):1–8, 1979. ISSN 0022-2852. URL https://www.sciencedirect.com/science/article/pii/0022285279900195.

[180] Jeffrey Cohn, Mario Motta, and Robert M. Parrish. Quantum filter diagonalization with compressed double-factorized hamiltonians. *PRX Quantum*, 2:040352, 2021. URL https://link.aps.org/doi/10.1103/PRXQuantum.2.040352.

[181] Dario Picozzi and Jonathan Tennyson. Symmetry-adapted encodings for qubit number reduction by point-group and other boolean symmetries. *Quantum Science and Technology*, 8(3):035026, 2023. URL https://dx.doi.org/10.1088/2058-9565/acd86c.

[182] N S Witte and Lloyd C L Hollenberg. Accurate calculation of ground-state energies in an analytic lanczos expansion. *Journal of Physics: Condensed Matter*, 9(9): 2031, 1997. URL https://dx.doi.org/10.1088/0953-8984/9/9/016.

[183] Kianna Wan, Mario Berta, and Earl T. Campbell. Randomized quantum algorithm for statistical phase estimation. *Phys. Rev. Lett.*, 129:030503, 2022. URL https://link.aps.org/doi/10.1103/PhysRevLett.129.030503.

[184] Thomas E O'Brien, Brian Tarasinski, and Barbara M Terhal. Quantum phase estimation of multiple eigenvalues for small-scale (noisy) experiments. *New Journal of Physics*, 21(2):023022, 2019. URL https://dx.doi.org/10.1088/1367-2630/aafb8e.

[185] Zhiyan Ding, Haoya Li, Lin Lin, HongKang Ni, Lexing Ying, and Ruizhe Zhang. Quantum Multiple Eigenvalue Gaussian filtered Search: an efficient and versatile quantum phase estimation method. *Quantum*, 8:1487, 2024. ISSN 2521-327X. URL https://doi.org/10.22331/q-2024-10-02-1487.